

PDF

Mathematics-I Suggestion (2025-2026)

For 1st Semester of Polytechnic

Top 300 Questions
with Complete Answer
From All Chapters



Prepared by **NatiTute** YouTube Channel

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এটা একটা পেইড pdf ebook যারা বইটি সংগ্রহ করছে তাদের কাছে অনুরোধ তোমরা এটা শেয়ার করো না, কারণ সবার কাছে pdf ফাইল টা পৌছে গেলে তোমরা তাদের থেকে এগিয়া থাকতে পারবে না এবং এই বই কেনার কোনো মর্যাদা থাকবে না.

1. The value of $\log_{\sqrt{3}} 27$ (use $\log_a a = 1$, rule):

(a) 6 (b) 3 (c) 1 (d) none

Solution:

$$\log_{\sqrt{3}} 27$$

$$= \log_{\sqrt{3}} 3^3 = \log_{\sqrt{3}} (\sqrt{3})^6 = 6 \cdot \log_{\sqrt{3}} \sqrt{3}$$

$$= 6 \cdot 1 = 6$$

Answer: (a)

2. The value of $2^{\log_2 5} + 9^{\log_3 \sqrt{3}}$ is-

(a) 9 (b) 7 (c) 8 (d) none of these

Solution:

$$2^{\log_2 5} + 9^{\log_3 \sqrt{3}}$$

$$= 5 + 3^{2 \log_3 \sqrt{3}}$$

$$= 5 + 3^{\log_3 (\sqrt{3})^2}$$

$$= 5 + 3^{\log_3 3} = 5 + 3 = 8$$

Answer: (c)

3. The equation $\log_a x + \log_a (1+x) = 0$ can be written as

a) $x^2 + x + 1 = 0$

b) $x^2 + x - 1 = 0$

c) $x^2 - x - 1 = 0$

d) $x^2 - x + 1 = 0$

Solution:

$$\log_a x + \log_a (1+x) = 0$$

$$\Rightarrow \log_a x (1+x) = \log_a 1$$

$$\Rightarrow x(1+x) = 1$$

$$\Rightarrow x^2 + x - 1 = 0$$

Answer: (b)

4. If $\log_2 x^2 = 2$ then find the value of x .

Solution:

$$\log_2 x^2 = 2$$

$$\Rightarrow 2 \log_2 x = 2$$

$$\Rightarrow \log_2 x = 1$$

$$\Rightarrow \log_2 x = \log_2 2$$

$$\Rightarrow x = 2 \text{ (Ans)}$$

5. When $\log_2 3 = x$, the value of $\log_8 27 = ?$

Solution:

$$\log_8 27 = \frac{\log 27}{\log 8} = \frac{\log 3^3}{\log 2^3}$$

$$= \frac{3 \log 3}{3 \log 2}$$

$$= \log_2 3 = x \text{ (Ans)}$$

6. Find the value of $\log_{\sqrt{7}} 343$

Solution:

$$\log_{\sqrt{7}} 343$$

$$= \log_{\sqrt{7}} 7^3$$

$$= \log_{\sqrt{7}} (\sqrt{7})^6$$

$$= 6 \log_{\sqrt{7}} \sqrt{7} = 6 \cdot 1 = 6 \text{ (Ans)}$$

7. Find the value of $\log_3 \log_2 \log_2 256$.

Solution:

$$\log_3 \log_2 \log_2 256$$

$$= \log_3 \log_2 \log_2 2^8$$

$$= \log_3 \log_2 8 \log_2 2$$

$$= \log_3 \log_2 (2^3 \cdot 1)$$

$$= \log_3 3 \log_2 2$$

$$= \log_3 (3 \cdot 1) = \log_3 3 = 1 \text{ (Ans)}$$

8. Find the value of $\log_b a \times \log_c b \times \log_d c$.

Solution:

$$\log_b a \times \log_c b \times \log_d c$$

$$= \frac{\log a}{\log b} \times \frac{\log b}{\log c} \times \frac{\log c}{\log d}$$

$$= \frac{\log a}{\log d} = \log_d a \text{ (Ans)}$$

9. If $2 \log_{\sqrt{2}} 2 = a$ then a is ____.

Solution:

$$2 \log_{\sqrt{2}} 2 = a$$

$$\Rightarrow \log_{\sqrt{2}} 2^2 = a$$

$$\Rightarrow a = \log_{\sqrt{2}} (\sqrt{2})^4$$

$$\Rightarrow a = 4 \log_{\sqrt{2}} \sqrt{2} = 4 \cdot 1 = 4 \text{ (Ans)}$$

10. If $\log_a x = 0.3$ and $\log_a 3 = 0.4$ then find the value of $\log_3 x$.

Solution:

$$\log_3 x = \frac{\log_a x}{\log_a 3} = \frac{0.3}{0.4} = 0.75 \text{ (Ans)}$$

11. Find the value of $\log_3 \log_2 \log_{\sqrt{3}} 81$.

Solution:

$$\log_3 \log_2 \log_{\sqrt{3}} 81$$

$$= \log_3 \log_2 \log_{\sqrt{3}} (\sqrt{3})^8$$

$$= \log_3 \log_2 8 \log_{\sqrt{3}} \sqrt{3}$$

$$= \log_3 \log_2 2^3 \cdot 1$$

$$= \log_3 3 \log_2 2 = 1 \cdot 1 = 1 \text{ (Ans)}$$

12. If $\log_{10} a + \log_{10} b = \log_{10}(a + b)$, then $b = ?$

Solution:

$$\begin{aligned}\log_{10} a + \log_{10} b &= \log_{10}(a + b) \\ \Rightarrow \log_{10} ab &= \log_{10}(a + b) \\ \Rightarrow ab &= a + b \\ \Rightarrow ab - b &= a \\ \Rightarrow b(a - 1) &= a \\ \Rightarrow b &= \frac{a}{a-1} \quad (\text{Ans})\end{aligned}$$

13. Simplify $\log_5 5 \log_4 9 \log_3 2$.

Solution:

$$\begin{aligned}\log_5 5 \log_4 9 \log_3 2 &= 1 \cdot \frac{\log 9}{\log 4} \cdot \log_3 2 \\ &= \frac{\log 3^2}{\log 2^2} \cdot \log_3 2 \\ &= \frac{2 \log 3}{2 \log 2} \cdot \frac{\log 2}{\log 3} = 1 \quad (\text{Ans})\end{aligned}$$

14. If $\log_e 2 \log_x 625 = \log_{10} 16 \log_e 10$ then find the value of x .

Solution:

$$\begin{aligned}\log_e 2 \log_x 625 &= \log_{10} 16 \log_e 10 \\ \Rightarrow \log_e 2 \log_x 5^4 &= \log_{10} 2^4 \log_e 10 \\ \Rightarrow \log_e 2 \times 4 \log_x 5 &= 4 \log_{10} 2 \log_e 10 \\ \Rightarrow \log_e 2 \times 4 \log_x 5 &= 4 \log_e 2 \\ \Rightarrow \log_x 5 &= \frac{4 \log_e 2}{4 \log_e 2} = 1 \\ \Rightarrow \log_5 x &= 1 \\ \Rightarrow \log_5 x &= \log_5 5 \\ \Rightarrow x &= 5 \quad (\text{Ans})\end{aligned}$$

15. If $\log_5 5 \log_2 x \log_{10} 2 = 2$ then $x = ?$

Solution:

$$\begin{aligned}\log_5 5 \log_2 x \log_{10} 2 &= 2 \\ \Rightarrow 1 \cdot \frac{\log x}{\log 2} \cdot \frac{\log 2}{\log 10} &= 2 \\ \Rightarrow \frac{\log x}{\log 10} &= 2 \\ \Rightarrow \log_{10} x &= 2 \\ \Rightarrow x &= 10^2 = 100 \quad (\text{Ans})\end{aligned}$$

16. If $\log_2 \log_x 81 = 1$ then find x .

Solution:

$$\begin{aligned}\log_2 \log_x 81 &= 1 \\ \Rightarrow \log_2 \log_x 81 &= \log_2 2 \\ \Rightarrow \log_x 81 &= 2 \\ \Rightarrow x^2 &= 81 \\ \Rightarrow x^2 &= 9^2 \\ \Rightarrow x &= \pm 9 \quad (\text{Ans})\end{aligned}$$

17. If $\log_x \log_2 \log_3 81 = 1$ then find x .

Solution:

$$\begin{aligned}\log_x \log_2 \log_3 81 &= 1 \\ \Rightarrow \log_x \log_2 \log_3 3^4 &= \log_x x \\ \Rightarrow \log_2 4 \log_3 3 &= x \\ \Rightarrow \log_2 2^2 \cdot 1 &= x \\ \Rightarrow 2 \log_2 2 &= x \\ \Rightarrow x &= 2 \cdot 1 = 2 \quad (\text{Ans})\end{aligned}$$

18. If $\log_{16} x + \log_4 x + \log_2 x = 14$ then $x = ?$

Solution:

$$\begin{aligned}\log_{16} x + \log_4 x + \log_2 x &= 14 \\ \Rightarrow \frac{\log x}{\log 16} + \frac{\log x}{\log 4} + \frac{\log x}{\log 2} &= 14 \\ \Rightarrow \log x \left(\frac{1}{\log 2^4} + \frac{1}{\log 2^2} + \frac{1}{\log 2} \right) &= 14 \\ \Rightarrow \log x \left(\frac{1}{4 \log 2} + \frac{1}{2 \log 2} + \frac{1}{\log 2} \right) &= 14 \\ \Rightarrow \frac{\log x}{\log 2} \left(\frac{1}{4} + \frac{1}{2} + 1 \right) &= 14 \\ \Rightarrow \log_2 x \left(\frac{1+2+4}{4} \right) &= 14 \\ \Rightarrow \log_2 x \times \left(\frac{7}{4} \right) &= 14 \\ \Rightarrow \log_2 x &= 14 \times \frac{4}{7} \\ \Rightarrow \log_2 x &= 8 \\ \Rightarrow x &= 2^8 = 256 \quad (\text{Ans})\end{aligned}$$

19. Find the value of x if

$$\log(x - 2) + \log(x - 3) = \log 2$$

Solution:

$$\begin{aligned}\log(x - 2) + \log(x - 3) &= \log 2 \\ \Rightarrow \log(x - 2)(x - 3) &= \log 2 \\ \Rightarrow (x - 2)(x - 3) &= 2 \\ \Rightarrow x^2 - 3x - 2x + 6 &= 2 \\ \Rightarrow x^2 - 5x + 4 &= 0 \\ \Rightarrow x^2 - 4x - x + 4 &= 0 \\ \Rightarrow x(x - 4) - 1(x - 4) &= 0 \\ \Rightarrow (x - 4)(x - 1) &= 0 \\ \therefore x &= 4 \text{ or } x = 1\end{aligned}$$

Since log of a negative number is not defined
so, $(x - 3) > 0$

$$\Rightarrow x > 3 \quad \therefore x = 4 \quad (\text{Ans})$$

20. Evaluate: $\log_{y^3} x \times \log_{z^3} y \times \log_{x^3} z$

Solution:

$$\begin{aligned}\log_{y^3} x \times \log_{z^3} y \times \log_{x^3} z &= \frac{\log x}{\log y^3} \times \frac{\log y}{\log z^3} \times \frac{\log z}{\log x^3} \\ &= \frac{\log x}{3 \log y} \times \frac{\log y}{3 \log z} \times \frac{\log z}{3 \log x} \\ &= \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27} \quad (\text{Ans})\end{aligned}$$

21. If $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b}$, prove that $x^{b+c} \cdot y^{c+a} \cdot z^{a+b} = 1$.

Solution:

$$\begin{aligned} \text{Suppose, } \frac{\log x}{b-c} &= \frac{\log y}{c-a} = \frac{\log z}{a-b} = k \\ \therefore \log x &= k(b-c) \\ \Rightarrow (b+c) \log x &= k(b+c)(b-c) \\ \Rightarrow \log x^{b+c} &= k(b^2 - c^2) \\ \text{and, } \log y &= k(c-a) \\ \Rightarrow (c+a) \log y &= k(c+a)(c-a) \\ \Rightarrow \log y^{c+a} &= k(c^2 - a^2) \\ \text{and, } \log z &= k(a-b) \\ \Rightarrow (a+b) \log z &= k(a+b)(a-b) \\ \Rightarrow \log z^{a+b} &= k(a^2 - b^2) \\ \therefore \log x^{b+c} + \log y^{c+a} + \log z^{a+b} &= k(b^2 - c^2) + k(c^2 - a^2) + k(a^2 - b^2) \\ &= k(b^2 - c^2 + c^2 - a^2 + a^2 - b^2) = k \times 0 = 0 \\ \therefore \log x^{b+c} + \log y^{c+a} + \log z^{a+b} &= 0 \\ \Rightarrow \log(x^{b+c} \cdot y^{c+a} \cdot z^{a+b}) &= \log 1 \\ \Rightarrow x^{b+c} \cdot y^{c+a} \cdot z^{a+b} &= 1 \text{ (Proved)} \end{aligned}$$

22. Find the value of

$$\log \tan 1^\circ + \log \tan 2^\circ + \dots + \log \tan 89^\circ$$

Solution:

$$\begin{aligned} \log \tan 1^\circ + \log \tan 2^\circ + \dots + \log \tan 89^\circ \\ &= \log[\tan 1^\circ \cdot \tan 2^\circ \dots \tan 89^\circ] \\ &= \log[\tan 1^\circ \cdot \tan 2^\circ \dots \tan(90^\circ - 1^\circ)] \\ &= \log[\tan 1^\circ \cdot \tan 2^\circ \dots \tan 45^\circ \dots \cot 1^\circ] \\ &= \log[1.1.1 \dots 1] = \log 1 = 0 \text{ (Ans)} \end{aligned}$$

23. Show that, $\frac{1}{\log_a bc+1} + \frac{1}{\log_b ca+1} + \frac{1}{\log_c ab+1} = 1$

Solution:

L.H.S

$$\begin{aligned} \frac{1}{\log_a bc+1} + \frac{1}{\log_b ca+1} + \frac{1}{\log_c ab+1} \\ &= \frac{1}{\log_a bc + \log_a a} + \frac{1}{\log_b ca + \log_b b} + \frac{1}{\log_c ab + \log_c c} \\ &= \frac{1}{\log_a abc} + \frac{1}{\log_b abc} + \frac{1}{\log_c abc} \\ &= \log_{abc} a + \log_{abc} b + \log_{abc} c \\ &= \log_{abc} abc = 1 = \text{R.H.S (Proved)} \end{aligned}$$

24. Find the value of $(\log_3 63 - \log_{27} 343)$.

Solution:

$$\begin{aligned} \log_3 63 - \log_{27} 343 \\ &= \log_3 (9 \times 7) - \frac{\log 343}{\log 27} \\ &= \log_3 9 + \log_3 7 - \frac{\log 7^3}{\log 3^3} \\ &= \log_3 3^2 + \log_3 7 - \frac{3 \log 7}{3 \log 3} \\ &= 2 \log_3 3 + \log_3 7 - \log_3 7 \\ &= 2.1 = 2 \text{ (Ans)} \end{aligned}$$

25. If $a^x = bc$, $b^y = ca$, $c^z = ab$ show that

$$\frac{x}{x+1} + \frac{y}{y+1} + \frac{z}{z+1} = 2$$

Solution:

$$\begin{aligned} a^x &= bc \\ \therefore x &= \log_a bc \\ \Rightarrow x+1 &= \log_a bc + 1 \\ \Rightarrow x+1 &= \log_a bc + \log_a a \\ \Rightarrow x+1 &= \log_a abc \\ \Rightarrow \frac{x}{x+1} &= \frac{\log_a bc}{\log_a abc} = \log_{(abc)} bc \end{aligned}$$

Again, $b^y = ca$

$$\therefore y = \log_b ca$$

$$\Rightarrow y+1 = \log_b ca + 1$$

$$\Rightarrow y+1 = \log_b ca + \log_b b$$

$$\Rightarrow y+1 = \log_b abc$$

$$\Rightarrow \frac{y}{y+1} = \frac{\log_b ca}{\log_b abc} = \log_{(abc)} ca$$

Again, $c^z = ab$

$$\therefore z = \log_c ab$$

$$\Rightarrow z+1 = \log_c ab + 1$$

$$\Rightarrow z+1 = \log_c ab + \log_c c$$

$$\Rightarrow z+1 = \log_c abc$$

$$\Rightarrow \frac{z}{z+1} = \frac{\log_c ab}{\log_c abc} = \log_{(abc)} ab$$

Now, L.H.S

$$\begin{aligned} &= \frac{x}{x+1} + \frac{y}{y+1} + \frac{z}{z+1} \\ &= \log_{(abc)} bc + \log_{(abc)} ca + \log_{(abc)} ab \\ &= \log_{(abc)} (bc \cdot ca \cdot ab) \\ &= \log_{(abc)} (abc)^2 \\ &= 2 \log_{(abc)} (abc) = 2.1 = 2 = \text{R.H.S (Proved)} \end{aligned}$$

26. Find the logarithm of 324 to the base $3\sqrt{2}$.

Solution:

$$\begin{aligned} \text{The logarithm of 324 to the base } 3\sqrt{2} \\ &= \log_{3\sqrt{2}} 324 \\ &= \log_{3\sqrt{2}} (3\sqrt{2})^4 = 4 \log_{3\sqrt{2}} (3\sqrt{2}) = 4.1 = 4 \text{ (Ans)} \end{aligned}$$

27. Find the logarithm of 729 to the base 3.

Solution:

$$\begin{aligned} \text{The logarithm of 729 to the base 3} \\ &= \log_3 729 \\ &= \log_3 (3)^6 = 6 \log_3 3 = 6.1 = 6 \text{ (Ans)} \end{aligned}$$

28. If $\frac{1}{\log_3 x} = \frac{1}{9}$, then find the value of x .

Solution:

$$\begin{aligned} \frac{1}{\log_3 x} &= \frac{1}{9} \\ \Rightarrow \log_3 x &= 9 \\ \Rightarrow x &= 3^9 = 19683 \text{ (Ans)} \end{aligned}$$

29. If $\frac{1}{\log_x 4} + 1 = \frac{2}{\log_2 4}$, then find the value of x .

Solution:

$$\begin{aligned}\frac{1}{\log_x 4} + 1 &= \frac{2}{\log_2 4} \\ \Rightarrow \log_4 x + \log_4 4 &= 2 \log_4 2 \\ \Rightarrow \log_4 (4x) &= \log_4 (2^2) \\ \Rightarrow 4x &= 2^2 \\ \Rightarrow 4x &= 4 \\ \Rightarrow x &= 1 \quad (\text{Ans})\end{aligned}$$

30. If $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b}$, prove that $x^a \cdot y^b \cdot z^c = 1$.

Solution:

$$\begin{aligned}\text{Suppose, } \frac{\log x}{b-c} &= \frac{\log y}{c-a} = \frac{\log z}{a-b} = k \\ \therefore \log x &= k(b-c) \\ \Rightarrow a \log x &= ka(b-c) \\ \Rightarrow \log x^a &= k(ab-ca) \\ \text{and, } \log y &= k(c-a) \\ \Rightarrow b \log y &= kb(c-a) \\ \Rightarrow \log y^b &= k(bc-ab) \\ \text{and, } \log z &= k(a-b) \\ \Rightarrow c \log z &= kc(a-b) \\ \Rightarrow \log z^c &= k(ca-bc) \\ \therefore \log x^a + \log y^b + \log z^c &= k(ab-ca) + k(bc-ab) + k(ca-bc) \\ &= k(ab-ca+bc-ab+ca-bc) = k \times 0 = 0 \\ \therefore \log x^a + \log y^b + \log z^c &= 0 \\ \Rightarrow \log(x^a \cdot y^b \cdot z^c) &= \log 1 \\ \Rightarrow x^a \cdot y^b \cdot z^c &= 1 \quad (\text{Proved})\end{aligned}$$

31. Prove that

$$2 \log(a+b) = 2 \log a + \log \left(1 + 2 \frac{b}{a} + \frac{b^2}{a^2}\right)$$

Solution:

L.H.S.

$$\begin{aligned}2 \log(a+b) &= \log(a+b)^2 \\ &= \log \left\{ a \left(1 + \frac{b}{a}\right) \right\}^2 \\ &= \log a^2 \left(1 + \frac{b}{a}\right)^2 \\ &= \log a^2 + \log \left(1 + \frac{b}{a}\right)^2 \\ &= 2 \log a + \log \left\{ 1^2 + 2 \cdot 1 \cdot \frac{b}{a} + \left(\frac{b}{a}\right)^2 \right\} \\ &= 2 \log a + \log \left(1 + 2 \frac{b}{a} + \frac{b^2}{a^2}\right) \quad (\text{Proved})\end{aligned}$$

32. If $\frac{\log x}{y+z} = \frac{\log y}{z+x} = \frac{\log z}{x+y}$, prove that $\left(\frac{x}{y}\right)^z \left(\frac{y}{z}\right)^x \left(\frac{z}{x}\right)^y = 1$

Solution:

$$\begin{aligned}\text{Let, } \frac{\log x}{y+z} &= \frac{\log y}{z+x} = \frac{\log z}{x+y} = k \\ \therefore \log x &= k(y+z) \\ \Rightarrow (y-z) \log x &= k(y-z)(y+z) \\ \Rightarrow \log x^{(y-z)} &= k(y^2 - z^2) \\ \Rightarrow \log \left(\frac{x^y}{x^z}\right) &= k(y^2 - z^2) \\ \text{and, } \log y &= k(z+x) \\ \Rightarrow (z-x) \log y &= k(z-x)(z+x) \\ \Rightarrow \log y^{(z-x)} &= k(z^2 - x^2) \\ \Rightarrow \log \left(\frac{y^z}{y^x}\right) &= k(z^2 - x^2) \\ \text{and, } \log z &= k(x+y) \\ \Rightarrow (x-y) \log z &= k(x-y)(x+y) \\ \Rightarrow \log z^{(x-y)} &= k(x^2 - y^2) \\ \Rightarrow \log \left(\frac{z^x}{z^y}\right) &= k(x^2 - y^2) \\ \therefore \log \left(\frac{x^y}{x^z}\right) + \log \left(\frac{y^z}{y^x}\right) + \log \left(\frac{z^x}{z^y}\right) &= k(y^2 - z^2) + k(z^2 - x^2) + k(x^2 - y^2) \\ &= k(y^2 - z^2 + z^2 - x^2 + x^2 - y^2) \\ &= k \times 0 = 0 \\ \therefore \log \left(\frac{x^y}{x^z}\right) + \log \left(\frac{y^z}{y^x}\right) + \log \left(\frac{z^x}{z^y}\right) &= 0 \\ \Rightarrow \log \left(\frac{x^y}{x^z} \times \frac{y^z}{y^x} \times \frac{z^x}{z^y}\right) &= \log 1 \\ \Rightarrow \frac{x^y}{x^z} \times \frac{y^z}{y^x} \times \frac{z^x}{z^y} &= 1 \\ \Rightarrow \left(\frac{y}{x}\right)^z \left(\frac{z}{y}\right)^x \left(\frac{x}{z}\right)^y &= 1 \\ \Rightarrow \left(\frac{x}{y}\right)^z \left(\frac{y}{z}\right)^x \left(\frac{z}{x}\right)^y &= 1 \quad (\text{Proved})\end{aligned}$$

33. If $x = \log_a bc$, $y = \log_b ca$, $z = \log_c ab$ then show that $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} = 1$.

Solution:

L.H.S.

$$\begin{aligned}\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} &= \frac{1}{\log_a bc + 1} + \frac{1}{\log_b ca + 1} + \frac{1}{\log_c ab + 1} \\ &= \frac{1}{\log_a bc + \log_a a} + \frac{1}{\log_b ca + \log_b b} + \frac{1}{\log_c ab + \log_c c} \\ &= \frac{1}{\log_a abc} + \frac{1}{\log_b abc} + \frac{1}{\log_c abc} \\ &= \log_{abc} a + \log_{abc} b + \log_{abc} c \\ &= \log_{abc} abc = 1 = \text{R.H.S.} \quad (\text{Proved})\end{aligned}$$

1. If $i = \sqrt{-1}$, the value of $i^{197} = ?$

Solution:

$$i = \sqrt{-1} \therefore i^2 = -1 \Rightarrow i^4 = 1$$

now, i^{197}

$$= i^{(4 \times 49 + 1)}$$

$$= (i^4)^{49} \cdot i = (1)^{49} \cdot i = i \text{ (Ans)}$$

2. If x, y are real and $(x + 3i)$ and $(-2 + iy)$ are conjugate to each other, find the value of x and y .

Solution:

Conjugate of $(x + 3i)$ is $(x - 3i)$

$$\therefore x - 3i = -2 + iy$$

Comparing both side we get,

$$x = -2, y = -3 \text{ (Ans)}$$

3. Find the square root of $(3 + 4i)$.

Solution:

$$3 + 4i$$

$$= 4 - 1 + 4i$$

$$= 2^2 + 2 \cdot 2 \cdot i + i^2$$

$$= (2 + i)^2$$

$$\therefore \text{The square root of } (3 + 4i)$$

$$= \pm(2 + i) \text{ (Ans)}$$

4. Find the square root of $\frac{7-24i}{3+4i}$.

Solution:

$$\frac{7-24i}{3+4i} = \frac{4^2 - 2 \cdot 4 \cdot 3i + (3i)^2}{2^2 + 2 \cdot 2 \cdot i + i^2} = \frac{(4-3i)^2}{(2+i)^2}$$

$$\therefore \text{Square root of } \frac{7-24i}{3+4i}$$

$$= \pm \frac{4-3i}{2+i}$$

$$= \pm \frac{(4-3i)(2-i)}{(2+i)(2-i)}$$

$$= \pm \frac{8-4i-6i-3}{4+1}$$

$$= \pm \frac{5-10i}{5} = \pm \frac{5(1-2i)}{5} = \pm(1 - 2i) \text{ (Ans)}$$

5. Find the modulus of $\frac{7-24i}{3+4i}$.

Solution:

$$\left| \frac{7-24i}{3+4i} \right|$$

$$= \frac{|7-24i|}{|3+4i|}$$

$$= \frac{\sqrt{7^2 + (-24)^2}}{\sqrt{3^2 + 4^2}}$$

$$= \sqrt{\frac{49+576}{9+16}} = \sqrt{\frac{625}{25}} = \sqrt{25} = 5 \text{ (Ans.)}$$

6. Find the modulus of $(a - ib)^2$.

Solution:

$$(a - ib)^2$$

$$= a^2 - 2 \cdot a \cdot ib + i^2 b^2$$

$$= a^2 - 2abi - b^2 \quad [\because i^2 = -1]$$

$$= (a^2 - b^2) + i(-2ab) \text{ [(x+iy) form]}$$

$\therefore$ Modulus of this complex number is-

$$\sqrt{(a^2 - b^2)^2 + (-2ab)^2}$$

$$= \sqrt{(a^2 - b^2)^2 + 4a^2 b^2}$$

$$= \sqrt{(a^2 + b^2)^2} = a^2 + b^2 \text{ (Ans)}$$

7. Find the argument of $\sqrt{12} + 6 \left\{ \frac{(1-i)}{(1+i)} \right\}$

Solution:

$$\sqrt{12} + 6 \left\{ \frac{(1-i)}{(1+i)} \right\}$$

$$= \sqrt{12} + 6 \left\{ \frac{(1-i)^2}{(1+i)(1-i)} \right\}$$

$$= \sqrt{12} + 6 \left\{ \frac{1-2i+i^2}{1-i^2} \right\}$$

$$= \sqrt{12} + 6 \left\{ \frac{-2i}{2} \right\}$$

$$= \sqrt{12} - 6i$$

$$\therefore \text{Arguments} = \tan^{-1} \left(\frac{-6}{\sqrt{12}} \right)$$

$$= -\tan^{-1} \left(\frac{6\sqrt{12}}{12} \right)$$

$$= -\tan^{-1} \left(\frac{2\sqrt{3}}{2} \right)$$

$$= -\tan^{-1}(\sqrt{3}) = -\frac{\pi}{3} \text{ (Ans)}$$

8. The smallest integer for which $\left(\frac{1+i}{1-i} \right)^n = 1, n = ?$

Solution:

$$\left(\frac{1+i}{1-i} \right)^n = 1$$

$$\text{or } \left\{ \frac{(1+i)^2}{(1-i)(1+i)} \right\}^n = 1$$

$$\text{or } \left\{ \frac{1^2 + 2i + i^2}{1^2 - i^2} \right\}^n = 1$$

$$\text{or } \left\{ \frac{2i}{2} \right\}^n = 1$$

$$\text{or } \{i\}^n = 1$$

$$\text{As, } i^4 = 1 \therefore n = 4 \text{ (Ans)}$$

9. If $i = \sqrt{-1}$ then find the value of $(\sqrt{i} + \sqrt{-i})$.

Solution:

$$\begin{aligned} & \sqrt{i} + \sqrt{-i} \\ &= \frac{1}{\sqrt{2}}(\sqrt{2}\sqrt{i} + \sqrt{2}\sqrt{-i}) \\ &= \frac{1}{\sqrt{2}}(\sqrt{2i} + \sqrt{-2i}) \\ &= \frac{1}{\sqrt{2}}(\sqrt{1+2i+i^2} + \sqrt{1-2i+i^2}) \\ &= \frac{1}{\sqrt{2}}(\sqrt{(1+i)^2} + \sqrt{(1-i)^2}) \\ &= \frac{1}{\sqrt{2}}(1+i+1-i) \\ &= \frac{1}{\sqrt{2}} \times 2 = \sqrt{2} \quad (\text{Ans}) \end{aligned}$$

10. Find the value of $(1 - \omega + \omega^2)(1 + \omega - \omega^2)$

Solution:

$$\begin{aligned} & \text{We know, } 1 + \omega + \omega^2 = 0 \\ & \therefore 1 + \omega = -\omega^2 \text{ \& } 1 + \omega^2 = -\omega \\ & \text{So, } (1 - \omega + \omega^2)(1 + \omega - \omega^2) \\ &= (-\omega - \omega)(-\omega^2 - \omega^2) \\ &= (-2\omega)(-2\omega^2) \\ &= 4\omega^3 = 4.1 = 4 \quad (\text{Ans}) \end{aligned}$$

11. Find the value of $(1 + \omega)^3 - (1 + \omega^2)^3$.

Solution:

$$\begin{aligned} & \text{We know, } 1 + \omega + \omega^2 = 0 \\ & \therefore 1 + \omega = -\omega^2 \text{ \& } 1 + \omega^2 = -\omega \\ & \text{So, } (1 + \omega)^3 - (1 + \omega^2)^3 \\ &= \{-\omega^2\}^3 - \{-\omega\}^3 \\ &= -(\omega^3)^2 + \omega^3 \\ &= -1^2 + 1 [\because \omega^3 = 1] \\ &= -1 + 1 = 0 \quad (\text{Ans}) \end{aligned}$$

12. Find the value of $(1 - \omega + \omega^2)^5 + (1 + \omega - \omega^2)^5$

Solution:

$$\begin{aligned} & \text{We know, } 1 + \omega + \omega^2 = 0 \\ & \therefore 1 + \omega = -\omega^2 \text{ \& } 1 + \omega^2 = -\omega \\ & \text{So, } (1 - \omega + \omega^2)^5 + (1 + \omega - \omega^2)^5 \\ &= (-\omega - \omega)^5 + (-\omega^2 - \omega^2)^5 \\ &= (-2\omega)^5 + (-2\omega^2)^5 \\ &= -32\omega^5 - 32\omega^{10} \\ &= -32\omega^3\omega^2 - 32(\omega^3)^3\omega \\ &= -32.1.\omega^2 - 32.1^3\omega \quad \because [\omega^3 = 1] \\ &= -32(\omega^2 + \omega) \\ &= -32(-1) = 32 \quad (\text{Ans.}) \end{aligned}$$

13. The value of the expression $\omega^2(1 + i\omega)(i\omega - 1)$ is-

- (a) 1
(b) -2
(c) -1
(d) 0

Solution:

$$\begin{aligned} & \omega^2(1 + i\omega)(i\omega - 1) \\ &= \omega^2(i^2\omega^2 - 1) \\ &= \omega^2(-\omega^2 - 1) \\ &= -\omega^4 - \omega^2 \\ &= -\omega - \omega^2 = -(\omega + \omega^2) = -(-1) = 1 \\ & \text{Answer: (a)} \end{aligned}$$

14. Find the amplitude of complex number $\frac{1}{1-i}$

Solution:

$$\begin{aligned} & \text{Let, } z = \frac{1}{1-i} = \frac{1+i}{(1-i)(1+i)} = \frac{1+i}{1+1} = \frac{1+i}{2} \\ & \Rightarrow z = \frac{1}{2} + \frac{1}{2}i \\ & \therefore \text{amp}(z) = \tan^{-1}\left(\frac{\frac{1}{2}}{\frac{1}{2}}\right) \\ &= \tan^{-1} 1 = \frac{\pi}{4} \quad (\text{Ans.}) \end{aligned}$$

15. If $(2 + i\sqrt{3})$ is a root of equation $x^2 - px + q = 0$, where p and q are real, then find the value of p and q .

Solution:

$$\begin{aligned} & \text{Let, one root of the equation } \alpha = 2 + i\sqrt{3} \\ & \therefore \text{other root } \beta = 2 - i\sqrt{3} \\ & \text{So, } \alpha + \beta = 2 + i\sqrt{3} + 2 - i\sqrt{3} = 4 \\ & \text{And, } \alpha\beta = (2 + i\sqrt{3})(2 - i\sqrt{3}) \\ &= 2^2 - (i\sqrt{3})^2 = 4 + 3 = 7 \\ & \text{Comparing } x^2 - px + q = 0 \\ & \text{With } x^2 - (\alpha + \beta)x + \alpha\beta = 0 \\ & \text{We get } p = \alpha + \beta = 4, q = \alpha\beta = 7 \\ & \text{So, } p = 4, q = 7 \quad (\text{Ans}) \end{aligned}$$

16. Form a quadratic equation with real co-efficients, whose one root is $(2 - 3i)$.

Solution:

∵ One root of equation is $(2 - 3i)$
 ∵ Other root will be $(2 + 3i)$
 ∵ Sum of roots = $2 - 3i + 2 + 3i = 4$
 and product of roots = $(2 - 3i)(2 + 3i)$

$$= 2^2 - (3i)^2$$

$$= 4 + 9 = 13$$

 ∵ The required equation will be,

$$x^2 - 4x + 13 = 0 \quad (\text{Ans})$$

17. $x = \cos \theta + i \sin \theta$, then $x^m + \frac{1}{x^m} = ?$

Solution:

$x = \cos \theta + i \sin \theta$
 By De Moivre's theorem we can say,
 $x^m = \cos m\theta + i \sin m\theta$
 And $x^{-m} = \cos m\theta - i \sin m\theta$

$$\therefore x^m + x^{-m}$$

$$= \cos m\theta + i \sin m\theta + \cos m\theta - i \sin m\theta$$

$$= 2 \cos m\theta$$

$$\therefore x^m + x^{-m} = 2 \cos m\theta$$

$$\Rightarrow x^m + \frac{1}{x^m} = 2 \cos m\theta \quad (\text{Ans})$$

18. If $\frac{1-ix}{1+ix} = a - ib$ then prove that $a^2 + b^2 = 1$

Solution:

$$\frac{1-ix}{1+ix} = a - ib$$

$$\Rightarrow 1 - ix = (a - ib)(1 + ix)$$

$$\Rightarrow 1 - ix = a + aix - ib - i^2bx$$

$$\Rightarrow 1 - ix = (a + bx) - i(b - ax)$$

 Since x is real, equating real and imaginary parts we get,

$$a + bx = 1$$

$$\Rightarrow bx = 1 - a, \Rightarrow x = \frac{1-a}{b}$$

 and, $b - ax = x$

$$\Rightarrow ax + x = b$$

$$\Rightarrow x(a + 1) = b, \Rightarrow x = \frac{b}{a+1}$$

$$\therefore \frac{1-a}{b} = \frac{b}{a+1}$$

$$\Rightarrow b^2 = (1+a)(1-a)$$

$$\Rightarrow b^2 = 1 - a^2$$

$$\Rightarrow a^2 + b^2 = 1 \quad (\text{Proved})$$

19. If $x = a + b$, $y = a\omega^2 + b\omega$ and $z = a\omega + b\omega^2$ show that $x^3 + y^3 + z^3 = 3(a^3 + b^3)$ where ω and ω^2 are cube root of unity.

Solution:

$x = a + b$, $y = a\omega^2 + b\omega$ and $z = a\omega + b\omega^2$

$$\therefore x + y + z$$

$$= a + b + a\omega^2 + b\omega + a\omega + b\omega^2$$

$$= a(1 + \omega + \omega^2) + b(1 + \omega + \omega^2)$$

$$= a \cdot 0 + b \cdot 0 = 0$$

$$\therefore x^3 + y^3 + z^3$$

$$= 3xyz \quad [\because x + y + z = 0]$$

$$= 3(a + b)(a\omega^2 + b\omega)(a\omega + b\omega^2)$$

$$= 3(a + b)(a^2\omega^3 + ab\omega^4 + ab\omega^2 + b^2\omega^3)$$

$$= 3(a + b)\{a^2 \cdot 1 + ab\omega + ab\omega^2 + b^2 \cdot 1\}$$

$$= 3(a + b)\{a^2 + ab(\omega + \omega^2) + b^2\}$$

$$= 3(a + b)(a^2 - ab + b^2)$$

$$= 3(a^3 + b^3) \quad (\text{Proved})$$

20. If $\sqrt[3]{x + iy} = a + ib$, then show that

$$\frac{x}{a} + \frac{y}{b} = 4(a^2 - b^2)$$

Solution:

$$\sqrt[3]{x + iy} = a + ib$$

 Or $x + iy = (a + ib)^3$
 Or $x + iy = a^3 + 3a^2ib + 3ai^2b^2 + i^3b^3$
 Or $x + iy = a^3 + 3a^2ib - 3ab^2 - ib^3$
 Or $x + iy = (a^3 - 3ab^2) + i(3a^2b - b^3)$

$$\therefore x = a^3 - 3ab^2$$

 Or $x = a(a^2 - 3b^2)$
 Or, $\frac{x}{a} = a^2 - 3b^2$
 And, $y = 3a^2b - b^3$
 Or, $y = b(3a^2 - b^2)$
 Or, $\frac{y}{b} = 3a^2 - b^2$

$$\therefore \frac{x}{a} + \frac{y}{b} = a^2 - 3b^2 + 3a^2 - b^2$$

$$= 4a^2 - 4b^2$$

$$= 4(a^2 - b^2) \quad (\text{Proved})$$

21. Find the value of $\left(\frac{i^{2021} + i^{2020}}{i^{2021} + i^{2000}}\right)$

Solution:

$$\frac{i^{2021} + i^{2020}}{i^{2021} + i^{2000}}$$

$$= \frac{i^{2020}(i + 1)}{i^{2000}(i^{21} + 1)}$$

$$= \frac{i^{20}(1 + i)}{(i^4)^5 \cdot i + 1}$$

$$= \frac{(i^4)^5(1 + i)}{1^5 \cdot i + 1} = \frac{1 + i}{1 + i} = 1 \quad (\text{Ans})$$

22. If $x + iy = \sqrt{\frac{a+ib}{c+id}}$, then show that,

$$(x^2 + y^2)^2 = \frac{a^2 + b^2}{c^2 + d^2}$$

Solution:

$$x + iy = \sqrt{\frac{a+ib}{c+id}}$$

$$\text{Or } (x + iy)^2 = \frac{a+ib}{c+id}$$

$$\text{Or, } x^2 + 2xyi + i^2y^2 = \frac{(a+ib)(c-id)}{(c+id)(c-id)}$$

$$\text{Or, } x^2 - y^2 + 2xyi = \frac{ac - aid + ibc - i^2bd}{c^2 - i^2d^2}$$

$$\text{Or, } x^2 - y^2 + 2xyi = \frac{(ac+bd) + i(bc-ad)}{c^2 + d^2}$$

$$\text{Or, } x^2 - y^2 + 2xyi = \frac{ac+bd}{c^2 + d^2} + i \frac{bc-ad}{c^2 + d^2}$$

$$\therefore x^2 - y^2 = \frac{ac+bd}{c^2 + d^2}, 2xy = \frac{bc-ad}{c^2 + d^2}$$

Now,

$$(x^2 + y^2)^2$$

$$= (x^2 - y^2)^2 + 4x^2y^2$$

$$= \left(\frac{ac+bd}{c^2 + d^2}\right)^2 + \left(\frac{bc-ad}{c^2 + d^2}\right)^2$$

$$= \frac{a^2c^2 + 2acbd + b^2d^2 + b^2c^2 - 2bcad + a^2d^2}{(c^2 + d^2)^2}$$

$$= \frac{a^2(c^2 + d^2) + b^2(c^2 + d^2)}{(c^2 + d^2)^2}$$

$$= \frac{(a^2 + b^2)(c^2 + d^2)}{(c^2 + d^2)^2} = \frac{a^2 + b^2}{c^2 + d^2} \text{ (Proved)}$$

23. If $z_1 = 3 + 5i$, $z_2 = 1 - i$, verify the relation

$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$$

Solution:

$$\text{L.H.S, } \left|\frac{z_1}{z_2}\right|$$

$$= \left|\frac{3+5i}{1-i}\right|$$

$$= \left|\frac{(3+5i)(1+i)}{(1-i)(1+i)}\right|$$

$$= \left|\frac{3+3i+5i+5i^2}{1^2 - i^2}\right|$$

$$= \left|\frac{3+8i-5}{1+1}\right|$$

$$= \left|\frac{-2+8i}{2}\right|$$

$$= \left|\frac{2(-1+4i)}{2}\right|$$

$$= |-1 + 4i| = \sqrt{(-1)^2 + 4^2} = \sqrt{17}$$

R.H.S

$$\frac{|z_1|}{|z_2|} = \frac{|3+5i|}{|1-i|} = \frac{\sqrt{3^2+5^2}}{\sqrt{1^2+(-1)^2}} = \frac{\sqrt{34}}{2} = \sqrt{17}$$

$$\therefore \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|} \text{ OK.}$$

24. Find the value of $\omega^{2022} + \omega^{2023} + \omega^{2024}$

Solution:

$$\omega^{2022} + \omega^{2023} + \omega^{2024}$$

$$= \omega^{2022}(1 + \omega + \omega^2)$$

$$= \omega^{2022} \times 0 \quad [\because 1 + \omega + \omega^2 = 0]$$

$$= 0 \text{ (Ans)}$$

25. Find the value of

$$(\omega^{2021} + \omega^{2020} - 1)(\omega^{2021} + \omega^{2020} + 1)$$

Solution:

$$\omega^{2021} = \omega^{2019+2} = (\omega^3)^{673} \cdot \omega^2 = 1 \cdot \omega^2 = \omega^2$$

$$\omega^{2020} = \omega^{2019+1} = (\omega^3)^{673} \cdot \omega = 1 \cdot \omega = \omega$$

$$\therefore (\omega^{2021} + \omega^{2020} - 1)(\omega^{2021} + \omega^{2020} + 1)$$

$$= (\omega^2 + \omega - 1)(\omega^2 + \omega + 1)$$

$$= (\omega^2 + \omega - 1) \times 0 \quad [\because \omega^2 + \omega + 1 = 0]$$

$$= 0 \text{ (Ans)}$$

26. If $x + \frac{1}{x} = 2 \cos \frac{\pi}{5}$, using De Moivre's theorem, prove

$$\text{that } x^5 + \frac{1}{x^5} = -2$$

Solution:

$$\text{Given, } x + \frac{1}{x} = 2 \cos \frac{\pi}{5}$$

$$\Rightarrow x^2 + 1 = 2x \cos \frac{\pi}{5}$$

$$\Rightarrow x^2 - 2x \cos \frac{\pi}{5} + 1 = 0$$

$$\therefore x = \frac{2 \cos \frac{\pi}{5} \pm \sqrt{4 \cos^2 \frac{\pi}{5} - 4.1.1}}{2.1}$$

$$\Rightarrow x = \frac{2 \cos \frac{\pi}{5} \pm 2 \sqrt{-(1 - \cos^2 \frac{\pi}{5})}}{2}$$

$$\Rightarrow x = \frac{2[\cos \frac{\pi}{5} \pm \sqrt{-1} \sin \frac{\pi}{5}]}{2}$$

$$\Rightarrow x = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \text{ (Taking +ve sign only)}$$

$$\therefore x^5 + \frac{1}{x^5} = x^5 + x^{-5}$$

$$= \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}\right)^5 + \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}\right)^{-5}$$

$$= \cos 5 \times \frac{\pi}{5} + i \sin 5 \times \frac{\pi}{5} + \cos 5 \times \frac{\pi}{5} - i \sin 5 \times \frac{\pi}{5}$$

$$= 2 \cos \pi = 2 \cdot (-1) = -2 \text{ (Proved)}$$

27. If $x + \frac{1}{x} = 2 \cos \theta$ and $y + \frac{1}{y} = 2 \cos \phi$, using De

Moivre's theorem, show that

$$x^m y^n + \frac{1}{x^m y^n} = 2 \cos(m\theta + n\phi)$$

Solution:

$$\text{Given, } x + \frac{1}{x} = 2 \cos \theta$$

$$\Rightarrow x^2 + 1 = 2x \cos \theta$$

$$\Rightarrow x^2 - 2x \cos \theta + 1 = 0$$

$$\therefore x = \frac{2 \cos \theta \pm \sqrt{4 \cos^2 \theta - 4.1.1}}{2.1}$$

$$\Rightarrow x = \frac{2 \cos \theta \pm 2\sqrt{-(1-\cos^2 \theta)}}{2}$$

$$\Rightarrow x = \frac{2[\cos \theta \pm \sqrt{1-\sin^2 \theta}]}{2}$$

$$\Rightarrow x = \cos \theta + i \sin \theta \quad (\text{Taking +ve sign only})$$

$$\text{Similarly } y = \cos \phi + i \sin \phi$$

$$\therefore x^m y^n = (\cos \theta + i \sin \theta)^m (\cos \phi + i \sin \phi)^n$$

By De Moivre's Theorem,

$$\Rightarrow x^m y^n = (\cos m\theta + i \sin m\theta)(\cos n\phi + i \sin n\phi)$$

$$\Rightarrow x^m y^n = \cos(m\theta + n\phi) + i \sin(m\theta + n\phi)$$

$$\therefore x^{-m} y^{-n} = \cos(m\theta + n\phi) - i \sin(m\theta + n\phi)$$

$$\therefore x^m y^n + x^{-m} y^{-n}$$

$$= \cos(m\theta + n\phi) + i \sin(m\theta + n\phi)$$

$$+ \cos(m\theta + n\phi) - i \sin(m\theta + n\phi)$$

$$= 2 \cos(m\theta + n\phi)$$

$$\therefore x^m y^n + x^{-m} y^{-n} = 2 \cos(m\theta + n\phi)$$

$$\Rightarrow x^m y^n + \frac{1}{x^m y^n} = 2 \cos(m\theta + n\phi) \quad (\text{Proved})$$

28. If $z = 2 + i\sqrt{3}$ then $z \cdot \bar{z} = ?$

(a) 7

(b) 8

(c) $2 - i\sqrt{3}$

(d) 13

Solution:

$$z = 2 + i\sqrt{3}$$

$$\Rightarrow \bar{z} = 2 - i\sqrt{3}$$

$$\therefore z \cdot \bar{z} = (2 + i\sqrt{3})(2 - i\sqrt{3})$$

$$= 2^2 - (i\sqrt{3})^2$$

$$= 4 + 3 = 7$$

Answer: (a)

29. If $1, \omega, \omega^2$ are the cube roots of unity, then the value of $(3 + \omega^2 + \omega^4)$ is-

(a) 0

(b) 1

(c) 2

(d) 3

Solution:

$$3 + \omega^2 + \omega^4$$

$$= 3 + \omega^2 + \omega^3 \cdot \omega$$

$$= 3 + \omega^2 + \omega \quad [\because \omega^3 = 1]$$

$$= 3 - 1 \quad [\because 1 + \omega + \omega^2 = 0]$$

$$= 2$$

Answer: (c)

30. If $|z - 5| = 5$ where $z = x + 3i$, x and y being real, then find the locus of point (x, y) .

Solution:

$$\text{Given, } |z - 5| = 5$$

$$\Rightarrow |x + 3i - 5| = 5 \quad [\because z = x + 3i]$$

$$\Rightarrow |(x - 5) + 3i| = 5$$

$$\Rightarrow \sqrt{(x - 5)^2 + 3^2} = 5$$

$$\Rightarrow (x - 5)^2 + 3^2 = 5^2$$

$$\Rightarrow x^2 - 10x + 25 + 9 = 25$$

$$\Rightarrow x^2 - 10x + 9 = 0$$

So, the required locus of point (x, y) is

$$x^2 - 10x + 9 = 0 \quad (\text{Ans})$$

31. The value of $(i^{2023} + i^{2024} + i^{2025} + i^{2026})$

where $i = \sqrt{-1}$.

(a) 1

(b) 2

(c) 0

(d) 4

Ans: (c)

Solution:

$$i^{2023} + i^{2024} + i^{2025} + i^{2026}$$

$$= i^{2023} (1 + i + i^2 + i^3)$$

$$= i^{2023} (1 + i - 1 - i)$$

$$= i^{2023} \times 0 = 0 \quad (\text{Ans})$$

1. If one root of the equation $3x^2 - 5x + c = 0$ is 1 then the other root is:

(a) 3 (b) $\frac{3}{2}$ (c) $\frac{2}{3}$ (d) None

Solution:

$$3x^2 - 5x + c = 0$$

Given, one root of the equation is 1.

Let, other root is α .

$$\therefore \alpha + 1 = \frac{5}{3}$$

$$\Rightarrow \alpha = \frac{5}{3} - 1, \Rightarrow \alpha = \frac{2}{3}$$

Answer: (c)

2. The roots of the equation $x + \frac{1}{x} = 2$ are-

(a) 1, 1

(b) 1, -1

(c) -1, -1

(d) -1, -5

Solution:

$$x + \frac{1}{x} = 2$$

$$\Rightarrow x^2 + 1 = 2x \quad [\text{Multiplying } x \text{ both side}]$$

$$\Rightarrow x^2 - 2x + 1 = 0$$

$$\Rightarrow x^2 - x - x + 1 = 0$$

$$\Rightarrow x(x - 1) - 1(x - 1) = 0$$

$$\Rightarrow (x - 1)(x - 1) = 0$$

$$\Rightarrow x - 1 = 0 \text{ \& } x - 1 = 0$$

$$\therefore \text{Both value of } x = 1$$

Answer: (a)

3. If the roots of the equation $ax^2 + bx + c = 0$ are reciprocal to each other then-

(a) $a = b$

(b) $b = c$

(c) $c = a$

(d) None of these

Solution:

Since roots of the equation $ax^2 + bx + c = 0$ are reciprocal to each other then, product of roots = 1

$$\Rightarrow \frac{c}{a} = 1, \Rightarrow c = a$$

Answer: (c)

4. If one root of $3x^2 + px + 3 = 0$ is square of the other then $p = ?$

(a) $\frac{2}{3}$ (b) 3 (c) 1 (d) -6

Solution:

Let, a and a^2 be the roots of given equation.

$$\therefore a \cdot a^2 = \frac{3}{3} = 1$$

$$\Rightarrow a^3 = 1, \Rightarrow a = 1$$

$$\text{and } a + a^2 = -\frac{p}{3}$$

$$\Rightarrow 1 + 1^2 = -\frac{p}{3}$$

$$\Rightarrow -\frac{p}{3} = 2, \Rightarrow p = -6$$

Answer: (d)

5. If one root of equation $x^2 - 6x + m = 0$ be double of the other, then m is-

(a) 4 (b) 6 (c) 8 (d) None

Solution:

Let, a and $2a$ be roots of $x^2 - 6x + m = 0$

$$\therefore a + 2a = 6$$

$$\Rightarrow 3a = 6, \Rightarrow a = 2$$

$$\text{and, } a \cdot 2a = m$$

$$\Rightarrow m = 2a^2 = 2 \cdot 2^2 = 8$$

Answer: (c)

6. The quadratic equation whose sum of roots is ' $-p$ ' and product of roots is ' q ', is-

(a) $x^2 - px + q = 0$

(b) $x^2 + px + q = 0$

(c) $x^2 + px - q = 0$

(d) $x^2 - px - q = 0$

Solution:

The required equation is -

$$x^2 - (\text{sum of roots})x + (\text{Product of roots}) = 0$$

$$\Rightarrow x^2 - (-p)x + q = 0$$

$$\Rightarrow x^2 + px + q = 0$$

Answer: (b)

7. The value of k , for which the roots of the equation $9x^2 - kx + 16 = 0$ are equal.

Solution:

Since roots are equal, $\therefore b^2 - 4ac = 0$

$$\Rightarrow (-k)^2 - 4 \cdot 9 \cdot 16 = 0$$

$$\Rightarrow k^2 = 4 \times 9 \times 16$$

$$\Rightarrow k^2 = (2 \times 3 \times 4)^2$$

$$\Rightarrow k^2 = (24)^2$$

$$\Rightarrow k = \pm 24 \quad (\text{Ans})$$

8. If one root of $5x^2 + 13x + k = 0$ is reciprocal of other then find the value of k .

Solution:

$$5x^2 + 13x + k = 0$$

Let, One root of above equation = α

$$\text{so, other root} = \frac{1}{\alpha}$$

$$\therefore \alpha \cdot \frac{1}{\alpha} = \frac{k}{5} \quad [\because \alpha\beta = \frac{c}{a}]$$

$$\Rightarrow \frac{k}{5} = 1, \Rightarrow k = 5 \quad (\text{Ans})$$

9. If the roots of the equation $x^2 - px + q = 0$ be 1:2 then, find the relation between p and q .

Solution:

The roots of equation $x^2 - px + q = 0$ be 1:2

Therefore roots of the equation are α and 2α .

$$\therefore \text{Addition of root } \alpha + 2\alpha = p$$

$$\Rightarrow 3\alpha = p, \Rightarrow \alpha = \frac{p}{3}$$

Multiplication of two root $\alpha \cdot 2\alpha = q$

$$\Rightarrow 2\alpha^2 = q$$

$$\Rightarrow 2\left(\frac{p}{3}\right)^2 = q$$

$$\Rightarrow 2p^2 = 9q \quad (\text{Ans})$$

10. If sum of the root of $ax^2 + 2x + 3a = 0$ is equal to their product, then $a = ?$

Solution:

$$ax^2 + 2x + 3a = 0$$

Let, one roots of above equation are α and β .

As sum of roots = Product of roots

$$\therefore \alpha + \beta = \alpha\beta$$

$$\Rightarrow -\frac{2}{a} = \frac{3a}{a}$$

$$\Rightarrow 3a = -2, \Rightarrow a = -\frac{2}{3} \quad (\text{Ans})$$

11. Find the maximum value of $(5 + 4x - 4x^2)$

Solution:

$$\text{Let, } y = 5 + 4x - 4x^2$$

$$\Rightarrow y = 6 - 1 - 4x^2 + 4x$$

$$\Rightarrow y = 6 - (4x^2 - 4x + 1)$$

$$\Rightarrow y = 6 - (2x - 1)^2$$

As minimum value of $(2x - 1)^2 = 0$

So, maximum value of y

$$= 6 - 0 = 6 \quad (\text{Ans})$$

12. Find the value of m . If the product of root of equation $3x^2 - 5mx + (m^2 + 5) = 0$ is 2.

Solution:

$$3x^2 - 5mx + (m^2 + 5) = 0$$

Let roots of above equation is α and β .

$$\therefore \alpha\beta = \frac{m^2+5}{3}$$

$$\Rightarrow 2 = \frac{m^2+5}{3}$$

$$\Rightarrow m^2 + 5 = 6$$

$$\Rightarrow m^2 = 1, \Rightarrow m = \pm 1 \quad (\text{Ans})$$

13. Write down the quadratic equation whose roots are $(3 + \sqrt{2})$ and $(3 - \sqrt{2})$

Solution:

The required equation is

$$x^2 - (3 + \sqrt{2} + 3 - \sqrt{2})x + (3 + \sqrt{2})(3 - \sqrt{2}) = 0$$

$$\Rightarrow x^2 - 6x + \{3^2 - (\sqrt{2})^2\} = 0$$

$$\Rightarrow x^2 - 6x + 7 = 0 \quad (\text{Ans})$$

14. If α and β are the roots of $x^2 - px + q = 0$ then find the value of $(\alpha^2 + \beta^2 + \alpha\beta)$.

Solution:

The equation is $x^2 - px + q = 0$

$$\therefore \alpha + \beta = p \text{ and } \alpha\beta = q$$

$$\text{So, } \alpha^2 + \beta^2 + \alpha\beta$$

$$= (\alpha + \beta)^2 - 2\alpha\beta + \alpha\beta$$

$$= p^2 - q \quad (\text{Ans})$$

15. If one root of equation $ax^2 + bx + c = 0$ is the square of the other, show that

$$b^3 + ac^2 + a^2c = 3abc$$

Solution:

Let us assume that α and α^2 be the roots of the equation $ax^2 + bx + c = 0$

$$\text{Then, } \alpha + \alpha^2 = -\frac{b}{a} \text{ And } \alpha \cdot \alpha^2 = \frac{c}{a} \text{ or, } \alpha^3 = \frac{c}{a}$$

$$\therefore (\alpha + \alpha^2)^3 = \left(-\frac{b}{a}\right)^3$$

$$\Rightarrow \alpha^3 + (\alpha^2)^3 + 3\alpha\alpha^2(\alpha + \alpha^2) = -\frac{b^3}{a^3}$$

$$\Rightarrow \alpha^3 + (\alpha^3)^2 + 3\alpha^3(\alpha + \alpha^2) = -\frac{b^3}{a^3}$$

$$\Rightarrow \frac{c}{a} + \left(\frac{c}{a}\right)^2 + 3\frac{c}{a}\left(-\frac{b}{a}\right) = -\frac{b^3}{a^3}$$

$$\Rightarrow \frac{c}{a} + \frac{c^2}{a^2} - 3\frac{bc}{a^2} = -\frac{b^3}{a^3}$$

$$\Rightarrow a^2c + ac^2 - 3abc = -b^3$$

$$\Rightarrow b^3 + a^2c + ac^2 = 3abc \quad (\text{Proved})$$

16. If α and β are the roots of $3x^2 - 6x + 4 = 0$ then find the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$.

Solution:

As α, β be the roots of $3x^2 - 6x + 4 = 0$

We have $\alpha + \beta = \frac{6}{3} = 2, \alpha\beta = \frac{4}{3}$

$$\therefore \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$$

$$= \frac{\alpha^2 + \beta^2}{\alpha\beta} + 2\frac{\beta + \alpha}{\alpha\beta} + 3 \times \frac{4}{3}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} + 2\frac{2}{\frac{4}{3}} + 4$$

$$= \frac{(2)^2 - 2 \times \frac{4}{3}}{\frac{4}{3}} + 3 + 4$$

$$= \frac{4 - \frac{8}{3}}{\frac{4}{3}} + 7$$

$$= \frac{12 - 8}{4} + 7 = \frac{4}{4} + 7 = 8 \quad (\text{Ans})$$

17. If α, β be the roots of equation $x^2 - 3x + 2 = 0$ find the equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

Solution:

Since α, β are the roots of $x^2 - 3x + 2 = 0$

Therefore, $\alpha + \beta = 3, \alpha\beta = 2$

$\therefore$ The equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is:

$$x^2 - \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)x + \frac{1}{\alpha} \times \frac{1}{\beta} = 0$$

$$\Rightarrow x^2 - \left(\frac{\beta + \alpha}{\alpha\beta}\right)x + \frac{1}{2} = 0$$

$$\Rightarrow x^2 - \left(\frac{3}{2}\right)x + \frac{1}{2} = 0$$

$$\Rightarrow 2x^2 - 3x + 1 = 0 \quad (\text{Ans})$$

18. If α, β be the roots of equation $x^2 + 3x + 4 = 0$ find the equation whose roots are $(\alpha - \beta)^2$ and $(\alpha + \beta)^2$.

Solution:

Since α, β are the roots of $x^2 + 3x + 4 = 0$

Therefore, $\alpha + \beta = -3, \alpha\beta = 4$

Now, $(\alpha + \beta)^2 + (\alpha - \beta)^2$

$$= (-3)^2 + \{(\alpha + \beta)^2 - 4\alpha\beta\}$$

$$= 9 + (-3)^2 - 4.4 = 18 - 16 = 2$$

And $(\alpha + \beta)^2 \cdot (\alpha - \beta)^2$

$$= (-3)^2 \cdot \{(\alpha + \beta)^2 - 4\alpha\beta\}$$

$$= 9\{(-3)^2 - 4.4\} = -63$$

$\therefore$ The equation whose roots are $(\alpha - \beta)^2$ and

$(\alpha + \beta)^2$ is:

$$x^2 - \{(\alpha + \beta)^2 + (\alpha - \beta)^2\}x + (\alpha + \beta)^2 \cdot (\alpha - \beta)^2 = 0$$

$$\Rightarrow x^2 - 2x - 63 = 0 \quad (\text{Ans})$$

19. If α, β are the roots of quadratic equation $ax^2 + bx + c = 0$ find the quadratic equation whose roots are $\left(\frac{1}{\alpha} + 1\right)$ and $\left(\frac{1}{\beta} + 1\right)$.

Solution:

Since, α, β be the roots of $ax^2 + bx + c = 0$

Then, $\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

Now, $\frac{1}{\alpha} + 1 + \frac{1}{\beta} + 1$

$$= \frac{\alpha + \beta}{\alpha\beta} + 2 = \frac{-\frac{b}{a}}{\frac{c}{a}} + 2 = 2 - \frac{b}{c}$$

And, $\left(\frac{1}{\alpha} + 1\right) \cdot \left(\frac{1}{\beta} + 1\right)$

$$= \frac{1}{\alpha\beta} + \frac{1}{\alpha} + \frac{1}{\beta} + 1$$

$$= \frac{1}{\frac{c}{a}} + \frac{\alpha + \beta}{\alpha\beta} + 1 = \frac{a}{c} - \frac{b}{c} + 1$$

$\therefore$ The required quadratic equation is

$$x^2 - \left(2 - \frac{b}{c}\right)x + \frac{a}{c} - \frac{b}{c} + 1 = 0$$

$$\text{Or, } cx^2 + (b - 2c)x + (a - b + c) = 0 \quad (\text{Ans})$$

20. Prove that if the roots of the equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ are equal then $ad = bc$

Solution:

Since roots of the equation are equal, discriminant of the equation is zero

$$\therefore B^2 - 4AC = 0$$

$$\Rightarrow \{-2(ac + bd)\}^2 - 4(a^2 + b^2)(c^2 + d^2) = 0$$

$$\Rightarrow 4(a^2c^2 + 2ac \cdot bd + b^2d^2) - 4(a^2c^2 +$$

$$a^2d^2 + b^2c^2 + b^2d^2) = 0$$

$$\Rightarrow a^2c^2 + 2ac \cdot bd + b^2d^2 - a^2c^2 - a^2d^2 -$$

$$b^2c^2 - b^2d^2 = 0$$

$$\Rightarrow 2acbd - a^2d^2 - b^2c^2 = 0$$

$$\Rightarrow (ad)^2 - 2adbc + (bc)^2 = 0$$

$$\Rightarrow (ad - bc)^2 = 0$$

$$\Rightarrow ad - bc = 0$$

$$\Rightarrow ad = bc \quad (\text{Proved})$$

21. If α, β be the roots of the equation $2x^2 - 4x + 3 = 0$ find the equation whose roots are α^2 and β^2

Solution: Given equation is $2x^2 - 4x + 3 = 0$

As α, β be the roots of the equation

$$\therefore \alpha + \beta = \frac{4}{2} = 2 \text{ and } \alpha\beta = \frac{3}{2}$$

$\therefore$ The required equation is

$$x^2 - (\alpha^2 + \beta^2)x + \alpha^2\beta^2 = 0$$

$$\text{Or, } x^2 - \{(\alpha + \beta)^2 - 2\alpha\beta\}x + (\alpha\beta)^2 = 0$$

$$\text{Or, } x^2 - \left\{2^2 - 2 \cdot \frac{3}{2}\right\}x + \left(\frac{3}{2}\right)^2 = 0$$

$$\text{Or, } x^2 - x + \frac{9}{4} = 0$$

$$\text{Or, } 4x^2 - 4x + 9 = 0 \quad (\text{Ans})$$

22. If the roots of equation

$$a(b - c)x^2 + b(c - a)x + c(a - b) = 0$$

Are equal, then show that $\frac{1}{a} + \frac{1}{c} = \frac{2}{b}$

Solution:

$$\text{Let, } p = a(b - c); q = b(c - a); r = c(a - b)$$

$$\therefore p + q + r$$

$$= a(b - c) + b(c - a) + c(a - b)$$

$$= ab - ac + bc - ab + ac - bc = 0$$

$$\text{Or, } p + q + r = 0$$

$$\text{Or, } p + r = -q \quad \text{Or, } (p + r)^2 = q^2$$

From question we get the equation is

$$px^2 + qx + r = 0$$

As the roots of the equation is equal then

$$q^2 - 4pr = 0$$

$$\text{Or, } (p + r)^2 - 4pr = 0$$

$$\text{Or, } (p - r)^2 = 0 \quad \text{Or, } p = r$$

$$\text{Or, } a(b - c) = c(a - b)$$

$$\text{Or, } ab - ac = ac - bc$$

$$\text{Or, } bc + ab = 2ac \quad \text{Or, } \frac{1}{a} + \frac{1}{c} = \frac{2}{b} \quad (\text{Proved})$$

23. If α, β are the roots of the quadratic equation

$2x^2 + x - 1 = 0$, find the equation whose roots are $(2\alpha + 1)$ and $(2\beta + 1)$.

Solution: The given equation is $2x^2 + x - 1 = 0$

$$\therefore \alpha + \beta = -\frac{1}{2} \text{ and } \alpha\beta = -\frac{1}{2}$$

$$\text{Now, } (2\alpha + 1) + (2\beta + 1)$$

$$= 2(\alpha + \beta) + 2 = 2 \times \left(-\frac{1}{2}\right) + 2 = 1$$

$$\text{And, } (2\alpha + 1)(2\beta + 1)$$

$$= 4\alpha\beta + 2\alpha + 2\beta + 1$$

$$= 4 \times \left(-\frac{1}{2}\right) + 2(\alpha + \beta) + 1$$

$$= -2 + 2 \times \left(-\frac{1}{2}\right) + 1 = -2$$

$$\text{Hence the required equation is } x^2 - x - 2 = 0$$

24. If one root of equation $x^2 + px + q = 0$ is the square of the other, prove that

$$p^3 - q(3p - 1) + q^2 = 0$$

Solution: Let one root of equation is α and other root is α^2

$$\text{The given equation is } x^2 + px + q = 0$$

$$\text{So, } \alpha \cdot \alpha^2 = q \text{ or } \alpha^3 = q$$

$$\text{And, } \alpha + \alpha^2 = -p$$

$$\text{Or, } \alpha(1 + \alpha) = -p$$

$$\text{Or, } \alpha^3(1 + \alpha)^3 = -p^3$$

$$\text{Or, } q(1 + 3\alpha + 3\alpha^2 + \alpha^3) = -p^3$$

$$\text{Or, } q(1 + 3(\alpha + \alpha^2) + q) = -p^3$$

$$\text{Or, } q(1 - 3p + q) = -p^3$$

$$\text{Or, } q - 3pq + q^2 = -p^3$$

$$\text{Or, } p^3 - q(3p - 1) + q^2 = 0 \quad (\text{Proved})$$

25. If α is a root of $4x^2 + 2x - 1 = 0$, prove that $(4\alpha^3 - 3\alpha)$ is the other root.

Solution:

$$\text{As } \alpha \text{ is a root of } 4x^2 + 2x - 1 = 0$$

$$\therefore 4\alpha^2 + 2\alpha - 1 = 0$$

$$\Rightarrow 4\alpha^2 = 1 - 2\alpha$$

Let, other root of $4x^2 + 2x - 1 = 0$ is β

$$\therefore \alpha + \beta = -\frac{2}{4}$$

$$\Rightarrow \beta = -\frac{1}{2} - \alpha$$

$$= -\frac{1}{2} + \alpha - 2\alpha$$

$$= -\frac{1}{2} + \frac{1}{2} \times 2\alpha - 2\alpha$$

$$= -\frac{1}{2}(1 - 2\alpha) - 2\alpha$$

$$= -\frac{1}{2}(4\alpha^2) - 2\alpha$$

$$= -2\alpha^2 - 2\alpha$$

$$= \alpha - 2\alpha^2 - 3\alpha$$

$$= \alpha(1 - 2\alpha) - 3\alpha$$

$$= \alpha(4\alpha^2) - 3\alpha$$

$$= 4\alpha^3 - 3\alpha \quad (\text{Proved})$$

26. If one root of the equation $x^2 + ax + 8 = 0$ is 4 and the roots of the equation $x^2 + ax + b = 0$ are equal, find the value of b .

Solution:

$$\therefore \text{One root of equation } x^2 + ax + 8 = 0 \text{ is } 4.$$

$$\therefore 4^2 + 4a + 8 = 0 \quad [\text{Putting } x = 4]$$

$$\Rightarrow 16 + 4a + 8 = 0$$

$$\Rightarrow 4a + 24 = 0$$

$$\Rightarrow 4a = -24, \Rightarrow a = -6$$

Now, the roots of the equation $x^2 + ax + b = 0$ are equal. So discriminant (D) of above equation will zero.

$$\therefore D = B^2 - 4AC = 0 \quad [\text{General theory}]$$

$$\Rightarrow a^2 - 4 \cdot 1 \cdot b = 0$$

$$\Rightarrow (-6)^2 - 4 \cdot 1 \cdot b = 0 \quad [\because a = -6]$$

$$\Rightarrow 36 - 4b = 0$$

$$\Rightarrow 4b = 36, \Rightarrow b = 9 \quad (\text{Ans})$$

{Prepared by [NatiTute](#) YouTube Channel}

1. The middle term/s in the expansion of $\left(x + \frac{1}{x}\right)^{10}$ is:

(a) 504 (b) 252 (c) $\frac{1}{x^4}$ (d) None

Solution:

Total no of term in the expansion of $\left(x + \frac{1}{x}\right)^{10}$ is

$$= 10 + 1 = 11 \text{ (Odd)}$$

$$\therefore \text{Middle term} = \frac{11+1}{2} = 6^{\text{th}} \text{ term}$$

$$= t_6 = t_{5+1} = {}^{10}C_5 \times x^{10-5} \times \frac{1}{x^5} = 252$$

Answer: (b)

2. The coefficient of x^3 in the expansion of

$(1 + 3x + 3x^2 + x^3)^{10}$ is-

(a) ${}^{10}C_3$

(b) ${}^{10}C_2$

(c) ${}^{30}C_3$

(d) ${}^{30}C_2$

Solution:

$$(1 + 3x + 3x^2 + x^3)^{10}$$

$$= \{(1 + x)^3\}^{10} = (1 + x)^{30}$$

Let x^3 occur in $(r + 1)^{\text{th}}$ term of $(1 + x)^{30}$

$$\text{Now, } t_{r+1} = {}^{30}C_r (1)^{30-r} (x)^r = {}^{30}C_r x^r$$

$$\therefore x^r = x^3, \Rightarrow r = 3$$

$$\therefore \text{Coefficient of } x^3 \text{ is } {}^{30}C_3.$$

Answer: (c)

3. Find the number of terms in the expansion of $(x + y)^7(x - y)^7$.

Solution:

$$(x + y)^7(x - y)^7$$

$$= \{(x + y)(x - y)\}^7$$

$$= (x^2 - y^2)^7$$

$\therefore$ Number of terms in the given expansion is-

$$7 + 1 = 8 \text{ (Ans)}$$

4. The fifth term in the expansion of $\left(1 - \frac{1}{x}\right)^{10}$ is:

(a) $\frac{210}{x^4}$ (b) $\frac{x^4}{210}$ (c) $\frac{1}{x^4}$ (d) None

Solution:

Fifth term in the expansion of $\left(1 - \frac{1}{x}\right)^{10}$

$$= t_5 = t_{4+1} = {}^{10}C_4 1^{10-4} \left(-\frac{1}{x}\right)^4$$

$$= 210 \times \frac{1}{x^4} = \frac{210}{x^4}$$

Answer: (a)

5. Find sixth term in the expansion of $\left(x - \frac{1}{x}\right)^9$.

Solution:

Sixth term in the expansion of $\left(x - \frac{1}{x}\right)^9$

$$= t_6 = t_{5+1} = {}^9C_5 x^{9-5} \left(-\frac{1}{x}\right)^5$$

$$= -{}^9C_5 x^4 \frac{1}{x^5}$$

$$= -{}^9C_5 \frac{1}{x}$$

$$= -\frac{9!}{5!(9-5)!} \times \frac{1}{x}$$

$$= -\frac{9 \times 8 \times 7 \times 6 \times 5!}{5! \times 4 \times 3 \times 2 \times 1} \times \frac{1}{x}$$

$$= -126 \times \frac{1}{x} \quad (\text{Ans})$$

6. Find the co-efficient of $\frac{1}{x^2}$ in the expansion of $\left(1 - \frac{1}{x}\right)^{10}$.

Solution:

Let $\frac{1}{x^2}$ occur in $(r + 1)^{\text{th}}$ term of $\left(1 - \frac{1}{x}\right)^{10}$

$$\text{Now, } t_{r+1} = {}^{10}C_r (1)^{10-r} \left(-\frac{1}{x}\right)^r$$

$$\therefore \frac{1}{x^r} = \frac{1}{x^2} \text{ or } r = 2$$

$\therefore$ The required co-efficient

$$= {}^{10}C_2 \times (1)^{10-2}$$

$$= \frac{10!}{2!(10-2)!}$$

$$= \frac{10 \times 9 \times 8!}{2 \times 8!} = 45 \quad (\text{Ans})$$

7. Find the Co-efficient of the middle term in the expansion of $\left(a - \frac{1}{2a}\right)^6$.

Solution:

Number of terms in the expansion of $\left(a - \frac{1}{2a}\right)^6$ is

$$= 6 + 1 = 7 \text{ i.e. odd number.}$$

$$\therefore \text{The middle term} = \frac{7+1}{2} = 4^{\text{th}} \text{ term.}$$

$$\text{Now, } t_4 = t_{3+1} = {}^6C_3 a^{6-3} \left(-\frac{1}{2a}\right)^3$$

$$= -\frac{6!}{3!(6-3)!} a^3 \frac{1}{8a^3}$$

$$= -\frac{6.5.4.3!}{3.2.1.3!} \cdot \frac{1}{8} = -\frac{5}{2} \quad (\text{Ans})$$

8. Find the co-efficient of 6th term in the expansion of $\left(x - \frac{1}{3x}\right)^{10}$.

Solution:

$$\begin{aligned} 6^{\text{th}} \text{ term of } \left(x - \frac{1}{3x}\right)^{10} & \text{ is} \\ = t_6 = t_{5+1} & = {}^{10}C_5 x^{10-5} \left(-\frac{1}{3x}\right)^5 \\ & = -\frac{10!}{5!(10-5)!} x^5 \frac{1}{243 x^5} \\ & = -\frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5!}{5!} \cdot \frac{1}{243} \\ & = -\frac{28}{27} \quad (\text{Ans.}) \end{aligned}$$

9. The fifth term in the expansion of $\left(x^2 - \frac{1}{x}\right)^n$ is independent of x . Find n .

Solution:

$$\begin{aligned} 5^{\text{th}} \text{ term in the expansion of } \left(x^2 - \frac{1}{x}\right)^n & \text{ is} \\ = t_5 = t_{4+1} & = {}^nC_4 \cdot (x^2)^{n-4} \cdot \left(-\frac{1}{x}\right)^4 \\ & = {}^nC_4 \cdot x^{2n-8} \cdot x^{-4} \\ & = {}^nC_4 \cdot x^{2n-12} \end{aligned}$$

As the 5th term is independent of x , then $(2n - 12)$ must be 0.

$$\therefore 2n - 12 = 0$$

$$\Rightarrow 2n = 12, \Rightarrow n = 6 \quad (\text{Ans})$$

10. Find the term independent of x in the expansion of $\left(x - \frac{3}{x^2}\right)^6$.

Solution:

Let $(r + 1)^{\text{th}}$ term in expansion of $\left(x - \frac{3}{x^2}\right)^6$ is independent of x .

$$\begin{aligned} \text{Now, } t_{r+1} & = {}^6C_r x^{6-r} \left(-\frac{3}{x^2}\right)^r \\ \therefore x^{6-r} \cdot x^{-2r} & = x^0 \end{aligned}$$

$$\Rightarrow x^{6-r-2r} = x^0$$

$$\Rightarrow 6 - 3r = 0$$

$$\Rightarrow 3r = 6 \text{ or, } r = 2$$

Hence the constant term is $= {}^6C_2 \cdot (-3)^2$

$$= \frac{6!}{2!(6-2)!} \cdot 9 = \frac{6 \cdot 5 \cdot 4!}{2 \cdot 4!} = 135 \quad (\text{Ans})$$

11. If the third term and the fourth term in the expansion of $\left(2x + \frac{1}{8}\right)^{10}$ are equal then find x .

Solution: As the third term and the fourth term in the expansion of $\left(2x + \frac{1}{8}\right)^{10}$ are equal.

$$\begin{aligned} \therefore t_3 & = t_4 \\ \Rightarrow t_{2+1} & = t_{3+1} \\ \Rightarrow {}^{10}C_2 \left(\frac{1}{8}\right)^{10-2} (2x)^2 & = {}^{10}C_3 \left(\frac{1}{8}\right)^{10-3} (2x)^3 \\ \Rightarrow \frac{10!}{2!(10-2)!} \cdot \frac{1}{8^8} \cdot 2^2 \cdot x^2 & = \frac{10!}{3!(10-3)!} \cdot \frac{1}{8^7} \cdot 2^3 \cdot x^3 \\ \Rightarrow \frac{1}{2!8!} \cdot \frac{1}{8} & = \frac{1}{3!7!} \cdot 2 \cdot x \\ \Rightarrow x & = \frac{3 \cdot 2 \cdot 7!}{2 \cdot 8 \cdot 7!} = \frac{3}{128} \quad (\text{Ans}) \end{aligned}$$

12. If the co-efficient of x^7 in the expansion of $\left(px^2 + \frac{1}{qx}\right)^{11}$ be equal to the co-efficient of x^{-7} in the expansion of $\left(px - \frac{1}{qx^2}\right)^{11}$, prove that $pq = 1$

Solution:

Let $(r + 1)^{\text{th}}$ term contains x^7 in the expansion of $\left(px^2 + \frac{1}{qx}\right)^{11}$

$$\text{Now, } t_{r+1} = {}^{11}C_r (px^2)^{11-r} \cdot \left(\frac{1}{qx}\right)^r$$

$$\therefore x^{22-2r} \cdot x^{-r} = x^7$$

$$\text{Or, } 22 - 3r = 7$$

$$\text{Or, } 3r = 15 \text{ or, } r = 5$$

$\therefore$ The co-efficient of x^7 is

$${}^{11}C_5 \cdot p^{11-5} \cdot \frac{1}{q^5}$$

$$= {}^{11}C_5 \cdot p^6 \cdot \frac{1}{q^5}$$

Again, Let $(u + 1)^{\text{th}}$ term contains x^{-7} in the expansion of $\left(px - \frac{1}{qx^2}\right)^{11}$

$$\text{Now, } t_{u+1} = {}^{11}C_u (px)^{11-u} \cdot \left(-\frac{1}{qx^2}\right)^u$$

$$\therefore x^{11-u} \cdot x^{-2u} = x^{-7}$$

$$\text{Or, } 11 - 3u = -7$$

$$\text{Or, } 3u = 18 \text{ or, } u = 6$$

$\therefore$ The co-efficient of x^{-7} is

$${}^{11}C_6 \cdot p^{11-6} \cdot \frac{1}{q^6} = {}^{11}C_6 \cdot p^5 \cdot \frac{1}{q^6}$$

$$\text{Hence, } {}^{11}C_6 \cdot p^6 \cdot \frac{1}{q^5} = {}^{11}C_6 \cdot p^5 \cdot \frac{1}{q^6}$$

$$\text{Or, } p = \frac{1}{q} \text{ or, } pq = 1 \quad (\text{Proved})$$

13. Prove that the middle term in the expansion of $(1+x)^{2n}$ is $\frac{1.3.5...(2n-1)}{n} 2^n x^n$

Solution:

∵ $2n$ is even number,

∴ Middle term of $(1+x)^{2n}$ is

$$\frac{2n}{2} + 1 = (n+1)\text{th term}$$

Now,

$$[\because t_{r+1} = nCr a^{n-r} x^r]$$

$$\Rightarrow t_{n+1} = 2nCn 1^{2n-n} x^n$$

$$= \frac{2n!}{n!} \cdot x^n$$

$$= \frac{1.2.3.4.5.6...(2n-2)(2n-1)2n}{n!} \cdot x^n$$

$$= \frac{1.3.5...(2n-1)}{n!} \cdot \frac{2.4.6...2n}{n!} \cdot x^n$$

$$= \frac{1.3.5...(2n-1)}{n!} \cdot \frac{2^n (1.2.3...n)}{n!} \cdot x^n$$

$$= \frac{1.3.5...(2n-1)}{n!} \cdot \frac{2^n n!}{n!} \cdot x^n$$

$$= \frac{1.3.5...(2n-1)}{n!} \cdot 2^n \cdot x^n \text{ (Proved)}$$

14. Prove that

$$(1+x+x^2+x^3+\dots\infty)(1-x+x^2-x^3+\dots\infty) \\ = (1+x^2+x^4+x^6+\dots\infty)$$

Solution:

LHS

$$(1+x+x^2+x^3+\dots\infty)(1-x+x^2-x^3+\dots\infty)$$

$$= (1-x)^{-1}(1+x)^{-1}$$

$$= \{(1-x)(1+x)\}^{-1}$$

$$= (1-x^2)^{-1}$$

$$= 1+x^2+x^4+x^6+\dots\infty \text{ (Proved)}$$

{Prepared by NatiTute YouTube Channel}

15. If the 4th term in the expansion of $(px + \frac{1}{x})^n$ is independent of x , find the value of n . Also calculate p if the 4th term be $\frac{5}{2}$.

Solution:

4th term in the expansion of $(px + \frac{1}{x})^n$

$$= t_4 = t_{3+1} = {}^nC_3 (px)^{n-3} \left(\frac{1}{x}\right)^3 \\ = {}^nC_3 p^{n-3} \cdot x^{n-3} \cdot x^{-3} \\ = {}^nC_3 p^{n-3} \cdot x^{n-6}$$

As 4th term is independent of x .

So power of x is 0.

$$\Rightarrow n-6=0, \Rightarrow n=6 \text{ (Ans)}$$

$$\text{Now, 4th term} = {}^6C_3 p^{6-3} = 20p^3$$

$$\text{But given 4th term} = \frac{5}{2}$$

$$\therefore 20p^3 = \frac{5}{2}$$

$$\Rightarrow p^3 = \frac{5}{2 \times 20}$$

$$\Rightarrow p = \sqrt[3]{\frac{1}{8}} = \sqrt[3]{\left(\frac{1}{2}\right)^3} = \frac{1}{2} \text{ (Ans)}$$

16. Find the co-efficient of x in the expansion of

$$(1-2x^3+3x^5)\left(1+\frac{1}{x}\right)^{10}$$

Answer:

We have to find out the co-efficient of $\frac{1}{x^2}$ and $\frac{1}{x^4}$ in the expression of $\left(1+\frac{1}{x}\right)^{10}$

$$\text{From binomial theorem we get, } \left(1+\frac{1}{x}\right)^{10} \\ = 1 + {}^{10}C_1 \cdot \frac{1}{x} + {}^{10}C_2 \cdot \frac{1}{x^2} + {}^{10}C_3 \cdot \frac{1}{x^3} + {}^{10}C_4 \cdot \frac{1}{x^4} + \dots$$

$$\text{So, the co-efficient of } \frac{1}{x^2} = 10C_2$$

$$\text{And, the co-efficient of } \frac{1}{x^4} = 10C_4$$

∴ Co-efficient of x in the final expression is

$$-2 \times 10C_2 + 3 \times 10C_4$$

$$= 3 \frac{10!}{4!(10-4)!} - 2 \frac{10!}{2!(10-2)!}$$

$$= 3 \cdot \frac{10.9.8.7.6!}{4.3.2.1.6!} - 2 \cdot \frac{10.9.8!}{2.1.8!}$$

$$= 630 - 90 = 540 \text{ (Ans)}$$

1. If $\vec{a} = 2\hat{i} - 3\hat{j} - 6\hat{k}$, then the value of $|\vec{a}|$:

(a) 14 (b) 7 (c) $\sqrt{7}$ (d) None

Solution:

$$\text{Given, } \vec{a} = 2\hat{i} - 3\hat{j} - 6\hat{k}$$

$$\therefore |\vec{a}| = \sqrt{2^2 + (-3)^2 + (-6)^2} \\ = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

Answer: (b)

2. If $\vec{a} = 6\hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} - 6\hat{k}$, then the value of $\vec{a} \cdot \vec{b}$:

(a) 0 (b) 15 (c) 7 (d) None

Solution:

$$\vec{a} \cdot \vec{b} = (6\hat{i} + 3\hat{j} - 2\hat{k}) \cdot (2\hat{i} - 3\hat{j} - 6\hat{k}) \\ = 12 - 9 + 12 = 15$$

Answer: (b)

3. If for the vectors $\vec{a}$ and $\vec{b}$, $|\vec{a}| = 1$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = \sqrt{3}$, then between the vectors $\vec{a}$ and $\vec{b}$ is-

(a) 90°

(b) 60°

(c) 45°

(d) 30°

Solution:

Let, angle between $\vec{a}$ and $\vec{b}$ is θ .

$$\therefore \vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cos \theta$$

$$\Rightarrow \sqrt{3} = 1 \cdot 2 \cdot \cos \theta$$

$$\Rightarrow \cos \theta = \frac{\sqrt{3}}{2} = \cos 30^\circ$$

$$\Rightarrow \theta = 30^\circ$$

Answer: (d)

4. If the vectors $2\hat{i} - 3\hat{j} + \hat{k}$ and $m\hat{i} - \hat{j} + m\hat{k}$ are perpendicular to each other, then the value of m is-

(a) 1

(b) -1

(c) 2

(d) -2

Solution:

As the vectors $2\hat{i} - 3\hat{j} + \hat{k}$ and $m\hat{i} - \hat{j} + m\hat{k}$ are perpendicular to each other, then

$$(2\hat{i} - 3\hat{j} + \hat{k}) \cdot (m\hat{i} - \hat{j} + m\hat{k}) = 0$$

$$\Rightarrow 2m + 3 + m = 0$$

$$\Rightarrow 3m = -3$$

$$\Rightarrow m = -1$$

Answer: (b)

5. The value of $\hat{k} \cdot (\hat{i} \times \hat{j})$ is-

(a) 1

(b) 0

(c) -1

(d) none of these

Solution:

$$\hat{k} \cdot (\hat{i} \times \hat{j}) = \hat{k} \cdot \hat{k} = 1$$

Answer: (a)

6. What is the unit vector in the direction of the vector $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$.

Solution:

$$\text{Unit vector } \hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{2\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{2^2 + (-2)^2 + 1^2}} \\ = \frac{2\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{9}} \\ = \frac{1}{3}(2\hat{i} - 2\hat{j} + \hat{k}) \quad (\text{Ans})$$

7. If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = -2\hat{i} + 4\hat{j} - 2\hat{k}$, then find (a) $\vec{a} \cdot \vec{b}$ and (b) $\vec{a} \times \vec{b}$. Mention the geometrical understanding of the vectors $\vec{a}$ and $\vec{b}$, after the calculation of $\vec{a} \cdot \vec{b}$.

Solution:

$$\text{Given, } \vec{a} = \hat{i} + 2\hat{j} + 3\hat{k} \text{ and } \vec{b} = -2\hat{i} + 4\hat{j} - 2\hat{k}$$

$$(a) \vec{a} \cdot \vec{b} = (\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (-2\hat{i} + 4\hat{j} - 2\hat{k}) \\ = -2 + 8 - 6 = 0 \quad (\text{Ans})$$

$$(b) \vec{a} \times \vec{b}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ -2 & 4 & -2 \end{vmatrix} \\ = \hat{i}(-4 - 12) - \hat{j}(-2 + 6) + \hat{k}(4 + 4) \\ = -16\hat{i} - 4\hat{j} + 8\hat{k} \quad (\text{Ans})$$

As dot product of given two vectors, $\vec{a} \cdot \vec{b} = 0$

$\therefore \vec{a}$ and $\vec{b}$ vectors are perpendicular to each other.

8. What will be the position vector of the middle point joining AB? If their position vector $A(2, 3, 4)$ and $B(2, 5, 6)$.

Solution:

$$\text{Position vector of } A(2, 3, 4) = \vec{OA} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\text{Position vector of } B(2, 5, 6) = \vec{OB} = 2\hat{i} + 5\hat{j} + 6\hat{k}$$

(O is an arbitrary point)

$\therefore$ Position vector of middle point of AB

$$= \frac{\vec{OA} + \vec{OB}}{2} = \frac{(2\hat{i} + 3\hat{j} + 4\hat{k}) + (2\hat{i} + 5\hat{j} + 6\hat{k})}{2} \\ = \frac{4\hat{i} + 8\hat{j} + 10\hat{k}}{2} \\ = \frac{2(2\hat{i} + 4\hat{j} + 5\hat{k})}{2} \\ = 2\hat{i} + 4\hat{j} + 5\hat{k} \quad (\text{Ans})$$

9. If the vector $\vec{a}$ and $\vec{b}$ are collinear where $\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$ and $\vec{b} = 4\vec{i} + m\vec{j} + 2\vec{k}$ then $m = ?$
Solution:

As the vector $\vec{a}$ and $\vec{b}$ are collinear.

Let us assume, $\vec{a} = p\vec{b}$ (p is a constant)

$$\Rightarrow 2\vec{i} - 3\vec{j} + \vec{k} = p(4\vec{i} + m\vec{j} + 2\vec{k})$$

$$\Rightarrow 2\vec{i} - 3\vec{j} + \vec{k} = 4p\vec{i} + mp\vec{j} + 2pk\vec{k}$$

Comparing both side we get,

$$4p = 2 \Rightarrow p = \frac{2}{4} = \frac{1}{2}$$

$$\text{and, } mp = -3$$

$$\Rightarrow m \times \frac{1}{2} = -3 \quad [\because p = \frac{1}{2}]$$

$$\Rightarrow m = -6 \quad (\text{Ans})$$

10. The vectors $(\gamma\vec{i} + \gamma\vec{j} + 3\vec{k})$ and $(3\vec{i} + 3\vec{j} - 2\gamma\vec{k})$ are perpendicular to each other then $\gamma = ?$
Solution:

As the given two vector are perpendicular to each other.

$$\therefore (\gamma\vec{i} + \gamma\vec{j} + 3\vec{k}) \cdot (3\vec{i} + 3\vec{j} - 2\gamma\vec{k}) = 0$$

$$\Rightarrow 3\gamma + 3\gamma - 6\gamma = 0$$

$$\Rightarrow 0 = 0$$

$$\therefore \gamma = \text{all real value} \quad (\text{Ans.})$$

11. What is the angle between the vectors $(2\vec{i} + 4\vec{j} - 7\vec{k})$ and $(3\vec{i} + 2\vec{j} + 2\vec{k})$?
Solution:

$$\text{Now, } (2\vec{i} + 4\vec{j} - 7\vec{k}) \cdot (3\vec{i} + 2\vec{j} + 2\vec{k})$$

$$= 6 + 8 - 14 = 0$$

$$\text{So the angle between vectors is } \frac{\pi}{2} \quad (\text{Ans})$$

12. If $\vec{a} = 2\vec{i} + \vec{j} + \vec{k}$, $\vec{b} = \vec{i} - 2\vec{j} + 2\vec{k}$ and $\vec{c} = 3\vec{i} - 4\vec{j} + 2\vec{k}$ then what is the projection of $(\vec{a} + \vec{c})$ in direction of $\vec{b}$?
Solution:

$$\text{Now, } \vec{a} + \vec{c}$$

$$= 2\vec{i} + \vec{j} + \vec{k} + 3\vec{i} - 4\vec{j} + 2\vec{k}$$

$$= 5\vec{i} - 3\vec{j} + 3\vec{k}$$

$$\therefore \text{The projection of } (\vec{a} + \vec{c}) \text{ in direction of } \vec{b} \text{ is}$$

$$= \frac{(\vec{a} + \vec{c}) \cdot \vec{b}}{|\vec{b}|}$$

$$= \frac{(5\vec{i} - 3\vec{j} + 3\vec{k}) \cdot (\vec{i} - 2\vec{j} + 2\vec{k})}{|\vec{i} - 2\vec{j} + 2\vec{k}|}$$

$$= \frac{5 + 6 + 6}{\sqrt{1^2 + (-2)^2 + 2^2}} = \frac{17}{3} \quad (\text{Ans})$$

13. Show that the points whose position vectors are $(2\vec{i} + 4\vec{j} - \vec{k})$, $(4\vec{i} + 5\vec{j} + \vec{k})$ and $(3\vec{i} + 6\vec{j} - 3\vec{k})$ form a right angled triangle.
Solution:

Let the triangle is $\triangle ABC$ and

$$\text{position vector of } A = 2\vec{i} + 4\vec{j} - \vec{k}$$

$$\text{position vector of } B = 4\vec{i} + 5\vec{j} + \vec{k}$$

$$\text{position vector of } C = 3\vec{i} + 6\vec{j} - 3\vec{k}$$

and O is an arbitrary point.

$$\therefore \vec{AB} = \vec{OB} - \vec{OA}$$

$$= (4\vec{i} + 5\vec{j} + \vec{k}) - (2\vec{i} + 4\vec{j} - \vec{k})$$

$$= 2\vec{i} + \vec{j} + 2\vec{k}$$

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$= (3\vec{i} + 6\vec{j} - 3\vec{k}) - (4\vec{i} + 5\vec{j} + \vec{k})$$

$$= -\vec{i} + \vec{j} - 4\vec{k}$$

$$\vec{CA} = \vec{OA} - \vec{OC}$$

$$= (2\vec{i} + 4\vec{j} - \vec{k}) - (3\vec{i} + 6\vec{j} - 3\vec{k})$$

$$= -\vec{i} - 2\vec{j} + 2\vec{k}$$

Now,

$$\vec{AB} \cdot \vec{CA} = (2\vec{i} + \vec{j} + 2\vec{k}) \cdot (-\vec{i} - 2\vec{j} + 2\vec{k})$$

$$= -2 - 2 + 4$$

$$= 0$$

$$\therefore AB \perp CA$$

So, $\triangle ABC$ is a right angled triangle. (Proved)

14. $\vec{a} = 2\vec{i} - 2\vec{j} + \vec{k}$ and $\vec{b} = 3\vec{i} + 2\vec{j} + \vec{k}$, then $\vec{a} \cdot \vec{b} = ?$
Solution:

$$\vec{a} \cdot \vec{b} = (2\vec{i} - 2\vec{j} + \vec{k}) \cdot (3\vec{i} + 2\vec{j} + \vec{k})$$

$$= 6 - 4 + 1$$

$$= 3 \quad (\text{Ans})$$

15. So that the vectors corresponding to the positions of the points $(3, -2, 1)$ and $(2, 3, 0)$ are at right angle.
Solution:

Now, vector corresponding to points $(3, -2, 1)$ and

$$(2, 3, 0) \text{ is } (3\vec{i} - 2\vec{j} + \vec{k}) \text{ and } (2\vec{i} + 3\vec{j})$$

$$\text{Here, } (3\vec{i} - 2\vec{j} + \vec{k}) \cdot (2\vec{i} + 3\vec{j})$$

$$= 6 - 6 = 0$$

$\therefore$ The given vectors are right angle to each other.

16. Find the unit vector along the vector $(2\vec{i} + 3\vec{j} - \vec{k})$.
Solution:

$$\text{Unit vector of } \vec{a}, \hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

$$= \frac{2\vec{i} + 3\vec{j} - \vec{k}}{|2\vec{i} + 3\vec{j} - \vec{k}|}$$

$$= \frac{2\vec{i} + 3\vec{j} - \vec{k}}{\sqrt{2^2 + 3^2 + (-1)^2}}$$

$$= \frac{2\vec{i} + 3\vec{j} - \vec{k}}{\sqrt{14}} \quad (\text{Ans})$$

17. If the vectors $(2\mathbf{i} + a\mathbf{j} + \mathbf{k})$ and $(4\mathbf{i} - 2\mathbf{j} - 2\mathbf{k})$ are perpendicular to each other then find a .

Solution:

As the given vectors are perpendicular

$$\therefore (2\mathbf{i} + a\mathbf{j} + \mathbf{k}) \cdot (4\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}) = 0$$

$$\Rightarrow 8 - 2a - 2 = 0$$

$$\Rightarrow -2a = -6, \Rightarrow a = 3 \text{ (Ans)}$$

18. If $(\vec{a} \times \vec{b}) \cdot \vec{c} = 0$ for the vectors $\vec{a} = m\mathbf{i} + \mathbf{k}$, $\vec{b} = \mathbf{i} - 5\mathbf{j} + \mathbf{k}$ and $\vec{c} = -2\mathbf{j}$, then $m = ?$

Solution:

$$\text{Now, } \vec{a} \times \vec{b} = (m\mathbf{i} + \mathbf{k}) \times (\mathbf{i} - 5\mathbf{j} + \mathbf{k})$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ m & 0 & 1 \\ 1 & -5 & 1 \end{vmatrix}$$

$$= \mathbf{i}(0 + 5) - \mathbf{j}(m - 1) + \mathbf{k}(-5m - 0)$$

$$= 5\mathbf{i} - \mathbf{j}(m - 1) - 5m\mathbf{k}$$

$$\text{Given, } (\vec{a} \times \vec{b}) \cdot \vec{c} = 0$$

$$\Rightarrow \{5\mathbf{i} - \mathbf{j}(m - 1) - 5m\mathbf{k}\} \cdot (-2\mathbf{j}) = 0$$

$$\Rightarrow (m - 1)2 = 0$$

$$\Rightarrow (m - 1) = 0, \Rightarrow m = 1 \text{ (Ans)}$$

19. If $\vec{a} = -\mathbf{i} + 2\mathbf{j}$, $\vec{b} = 3\mathbf{i} - 2\mathbf{j}$ and $\vec{c} = 5\mathbf{j}$ then find $|\vec{a} + \vec{b} - 2\vec{c}|$.

Solution:

$$|\vec{a} + \vec{b} - 2\vec{c}|$$

$$= |(-\mathbf{i} + 2\mathbf{j}) + (3\mathbf{i} - 2\mathbf{j}) - 2(5\mathbf{j})|$$

$$= |-\mathbf{i} + 2\mathbf{j} + 3\mathbf{i} - 2\mathbf{j} - 10\mathbf{j}|$$

$$= |2\mathbf{i} - 10\mathbf{j}| = \sqrt{2^2 + (-10)^2} = \sqrt{104} \text{ (Ans)}$$

20. If the position vector of two points A and B are $(\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$ and $(4\mathbf{i} - 5\mathbf{j} + 6\mathbf{k})$ respectively, then find the unit vector in the direction of $\vec{AB}$.

Solution:

Assuming an arbitrary point O .

$$\text{We can say, } \vec{OA} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$$

$$\vec{OB} = 4\mathbf{i} - 5\mathbf{j} + 6\mathbf{k}$$

$$\therefore \vec{AB} = \vec{OB} - \vec{OA}$$

$$= (4\mathbf{i} - 5\mathbf{j} + 6\mathbf{k}) - (\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$$

$$= 3\mathbf{i} - 7\mathbf{j} + 9\mathbf{k}$$

So, unit vector along $\vec{AB}$

$$= \frac{3\mathbf{i} - 7\mathbf{j} + 9\mathbf{k}}{\sqrt{3^2 + (-7)^2 + 9^2}}$$

$$= \frac{3\mathbf{i} - 7\mathbf{j} + 9\mathbf{k}}{\sqrt{9 + 49 + 81}} = \frac{3\mathbf{i} - 7\mathbf{j} + 9\mathbf{k}}{\sqrt{139}} \text{ (Ans)}$$

21. The position vectors of A, B, C, D are given by the vectors $\mathbf{i} + \mathbf{j} + \hat{\mathbf{k}}, 2\mathbf{i} + 3\mathbf{j}, 3\mathbf{i} + 5\mathbf{j} - 2\hat{\mathbf{k}}$ and $\hat{\mathbf{k}} - \mathbf{j}$. Prove that $\vec{AB}$ and $\vec{CD}$ are parallel vectors.

Solution:

Assuming an arbitrary point O .

$$\text{We can say, } \vec{OA} = \mathbf{i} + \mathbf{j} + \hat{\mathbf{k}}$$

$$\vec{OB} = 2\mathbf{i} + 3\mathbf{j}$$

$$\vec{OC} = 3\mathbf{i} + 5\mathbf{j} - 2\hat{\mathbf{k}}$$

$$\vec{OD} = \hat{\mathbf{k}} - \mathbf{j}$$

$$\therefore \vec{AB} = \vec{OB} - \vec{OA}$$

$$= (2\mathbf{i} + 3\mathbf{j}) - (\mathbf{i} + \mathbf{j} + \hat{\mathbf{k}}) = \mathbf{i} + 2\mathbf{j} - \hat{\mathbf{k}}$$

$$\vec{CD} = \vec{OD} - \vec{OC}$$

$$= (\hat{\mathbf{k}} - \mathbf{j}) - (3\mathbf{i} + 5\mathbf{j} - 2\hat{\mathbf{k}})$$

$$= -3\mathbf{i} - 6\mathbf{j} + 3\hat{\mathbf{k}}$$

$$= -3(\mathbf{i} + 2\mathbf{j} - \hat{\mathbf{k}}) = -3\vec{AB}$$

$$\therefore \vec{AB} \text{ and } \vec{CD} \text{ are parallel vectors.}$$

22. If $\vec{a} = 2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$ and $\vec{b} = \mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ find their cross product.

Solution:

$$\vec{a} \times \vec{b} = (2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) \times (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k})$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -3 & -1 \\ 1 & 4 & -2 \end{vmatrix}$$

$$= \mathbf{i}(6 + 4) - \mathbf{j}(-4 + 1) + \mathbf{k}(8 + 3)$$

$$= 10\mathbf{i} + 3\mathbf{j} + 11\mathbf{k} \text{ (Ans.)}$$

23. A force $\vec{F} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$ acts at a point whose position vector is $(2\mathbf{i} - \mathbf{j})$, find the moment of $\vec{F}$ about the origin.

Solution:

Let, P is a point whose position vector $(2\mathbf{i} - \mathbf{j})$ and point O is origin and $OP = r$

$$\therefore \vec{r} = \vec{OP} = (2\mathbf{i} - \mathbf{j})$$

$$\text{Given, } \vec{F} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$$

$\therefore$ The required moment

$$\vec{M} = \vec{r} \times \vec{F} = (2\mathbf{i} - \mathbf{j}) \times (2\mathbf{i} + \mathbf{j} - \mathbf{k})$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 0 \\ 2 & 1 & -1 \end{vmatrix}$$

$$= \mathbf{i}(1 - 0) - \mathbf{j}(-2 - 0) + \mathbf{k}(2 + 2)$$

$$= \mathbf{i} + 2\mathbf{j} + 4\mathbf{k} \text{ (Ans)}$$

24. A force of 15 units acts in the direction of the vector $(i - 2j + 2k)$ and passes through a point $(2i - 2j + 2k)$. Find the moment of the force about the point $(i + j + k)$.

Solution:

Here force $\vec{F} = 15(i - 2j + 2k)$

Now, radius of moment,

$$\vec{r} = (2i - 2j + 2k) - (i + j + k) \\ = i - 3j + k$$

∴ Moment of $\vec{F}$ about the given point,

$$= \vec{F} \times \vec{r} = 15(i - 2j + 2k) \times (i - 3j + k) \\ = 15 \begin{vmatrix} i & j & k \\ 1 & -2 & 2 \\ 1 & -3 & 1 \end{vmatrix} \\ = 15[i(-2 + 6) - j(1 - 2) + k(-3 + 2)] \\ = 15(4i + j - k) \\ = 60i + 15j - 15k \text{ (Ans)}$$

25. Find a vector perpendicular to both $(2i + 3j - k)$ and $(3i - j + 2k)$.

Solution:

For any two vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b}$ represents a vector which is perpendicular to both $\vec{a}$ and $\vec{b}$

Let, $\vec{a} = 2i + 3j - k$ and $\vec{b} = 3i - j + 2k$

$$\therefore \vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 2 & 3 & -1 \\ 3 & -1 & 2 \end{vmatrix} \\ = i(6 - 1) - j(4 + 3) + k(-2 - 9) \\ = 5i - 7j - 11k$$

∴ The required vector is $5i - 7j - 11k$ (Ans)

26. If $\vec{a} = 4i - j + 3k$ and $\vec{b} = -2i + j - 2k$, find the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

Solution:

We know $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$.

$$\text{Now, } \vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 4 & -1 & 3 \\ -2 & 1 & -2 \end{vmatrix} \\ = i(2 - 3) - j(-8 + 6) + k(4 - 2) \\ = -i + 2j + 2k$$

So, unit vector along $(\vec{a} \times \vec{b})$

$$= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{-i + 2j + 2k}{|-i + 2j + 2k|} = \frac{-i + 2j + 2k}{\sqrt{(-1)^2 + 2^2 + 2^2}} \\ = \frac{-i + 2j + 2k}{\sqrt{1 + 4 + 4}} \\ = -\frac{1}{3}(-i + 2j + 2k) \text{ (Ans)}$$

27. If $\vec{a} = 2i - j + 2k$ and $\vec{b} = -3i + j - 3k$, find the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

Solution:

We know $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$.

$$\text{Now, } \vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 2 & -1 & 2 \\ -3 & 1 & -3 \end{vmatrix} \\ = i(3 - 2) - j(-6 + 6) + k(2 - 3) \\ = i - k$$

So, unit vector along $(\vec{a} \times \vec{b})$

$$= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{i - k}{|i - k|} = \frac{i - k}{\sqrt{(1)^2 + (-1)^2}} \\ = \frac{i - k}{\sqrt{2}} = \frac{1}{\sqrt{2}}(i - k) \text{ (Ans)}$$

28. If $(2i + 6j + 27k) \times (i + 3j + pk) = \vec{0}$ then find the value of p .

Solution:

$$(2i + 6j + 27k) \times (i + 3j + pk) = \vec{0}$$

$$\Rightarrow \begin{vmatrix} i & j & k \\ 2 & 6 & 27 \\ 1 & 3 & p \end{vmatrix} = 0$$

$$\Rightarrow i(6p - 81) - j(2p - 27) + k(6 - 6) = 0$$

$$\therefore 2p = 27 = 0$$

$$\Rightarrow p = \frac{27}{2} \text{ (Ans)}$$

29. If $(2i + 3j + 4k)$ and $(3i + 4j + 7k)$ are the two diagonals of a parallelogram, find its area.

Solution:

If $\vec{a}$ and $\vec{b}$ represents two diagonals of a parallelogram then its area $= |\vec{a} \times \vec{b}|$

$$\text{Now, } \vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 2 & 3 & 4 \\ 3 & 4 & 7 \end{vmatrix} \\ = i(21 - 16) - j(14 - 12) + k(8 - 9) \\ = 5i - 2j - k$$

$$\therefore \text{Area of parallelogram} = |\vec{a} \times \vec{b}| \\ = |5i - 2j - k| \\ = \sqrt{5^2 + (-2)^2 + (-1)^2} \\ = \sqrt{30} \text{ Sq. Unit (Ans)}$$

30. Find the area of the triangle formed by the points whose position vectors are $(3i + j)$, $(5i + 2j + k)$ and $(i - 2j + 3k)$.

Solution:

Let, the triangle is ABC

$$\therefore \overrightarrow{AB} = (5i + 2j + k) - (3i + j) \\ = 2i + j + k$$

$$\overrightarrow{BC} = (i - 2j + 3k) - (5i + 2j + k) \\ = -4i - 4j + 2k$$

$$\text{Now, } \overrightarrow{AB} \times \overrightarrow{BC} = \begin{vmatrix} i & j & k \\ 2 & 1 & 1 \\ -4 & -4 & 2 \end{vmatrix} \\ = i(2 + 4) - j(4 + 4) + k(-8 + 4) \\ = 6i - 8j - 4k$$

$$\therefore \text{Area of triangle} = \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{BC}| \\ = \frac{1}{2} |6i - 8j - 4k| \\ = \frac{1}{2} \sqrt{6^2 + (-8)^2 + (-4)^2} \\ = \frac{1}{2} \sqrt{36 + 64 + 16} \\ = \frac{1}{2} \sqrt{116} \\ = \frac{1}{2} \times 2\sqrt{29} = \sqrt{29} \text{ Sq. unit}$$

31. The vectors $(2i + 7j)$, $(2i - 5j + 6k)$ and $(-i + 2j - k)$ are coplanar or not.

Solution:

$$\text{Now, } \begin{vmatrix} 2 & 7 & 0 \\ 2 & -5 & 6 \\ -1 & 2 & -1 \end{vmatrix} \\ = 2(5 - 12) - 7(-2 + 6) + 0 \\ = -42 \neq 0$$

$\therefore$ The vectors are not coplanar

32. The forces $(2i + 2j + k)$ and $(3i + 4j - 2k)$ are acted on a particle produces a displacement from the point $(1, 2, 3)$ to $(5, 4, 1)$. Find the work done by the forces.

Solution:

$$\text{Total force } \vec{F} = (2i + 2j + k) + (3i + 4j - 2k) \\ = 5i + 6j - k$$

Now, position vector of $(1, 2, 3)$ is $(i + 2j + 3k)$

And position vector of $(5, 4, 1)$ is $(5i + 4j + k)$

$\therefore$ The displacement

$$\vec{d} = (5i + 4j + k) - (i + 2j + 3k) \\ = 4i + 2j - 2k$$

$$\therefore \text{The work done} = \vec{F} \cdot \vec{d} \\ = (5i + 6j - k) \cdot (4i + 2j - 2k) \\ = 20 + 12 + 2 \\ = 34 \text{ unit of work (Ans.)}$$

33. Force $\vec{F} = 2i - j - k$ is applied to an object making a displacement $\vec{d} = 3i + 2j - 5k$, find the work done.

Solution:

$$\text{Given, force } \vec{F} = 2i - j - k$$

$$\text{displacement } \vec{d} = 3i + 2j - 5k$$

Now, Work done $(W) = (\text{Force}) \cdot (\text{Displacement})$

$$\Rightarrow W = \vec{F} \cdot \vec{d} \\ = (2i - j - k) \cdot (3i + 2j - 5k) \\ = 6 - 2 + 5 = 9 \text{ unit (Ans)}$$

34. A particle acted on by a constant force of 22 kgf in the direction of the vector $(9i + 6j - 2k)$ is displaced from the point $(1, 2, 1)$ to the point $(5, 4, 2)$, find the work done if the displacements in meters.

Solution:

The unit vector in direction of $(9i + 6j - 2k)$ is

$$\frac{(9i + 6j - 2k)}{\sqrt{9^2 + 6^2 + (-2)^2}} = \frac{1}{11} (9i + 6j - 2k)$$

Hence, vector representing the force

$$\vec{F} = 22 \times \frac{1}{11} (9i + 6j - 2k) \\ = (18i + 12j - 4k) \text{ kgf} \\ = (18i + 12j - 4k) \times 9.8 \text{ N}$$

Displacement vector,

$$\vec{d} = (5i + 4j + 2k) - (i + 2j + k) = 4i + 2j + k$$

$$\therefore \text{Work done} = \vec{F} \cdot \vec{d} \\ = 9.8(18i + 12j - 4k) \cdot (4i + 2j + k) \\ = 9.8(72 + 24 - 4) = 9.8 \times 92 \\ = 901.6 \text{ N (Ans)}$$

35. A particle acted on by forces $(4i + j - 3k)$ and $(3i + j - k)$ is displaced from the point $A(1, 2, 3)$ to the point $B(5, 4, 1)$. Find the work done by the forces on the particle.

Solution:

Net force acted on the particle,

$$\vec{F} = (4i + j - 3k) + (3i + j - k) \\ = 7i + 2j - 4k$$

Let, O be an origin,

$$\therefore \overrightarrow{OA} = i + 2j + 3k \text{ and } \overrightarrow{OB} = 5i + 4j + k$$

$$\therefore \overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} \\ = (5i + 4j + k) - (i + 2j + 3k) \\ = 4i + 2j - 2k$$

$$\therefore \text{Work done} = (\text{Force}) \cdot (\text{Displacement}) \\ = \vec{F} \cdot \overrightarrow{AB} \\ = (7i + 2j - 4k) \cdot (4i + 2j - 2k) \\ = 28 + 4 + 8 = 40 \text{ units work (Ans.)}$$

36. Forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ of magnitude 14, 7 and 21 units which act in the direction of $2i - 3j + 6k$, $6i + 3j - 2k$ and $-3i + 6j - 2k$ respectively. Find the resultant force $\vec{F}$ of the forces. The resultant force $\vec{F}$ acts on a particle which is displaced from the point A(2, 3, 4) to the point B(3, 18, 8). Find the amount of work done.

Solution:

$$\begin{aligned}\vec{F}_1 &= 14 \times \frac{2i-3j+6k}{\sqrt{2^2+(-3)^2+6^2}} \\ &= 14 \times \frac{2i-3j+6k}{7} \\ &= 2(2i-3j+6k) = 4i-6j+12k \\ \vec{F}_2 &= 7 \times \frac{6i+3j-2k}{\sqrt{6^2+3^2+(-2)^2}} \\ &= 7 \times \frac{6i+3j-2k}{7} = 6i+3j-2k \\ \vec{F}_3 &= 21 \times \frac{(-3i+6j-2k)}{\sqrt{(-3)^2+6^2+(-2)^2}} \\ &= 21 \times \frac{(-3i+6j-2k)}{7} \\ &= 3(-3i+6j-2k) = -9i+18j-6k\end{aligned}$$

$$\begin{aligned}\therefore \vec{F} &= \vec{F}_1 + \vec{F}_2 + \vec{F}_3 \\ &= 4i-6j+12k + 6i+3j-2k - 9i+18j-6k \\ &= i+15j+4k \quad (\text{Ans})\end{aligned}$$

The position vectors of given two points are $(2i + 3j + 4k)$ and $(3i + 18j + 8k)$ respectively.

$$\begin{aligned}\therefore \text{Displacement } \vec{d} &= \vec{AB} = \vec{OB} - \vec{OA} \\ &= (3i + 18j + 8k) - (2i + 3j + 4k) \\ &= i + 15j + 4k\end{aligned}$$

$$\begin{aligned}\text{Hence work done by the forces} &= \vec{F} \cdot \vec{d} \\ &= (i + 15j + 4k) \cdot (i + 15j + 4k) \\ &= 1 + 225 + 16 = 242 \text{ Unit (Ans)}\end{aligned}$$

37. If $\vec{\alpha}$, $\vec{\beta}$ and $\vec{\gamma}$ are three vectors such that $\vec{\alpha} + \vec{\beta} + \vec{\gamma} = \vec{0}$ and $|\vec{\alpha}| = 2$, $|\vec{\beta}| = 4$, $|\vec{\gamma}| = 6$, then show that $\vec{\alpha} \cdot \vec{\beta} + \vec{\beta} \cdot \vec{\gamma} + \vec{\gamma} \cdot \vec{\alpha} = -28$

Solution:

$$\text{Given, } \vec{\alpha} + \vec{\beta} + \vec{\gamma} = \vec{0}$$

$$\text{and } |\vec{\alpha}| = 2, |\vec{\beta}| = 4, |\vec{\gamma}| = 6$$

$$\begin{aligned}\text{Now, } (\vec{\alpha} + \vec{\beta} + \vec{\gamma})^2 &= |\vec{\alpha}|^2 + |\vec{\beta}|^2 + |\vec{\gamma}|^2 + 2(\vec{\alpha} \cdot \vec{\beta} + \vec{\beta} \cdot \vec{\gamma} + \vec{\gamma} \cdot \vec{\alpha}) \\ \Rightarrow 0 &= 2^2 + 4^2 + 6^2 + 2(\vec{\alpha} \cdot \vec{\beta} + \vec{\beta} \cdot \vec{\gamma} + \vec{\gamma} \cdot \vec{\alpha}) \\ \Rightarrow 56 &+ 2(\vec{\alpha} \cdot \vec{\beta} + \vec{\beta} \cdot \vec{\gamma} + \vec{\gamma} \cdot \vec{\alpha}) = 0 \\ \Rightarrow 2(\vec{\alpha} \cdot \vec{\beta} + \vec{\beta} \cdot \vec{\gamma} + \vec{\gamma} \cdot \vec{\alpha}) &= -56 \\ \Rightarrow \vec{\alpha} \cdot \vec{\beta} + \vec{\beta} \cdot \vec{\gamma} + \vec{\gamma} \cdot \vec{\alpha} &= -28 \quad (\text{Proved})\end{aligned}$$

38. If $\vec{a}$ and $\vec{b}$ are unit vector and θ is the angle between them, then show that $\sin \frac{\theta}{2} = \frac{1}{2} |\vec{a} - \vec{b}|$.

Solution:

$$\text{We know, } \vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cos \theta$$

$$\Rightarrow \vec{a} \cdot \vec{b} = \cos \theta \quad [\because |\vec{a}| = 1, |\vec{b}| = 1]$$

$$\text{Now, } |\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b}$$

$$\Rightarrow |\vec{a} - \vec{b}|^2 = 1 + 1 - 2 \cos \theta$$

$$\Rightarrow |\vec{a} - \vec{b}|^2 = 2 - 2 \cos \theta$$

$$\Rightarrow |\vec{a} - \vec{b}|^2 = 2(1 - \cos \theta)$$

$$\Rightarrow |\vec{a} - \vec{b}|^2 = 2 \cdot 2 \sin^2 \frac{\theta}{2}$$

$$\Rightarrow |\vec{a} - \vec{b}|^2 = \left(2 \sin \frac{\theta}{2}\right)^2$$

$$\Rightarrow \sin \frac{\theta}{2} = \frac{1}{2} |\vec{a} - \vec{b}| \quad (\text{Proved})$$

39. If for two vectors $\vec{a}$ and $\vec{b}$, $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, find the angle between the vectors.

Solution:

$$\text{Given, } |\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$$

$$\Rightarrow |\vec{a} + \vec{b}|^2 = |\vec{a} - \vec{b}|^2$$

$$\Rightarrow (\vec{a})^2 + (\vec{b})^2 + 2\vec{a} \cdot \vec{b} = (\vec{a})^2 + (\vec{b})^2 - 2\vec{a} \cdot \vec{b}$$

$$\Rightarrow 2\vec{a} \cdot \vec{b} = -2\vec{a} \cdot \vec{b}$$

$$\Rightarrow 4\vec{a} \cdot \vec{b} = 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} = 0$$

$$\text{We know, } \vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cos \theta$$

$$\Rightarrow \cos \theta = 0 \quad [\because \vec{a} \cdot \vec{b} = 0]$$

$$\Rightarrow \cos \theta = \cos \frac{\pi}{2}$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

So, $\vec{a}$ and $\vec{b}$ are perpendicular to each other.

{Prepared by [NatiTute](#) YouTube Channel}

1. If $\cos 3x = \sin 2x$, then $x = ?$

- (a) 15°
(b) 18°
(c) 30°
(d) $22\frac{1}{2}^\circ$

Solution:

$$\begin{aligned}\cos 3x &= \sin 2x \\ \Rightarrow \cos 3x &= \cos(90^\circ - 2x) \\ \Rightarrow 3x &= 90^\circ - 2x \\ \Rightarrow 5x &= 90^\circ \\ \Rightarrow x &= 18^\circ \\ \text{Answer: (b)}\end{aligned}$$

2. The value of $\frac{\cos 8^\circ + \sin 8^\circ}{\cos 8^\circ - \sin 8^\circ}$:

- (a) $\tan 53^\circ$ (b) $\tan 43^\circ$ (c) $\cot 53^\circ$ (d) None

Solution:

$$\begin{aligned}\frac{\cos 8^\circ + \sin 8^\circ}{\cos 8^\circ - \sin 8^\circ} &= \frac{1 + \frac{\sin 8^\circ}{\cos 8^\circ}}{1 - \frac{\sin 8^\circ}{\cos 8^\circ}} = \frac{1 + \tan 8^\circ}{1 - \tan 8^\circ} = \frac{\tan 45^\circ + \tan 8^\circ}{1 - \tan 45^\circ \tan 8^\circ} \\ &= \tan(45^\circ + 8^\circ) = \tan 53^\circ \\ \text{Answer: (a)}\end{aligned}$$

3. $\tan x \tan 3x = 1$, then find the value of $\tan 2x$.

Solution:

$$\begin{aligned}\tan x \tan 3x &= 1 \\ \Rightarrow \tan 3x &= \frac{1}{\tan x} \\ \Rightarrow \tan 3x &= \cot x \\ \Rightarrow \tan 3x &= \tan\left(\frac{\pi}{2} - x\right) \\ \Rightarrow 3x &= \frac{\pi}{2} - x \\ \Rightarrow 4x &= \frac{\pi}{2} \\ \Rightarrow 2x &= \frac{\pi}{4} \\ \Rightarrow \tan 2x &= \tan \frac{\pi}{4} = 1 \text{ (Ans.)}\end{aligned}$$

4. If $\tan x \tan 5x = 1$, prove that $\tan 3x = 1$.

Solution:

$$\begin{aligned}\tan x \tan 5x &= 1 \\ \Rightarrow \tan 5x &= \frac{1}{\tan x} \\ \Rightarrow \tan 5x &= \cot x \\ \Rightarrow \tan 5x &= \tan\left(\frac{\pi}{2} - x\right) \\ \Rightarrow 5x &= \frac{\pi}{2} - x \\ \Rightarrow 6x &= \frac{\pi}{2} \\ \Rightarrow 3x &= \frac{\pi}{4} \\ \Rightarrow \tan 3x &= \tan \frac{\pi}{4} = 1 \text{ (Proved)}\end{aligned}$$

5. The value of $\cos 420^\circ$:

- (a) $\frac{1}{2}$ (b) $\sin 60^\circ$ (c) $\cos 30^\circ$ (d) None

Solution:

$$\begin{aligned}\cos 420^\circ &= \cos(90^\circ \times 4 + 60^\circ) = \cos 60^\circ = \frac{1}{2} \\ \text{Answer: (a)}\end{aligned}$$

6. Find the value of $\cos(-1170^\circ)$

Solution:

$$\begin{aligned}\cos(-1170^\circ) &= \cos 1170^\circ \\ &= \cos(90^\circ \times 13 + 0^\circ) \\ &= \sin 0 = 0 \text{ (Ans)}\end{aligned}$$

7. Find the value of $\cot(-870^\circ)$

Solution:

$$\begin{aligned}\cot(-870^\circ) &= -\cot 870^\circ \\ &= -\cot(9 \times 90^\circ + 60^\circ) \\ &= \tan 60^\circ = \sqrt{3} \text{ (Ans)}\end{aligned}$$

8. Find the value of $\sin 1755^\circ$.

Solution:

$$\begin{aligned}\sin 1755^\circ &= \sin(1710^\circ + 45^\circ) \\ &= \sin(19 \times 90^\circ + 45^\circ) \\ &= -\cos 45^\circ = -\frac{1}{\sqrt{2}} \text{ (Ans)}\end{aligned}$$

9. Find the value of

$$\sin 420^\circ \cos 390^\circ - \cos(-300^\circ) \sin(-330^\circ).$$

Solution:

$$\begin{aligned}\sin 420^\circ \cos 390^\circ - \cos(-300^\circ) \sin(-330^\circ) &= \sin 420^\circ \cos 390^\circ + \cos 300^\circ \sin 330^\circ \\ &= \sin(90^\circ \times 4 + 60^\circ) \cos(90^\circ \times 4 + 30^\circ) + \cos(90^\circ \times 3 + 30^\circ) \sin(90^\circ \times 3 + 60^\circ) \\ &= \sin 60^\circ \cos 30^\circ - \sin 30^\circ \cos 60^\circ \\ &= \sin(60^\circ - 30^\circ) = \sin 30^\circ = \frac{1}{2} \text{ (Ans)}\end{aligned}$$

10. If $\cos 2x = -\frac{1}{2}$, find the value of $\cos x$.

Solution:

$$\begin{aligned}\cos 2x &= -\frac{1}{2} \\ \text{or, } 2 \cos^2 x - 1 &= -\frac{1}{2} \\ \text{or, } 2 \cos^2 x &= 1 - \frac{1}{2} \\ \text{or, } 2 \cos^2 x &= \frac{1}{2} \\ \text{or, } \cos^2 x &= \frac{1}{4} \\ \text{or, } \cos x &= \frac{1}{2} \text{ (Ans.)}\end{aligned}$$

11. Find The value of $\tan\left(\frac{3\pi}{20}\right)\tan\left(\frac{5\pi}{20}\right)\tan\left(\frac{7\pi}{20}\right)$.

Solution:

$$\begin{aligned} & \tan\left(\frac{3\pi}{20}\right)\tan\left(\frac{5\pi}{20}\right)\tan\left(\frac{7\pi}{20}\right) \\ &= \tan\left(\frac{3 \times 180^\circ}{20}\right)\tan\left(\frac{5 \times 180^\circ}{20}\right)\tan\left(\frac{7 \times 180^\circ}{20}\right) \\ &= \tan 27^\circ \tan 45^\circ \tan 63^\circ \\ &= \tan 27^\circ \cdot 1 \cdot \tan(90^\circ - 27^\circ) \\ &= \tan 27^\circ \cot 27^\circ = 1 \quad (\text{Ans}) \end{aligned}$$

12. If $10\alpha = \frac{\pi}{2}$, then $\tan 3\alpha \tan 5\alpha \tan 7\alpha = ?$

Solution:

$$\text{Given, } 10\alpha = \frac{\pi}{2}$$

$$\Rightarrow 7\alpha = \frac{\pi}{2} - 3\alpha$$

$$\text{Now, } \tan 3\alpha \tan 5\alpha \tan 7\alpha$$

$$= \tan 3\alpha \tan\left(\frac{1}{2} \times 10\alpha\right) \tan\left(\frac{\pi}{2} - 3\alpha\right)$$

$$= \tan 3\alpha \tan\left(\frac{1}{2} \times \frac{\pi}{2}\right) \cot 3\alpha$$

$$= \tan 3\alpha \cot 3\alpha \tan\left(\frac{\pi}{4}\right)$$

$$= 1.1 = 1 \quad (\text{Ans})$$

13. Prove that $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

Solution:

L.H.S.

$$\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ$$

$$= \cos 20^\circ \cos 40^\circ \cdot \frac{1}{2} \cdot \cos 80^\circ$$

$$= \frac{1}{2 \sin 20^\circ} \sin 20^\circ \cos 20^\circ \cos 40^\circ \cos 80^\circ$$

$$= \frac{1}{4 \sin 20^\circ} 2 \sin 20^\circ \cos 20^\circ \cos 40^\circ \cos 80^\circ$$

$$= \frac{1}{8 \sin 20^\circ} 2 \sin 40^\circ \cos 40^\circ \cos 80^\circ$$

$$= \frac{1}{16 \sin 20^\circ} 2 \sin 80^\circ \cos 80^\circ$$

$$= \frac{1}{16 \sin 20^\circ} \sin 160^\circ$$

$$= \frac{1}{16 \sin 20^\circ} \sin(2 \times 90^\circ - 20^\circ)$$

$$= \frac{1}{16 \sin 20^\circ} \sin 20^\circ = \frac{1}{16} \quad (\text{Proved})$$

14. If $\tan \frac{\theta}{2} = \frac{3}{4}$, find the value of $\sin \theta$.

Solution:

$$\sin \theta = \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$$

$$= \frac{2 \times \frac{3}{4}}{1 + \left(\frac{3}{4}\right)^2} = \frac{\frac{3}{2}}{1 + \frac{9}{16}} = \frac{24}{16 + 9} = \frac{24}{25} \quad (\text{Ans})$$

15. Find the value of $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$.

Solution:

$$\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$$

$$= \frac{1}{2 \sin \frac{\pi}{7}} \left(2 \sin \frac{\pi}{7} \cos \frac{\pi}{7}\right) \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$$

$$= \frac{1}{4 \sin \frac{\pi}{7}} \left(2 \sin \frac{2\pi}{7} \cos \frac{2\pi}{7}\right) \cos \frac{4\pi}{7}$$

$$= \frac{1}{8 \sin \frac{\pi}{7}} \left(2 \sin \frac{4\pi}{7} \cos \frac{4\pi}{7}\right)$$

$$= \frac{1}{8 \sin \frac{\pi}{7}} \sin \frac{8\pi}{7}$$

$$= \frac{1}{8 \sin \frac{\pi}{7}} \sin \left(\pi + \frac{\pi}{7}\right)$$

$$= \frac{1}{8 \sin \frac{\pi}{7}} \left(-\sin \frac{\pi}{7}\right) = -\frac{1}{8} \quad (\text{Ans})$$

16. If $180^\circ < \theta < 270^\circ$ and $\sin \theta = -\frac{3}{5}$, find the value of $\cos \theta$.

Solution:

$$\text{Given, } \sin \theta = -\frac{3}{5}$$

$$\Rightarrow \sin^2 \theta = \frac{9}{25}$$

$$\Rightarrow 1 - \sin^2 \theta = 1 - \frac{9}{25}$$

$$\Rightarrow \cos^2 \theta = \frac{16}{25}$$

$$\Rightarrow \cos \theta = \pm \sqrt{\frac{16}{25}} = \pm \frac{4}{5}$$

As $180^\circ < \theta < 270^\circ$ where $\cos \theta$ is (-ve).

$$\therefore \cos \theta = -\frac{4}{5} \quad (\text{Ans})$$

17. If $\tan \theta = -\frac{1}{\sqrt{5}}$, $\sin \theta$ is negative then find the value of $\cos \theta$.

Solution:

$$\tan \theta = -\frac{1}{\sqrt{5}}$$

$$\Rightarrow \tan^2 \theta = \frac{1}{5}$$

$$\Rightarrow 1 + \tan^2 \theta = 1 + \frac{1}{5}$$

$$\Rightarrow \sec^2 \theta = \frac{6}{5}$$

$$\Rightarrow \cos^2 \theta = \frac{5}{6}$$

$$\Rightarrow \cos \theta = \pm \sqrt{\frac{5}{6}} \quad [\because \sin \theta, \tan \theta \text{ is } (-ve)]$$

$$\Rightarrow \cos \theta = -\sqrt{\frac{5}{6}} \quad (\text{Ans.})$$

18. If $\tan \theta = 2$, then find the value of $\cos 2\theta$.

Solution: Given, $\tan \theta = 2$

$$\begin{aligned}\text{We know, } \cos 2\theta &= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \\ &= \frac{1 - 2^2}{1 + 2^2} \\ &= \frac{1 - 4}{1 + 4} = \frac{-3}{5} \quad (\text{Ans})\end{aligned}$$

19. Find the maximum value of $(\sin \theta + \cos \theta)$.

Solution:

$$\begin{aligned}\sin \theta + \cos \theta &= \sqrt{(\sin \theta + \cos \theta)^2} \\ &= \sqrt{\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta} \\ &= \sqrt{1 + \sin 2\theta} \\ \text{Maximum value of } \sin 2\theta &= 1 \\ \therefore (\sin \theta + \cos \theta)_{\max} &= \sqrt{1 + 1} = \sqrt{2} \quad (\text{Ans})\end{aligned}$$

20. Find the greatest value of $(\sin x \cos x)$.

Solution:

$$\begin{aligned}\sin x \cos x &= \frac{1}{2} 2 \sin x \cos x = \frac{1}{2} \sin 2x \\ \text{As the greatest value of } \sin 2x &= 1 \\ \therefore \text{The greatest value of } \sin x \cos x &= \frac{1}{2} \quad (\text{Ans})\end{aligned}$$

21. Prove that $\tan A + \frac{1}{\tan A} = 2 \operatorname{cosec} 2A$.

Solution:

LHS

$$\begin{aligned}\tan A + \frac{1}{\tan A} &= \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} \\ &= \frac{\sin^2 A + \cos^2 A}{\sin A \cos A} \\ &= \frac{1}{\sin A \cos A} \\ &= \frac{2}{2 \sin A \cos A} \\ &= \frac{2}{\sin 2A} = 2 \operatorname{cosec} 2A = \text{RHS} \quad (\text{Proved})\end{aligned}$$

22. If $\tan(A + B) = \frac{1}{2}$ and $\tan(A - B) = \frac{1}{3}$, find the value of $\tan 2A$.

Solution:

$$\begin{aligned}\text{Given, } \tan(A + B) &= \frac{1}{2} \text{ and } \tan(A - B) = \frac{1}{3} \\ \text{Now, } \tan 2A &= \tan\{(A + B) + (A - B)\} \\ &= \frac{\tan(A + B) + \tan(A - B)}{1 - \tan(A + B) \tan(A - B)} \\ &= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}} \\ &= \frac{6(\frac{1}{2} + \frac{1}{3})}{6(1 - \frac{1}{6})} = \frac{3 + 2}{6 - 1} = \frac{5}{5} = 1 \quad (\text{Ans})\end{aligned}$$

23. Show that $\frac{\sin(x+y)}{\sin(x-y)} = \frac{\tan x + \tan y}{\tan x - \tan y}$.

Solution:

L.H.S

$$\begin{aligned}\frac{\sin(x+y)}{\sin(x-y)} &= \frac{\sin x \cos y + \cos x \sin y}{\sin x \cos y - \cos x \sin y} \\ &= \frac{\frac{\sin x \cos y}{\cos x \cos y} + \frac{\cos x \sin y}{\cos x \cos y}}{\frac{\sin x \cos y}{\cos x \cos y} - \frac{\cos x \sin y}{\cos x \cos y}} \\ &= \frac{\frac{\sin x \cos y}{\cos x \cos y} + \frac{\cos x \sin y}{\cos x \cos y}}{\frac{\sin x \cos y}{\cos x \cos y} - \frac{\cos x \sin y}{\cos x \cos y}} = \frac{\tan x + \tan y}{\tan x - \tan y} \quad (\text{Proved})\end{aligned}$$

24. Prove that $\sec \alpha + \tan \alpha = \tan\left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$.

Solution:

L.H.S.

$$\begin{aligned}\sec \alpha + \tan \alpha &= \frac{1}{\cos \alpha} + \frac{\sin \alpha}{\cos \alpha} = \frac{1 + \sin \alpha}{\cos \alpha} \\ &= \frac{\sin^2 \frac{\alpha}{2} + \cos^2 \frac{\alpha}{2} + 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}}{\cos^2 \frac{\alpha}{2} - \sin^2 \frac{\alpha}{2}} = \frac{(\sin \frac{\alpha}{2} + \cos \frac{\alpha}{2})^2}{(\cos \frac{\alpha}{2} + \sin \frac{\alpha}{2})(\cos \frac{\alpha}{2} - \sin \frac{\alpha}{2})} \\ &= \frac{\sin \frac{\alpha}{2} + \cos \frac{\alpha}{2}}{\cos \frac{\alpha}{2} - \sin \frac{\alpha}{2}} = \frac{1 + \tan \frac{\alpha}{2}}{1 - \tan \frac{\alpha}{2}} \\ &= \frac{\tan \frac{\pi}{4} + \tan \frac{\alpha}{2}}{1 - \tan \frac{\pi}{4} \tan \frac{\alpha}{2}} = \tan\left(\frac{\pi}{4} + \frac{\alpha}{2}\right) = \text{R.H.S.} \quad (\text{Proved})\end{aligned}$$

25. Prove that $\sin 3x \operatorname{cosec} x - \cos 3x \sec x = 2$.

Solution:

L.H.S.

$$\begin{aligned}\sin 3x \operatorname{cosec} x - \cos 3x \sec x &= \frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x} \\ &= \frac{\sin 3x \cos x - \cos 3x \sin x}{\sin x \cos x} \\ &= \frac{2 \sin(3x - x)}{2 \sin x \cos x} = \frac{2 \sin 2x}{\sin 2x} = 2 \quad (\text{Proved})\end{aligned}$$

26. If $\sin^4 x + \sin^2 x = 1$, show that

$$\cot^4 x + \cot^2 x = 1.$$

Solution:

$$\begin{aligned}\sin^4 x + \sin^2 x &= 1 \\ \Rightarrow \sin^4 x &= 1 - \sin^2 x = \cos^2 x \\ \text{LHS} &= \cot^4 x + \cot^2 x \\ &= \frac{\cos^4 x}{\sin^4 x} + \frac{\cos^2 x}{\sin^2 x} \\ &= \frac{\cos^4 x}{\cos^2 x} + \frac{\sin^4 x}{\sin^2 x} \quad [\because \sin^4 x = \cos^2 x] \\ &= \cos^2 x + \sin^2 x \\ &= 1 = \text{RHS} \quad (\text{Proved})\end{aligned}$$

27. Find the value of

$$\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \dots \cos 179^\circ.$$

Solution:

$$\begin{aligned}\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \dots \cos 179^\circ &= \cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 90^\circ \dots \cos 179^\circ \\ &= 0 \quad [\because \cos 90^\circ = 0] \quad (\text{Ans})\end{aligned}$$

28. Find the value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ$.

Solution:

$$\begin{aligned} & \tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ \\ &= \tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 45^\circ \dots \tan(90^\circ - 2^\circ) \tan(90^\circ - 1^\circ) \\ &= \tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 45^\circ \dots \cot 2^\circ \cot 1^\circ \\ &= 1 [\because \tan \theta \cot \theta = 1] \quad (\text{Ans}) \end{aligned}$$

29. Prove that, $\tan \frac{\pi}{12} \tan \frac{5\pi}{12} \tan \frac{7\pi}{12} \tan \frac{11\pi}{12} = 1$

Solution:

L.H.S

$$\begin{aligned} & \tan \frac{\pi}{12} \tan \frac{5\pi}{12} \tan \frac{7\pi}{12} \tan \frac{11\pi}{12} \\ &= \tan \frac{\pi}{12} \tan \left(\frac{\pi}{2} - \frac{\pi}{12} \right) \tan \left(\frac{\pi}{2} + \frac{\pi}{12} \right) \tan \left(\pi - \frac{\pi}{12} \right) \\ &= \tan \frac{\pi}{12} \cot \frac{\pi}{12} \left(-\cot \frac{\pi}{12} \right) \left(-\tan \frac{\pi}{12} \right) \\ &= 1.1 = 1 \quad (\text{Proved}) \end{aligned}$$

30. Prove that, $16 \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{16\pi}{15} = 1$

Solution:

L.H.S

$$\begin{aligned} & 16 \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{16\pi}{15} \\ &= \frac{8}{\sin \frac{2\pi}{15}} \cdot 2 \sin \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{16\pi}{15} \\ &= \frac{4}{\sin \frac{2\pi}{15}} \cdot 2 \sin \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{16\pi}{15} \\ &= \frac{2}{\sin \frac{2\pi}{15}} \cdot 2 \sin \frac{8\pi}{15} \cos \frac{16\pi}{15} \\ &= \frac{1}{\sin \frac{2\pi}{15}} \cdot 2 \sin \frac{16\pi}{15} \cos \frac{16\pi}{15} \\ &= \frac{1}{\sin \frac{2\pi}{15}} \cdot \sin \frac{32\pi}{15} \\ &= \frac{1}{\sin \frac{2\pi}{15}} \cdot \sin \left(2\pi + \frac{2\pi}{15} \right) \\ &= \frac{1}{\sin \frac{2\pi}{15}} \cdot \sin \frac{2\pi}{15} = 1 \quad (\text{Ans}) \end{aligned}$$

31. Prove that, $\frac{\cos 10^\circ - \sin 10^\circ}{\cos 10^\circ + \sin 10^\circ} = \tan 35^\circ$.

Solution:

$$\begin{aligned} & \frac{\cos 10^\circ - \sin 10^\circ}{\cos 10^\circ + \sin 10^\circ} \\ &= \frac{1 - \tan 10^\circ}{1 + \tan 10^\circ} \\ &= \frac{1 - \tan 10^\circ}{\tan 45^\circ + \tan 10^\circ} = \tan 35^\circ \quad (\text{Proved}) \end{aligned}$$

32. Find the value of $\tan 27^\circ + \tan 18^\circ + \tan 27^\circ \tan 18^\circ$

Solution:

$$\begin{aligned} & \text{Now, } \tan 45^\circ = \tan(27^\circ + 18^\circ) \\ & \Rightarrow 1 = \frac{\tan 27^\circ + \tan 18^\circ}{1 - \tan 27^\circ \tan 18^\circ} \\ & \Rightarrow \tan 27^\circ + \tan 18^\circ = 1 - \tan 27^\circ \tan 18^\circ \\ & \Rightarrow \tan 27^\circ + \tan 18^\circ + \tan 27^\circ \tan 18^\circ = 1 \quad (\text{Ans.}) \end{aligned}$$

33. Prove that, $\frac{\sin 2\theta}{1 + \cos 2\theta} \cdot \frac{\cos \theta}{1 + \cos \theta} = \tan \frac{\theta}{2}$

Solution:

$$\begin{aligned} & \frac{\sin 2\theta}{1 + \cos 2\theta} \cdot \frac{\cos \theta}{1 + \cos \theta} \\ &= \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta} \cdot \frac{\cos \theta}{1 + \cos \theta} \\ &= \frac{\sin \theta}{1 + \cos \theta} \\ &= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = \tan \frac{\theta}{2} \quad (\text{Proved}) \end{aligned}$$

34. If $\tan 25^\circ = a$, show that $\frac{\tan 155^\circ - \tan 115^\circ}{1 + \tan 155^\circ \tan 115^\circ} = \frac{1 - a^2}{2a}$.

Solution:

$$\begin{aligned} & \tan 155^\circ = \tan(2 \times 90^\circ - 25^\circ) = -\tan 25^\circ = -a \\ & \tan 115^\circ = \tan(90^\circ + 25^\circ) = -\cot 25^\circ = -\frac{1}{a} \\ & \text{LHS} = \frac{\tan 155^\circ - \tan 115^\circ}{1 + \tan 155^\circ \tan 115^\circ} \\ &= \frac{-a + \frac{1}{a}}{1 + (-a)\left(-\frac{1}{a}\right)} \\ &= \frac{\frac{-a^2 + 1}{a}}{\frac{a^2 - 1}{a}} = \frac{a^2 - 1}{2a} = \text{RHS} \quad (\text{Proved}) \end{aligned}$$

35. If $\cos 2\theta = -\frac{1}{2}$ then what is the value of $\cos \theta$

Solution:

$$\begin{aligned} & \cos 2\theta = -\frac{1}{2} \\ & \Rightarrow \cos 2\theta = \cos 120^\circ \\ & \Rightarrow 2\theta = 120^\circ \\ & \Rightarrow \theta = 60^\circ \\ & \Rightarrow \cos \theta = \cos 60^\circ \\ & \Rightarrow \cos \theta = \frac{1}{2} \quad (\text{Ans}) \end{aligned}$$

36. Find the value of $\left(\frac{1}{\sin 10^\circ} - \frac{\sqrt{3}}{\cos 10^\circ} \right)$.

Solution:

$$\begin{aligned} & \frac{1}{\sin 10^\circ} - \frac{\sqrt{3}}{\cos 10^\circ} \\ &= \frac{\cos 10^\circ - \sqrt{3} \sin 10^\circ}{\sin 10^\circ \cos 10^\circ} \\ &= \frac{2 \left(\frac{1}{2} \cos 10^\circ - \frac{\sqrt{3}}{2} \sin 10^\circ \right)}{\frac{1}{2} \times 2 \sin 10^\circ \cos 10^\circ} \\ &= \frac{4(\sin 30^\circ \cos 10^\circ - \cos 30^\circ \sin 10^\circ)}{\sin 20^\circ} \\ &= \frac{4 \sin(30^\circ - 10^\circ)}{\sin 20^\circ} = \frac{4 \sin 20^\circ}{\sin 20^\circ} = 4 \quad (\text{Ans}) \end{aligned}$$

37. Show that $\frac{1}{2 \sin 10^\circ} - 2 \sin 70^\circ = 1$

Solution:

$$\begin{aligned} & \text{LHS} = \frac{1}{2 \sin 10^\circ} - 2 \sin 70^\circ \\ &= \frac{1 - 2 \sin 70^\circ \sin 10^\circ}{2 \sin 10^\circ} \\ &= \frac{1 - 2[\cos(70^\circ - 10^\circ) - \cos(70^\circ + 10^\circ)]}{2 \sin 10^\circ} \\ &= \frac{1 - 2[\cos 60^\circ - \cos 80^\circ]}{2 \sin 10^\circ} \\ &= \frac{1 - 2\left[\frac{1}{2} - \cos(90^\circ - 10^\circ)\right]}{2 \sin 10^\circ} \\ &= \frac{1 - 1 + 2 \sin 10^\circ}{2 \sin 10^\circ} \\ &= \frac{2 \sin 10^\circ}{2 \sin 10^\circ} = 1 = \text{RHS} \quad (\text{Proved}) \end{aligned}$$

38. Find the value of $(\sin 105^\circ + \cos 105^\circ)$.

Solution:

$$\begin{aligned} & (\sin 105^\circ + \cos 105^\circ) \\ &= \sin(60^\circ + 45^\circ) + \cos(60^\circ + 45^\circ) \\ &= \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ \\ &\quad + \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ \\ &= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} \\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}} \\ &= \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \quad (\text{Ans}) \end{aligned}$$

39. What is the value of $\sin 75^\circ \sin 15^\circ$?

Solution:

$$\begin{aligned} & \sin 75^\circ \sin 15^\circ \\ &= \sin(90^\circ - 15^\circ) \sin 15^\circ \\ &= \sin 15^\circ \cos 15^\circ \\ &= \frac{1}{2} 2 \sin 15^\circ \cos 15^\circ \\ &= \frac{1}{2} \sin 30^\circ = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \quad (\text{Ans}) \end{aligned}$$

40. If $A + B = 45^\circ$ then prove that $(1 + \tan A)(1 + \tan B) = 2$ and $\tan\left(22\frac{1}{2}^\circ\right) = \sqrt{2} - 1$

Solution:

$$\begin{aligned} & A + B = 45^\circ \\ & \Rightarrow \tan(A + B) = \tan 45^\circ \\ & \Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = 1 \\ & \Rightarrow \tan A + \tan B = 1 - \tan A \tan B \\ & \Rightarrow \tan A + \tan A \tan B + 1 + \tan B = 2 \\ & \Rightarrow \tan A (1 + \tan B) + (1 + \tan B) = 2 \\ & \Rightarrow (1 + \tan A)(1 + \tan B) = 2 \quad (\text{Proved}) \end{aligned}$$

If $A = B$ then

$$\begin{aligned} & A + B = 45^\circ \\ & \Rightarrow A + A = 45^\circ \\ & \Rightarrow 2A = 45^\circ \\ & \Rightarrow A = 22\frac{1}{2}^\circ \end{aligned}$$

Now,

$$\begin{aligned} & (1 + \tan A)(1 + \tan B) = 2 \\ & \Rightarrow (1 + \tan A)(1 + \tan A) = 2 \\ & \Rightarrow (1 + \tan A)^2 = 2 \\ & \Rightarrow 1 + \tan A = \sqrt{2} \\ & \Rightarrow \tan A = \sqrt{2} - 1 \\ & \Rightarrow \tan 22\frac{1}{2}^\circ = \sqrt{2} - 1 \quad (\text{Proved}) \end{aligned}$$

41. Find the value of,

$$\cos^2\left(\frac{\pi}{8}\right) + \cos^2\left(\frac{3\pi}{8}\right) + \cos^2\left(\frac{5\pi}{8}\right) + \cos^2\left(\frac{7\pi}{8}\right)$$

Solution:

$$\begin{aligned} & \cos^2\left(\frac{\pi}{8}\right) + \cos^2\left(\frac{3\pi}{8}\right) + \cos^2\left(\frac{5\pi}{8}\right) + \cos^2\left(\frac{7\pi}{8}\right) \\ &= \cos^2\left(\frac{\pi}{8}\right) + \cos^2\left(\frac{3\pi}{8}\right) + \cos^2\left(\frac{\pi}{2} + \frac{\pi}{8}\right) + \cos^2\left(\frac{\pi}{2} + \frac{3\pi}{8}\right) \\ &= \cos^2\left(\frac{\pi}{8}\right) + \sin^2\left(\frac{\pi}{8}\right) + \cos^2\left(\frac{3\pi}{8}\right) + \sin^2\left(\frac{3\pi}{8}\right) \\ &= 1 + 1 = 2 \quad (\text{Ans}) \end{aligned}$$

42. If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, then prove that

$$\cos \theta - \sin \theta = \sqrt{2} \sin \theta$$

Solution:

$$\begin{aligned} & \cos \theta + \sin \theta = \sqrt{2} \cos \theta \\ & \Rightarrow (\cos \theta + \sin \theta)^2 = 2 \cos^2 \theta \\ & \Rightarrow \cos^2 \theta + \sin^2 \theta + 2 \cos \theta \sin \theta = 2(1 - \sin^2 \theta) \\ & \Rightarrow 1 + 2 \cos \theta \sin \theta = 2 - 2 \sin^2 \theta \\ & \Rightarrow -1 + 2 \cos \theta \sin \theta = -2 \sin^2 \theta \\ & \Rightarrow 1 - 2 \cos \theta \sin \theta = 2 \sin^2 \theta \\ & \Rightarrow \cos^2 \theta + \sin^2 \theta - 2 \cos \theta \sin \theta = 2 \sin^2 \theta \\ & \Rightarrow (\cos \theta - \sin \theta)^2 = 2 \sin^2 \theta \\ & \Rightarrow \cos \theta - \sin \theta = \sqrt{2} \sin \theta \quad (\text{Proved}) \end{aligned}$$

43. If $x \sin^3 a + y \cos^3 a = \sin a \cos a$ and $x \sin a = y \cos a$, prove that $x^2 + y^2 = 1$

Solution:

$$\text{Given, } x \sin a = y \cos a$$

$$\Rightarrow \frac{x}{\cos a} = \frac{y}{\sin a} = k \quad (\text{Let})$$

$$\therefore x = k \cos a \text{ and } y = k \sin a$$

$$\text{Now, } x \sin^3 a + y \cos^3 a = \sin a \cos a$$

$$\Rightarrow k \cos a \sin^3 a + k \sin a \cos^3 a = \sin a \cos a$$

$$\Rightarrow k \sin a \cos a (\sin^2 a + \cos^2 a) = \sin a \cos a$$

$$\Rightarrow k = 1$$

$$\therefore x^2 + y^2 = (k \cos a)^2 + (k \sin a)^2$$

$$= k^2 \cos^2 a + k^2 \sin^2 a$$

$$= k^2 (\cos^2 a + \sin^2 a)$$

$$= 1 \cdot 1 = 1 \quad (\text{Proved})$$

44. If A and B are positive acute angles and

$$\cos 2A = \frac{3 \cos 2B - 1}{3 - \cos 2B}. \text{ Show that } \tan A = \sqrt{2} \tan B.$$

Solution:

$$\cos 2A = \frac{3 \cos 2B - 1}{3 - \cos 2B}$$

$$\Rightarrow \frac{1}{\cos 2A} = \frac{3 \cos 2B - 1}{3 - \cos 2B}$$

$$\Rightarrow \frac{1 - \cos 2A}{1 + \cos 2A} = \frac{3 \cos 2B - 1}{3 - \cos 2B}$$

$$\Rightarrow \tan^2 A = \frac{4 - 4 \cos 2B}{2 + 2 \cos 2B}$$

$$\Rightarrow \tan^2 A = \frac{4}{2} \times \frac{1 - \cos 2B}{1 + \cos 2B}$$

$$\Rightarrow \tan^2 A = 2 \tan^2 B$$

$$\Rightarrow \tan A = \sqrt{2} \tan B \quad (\text{Proved})$$

45. If $\cos^2 \theta - \sin^2 \theta = \tan^2 \alpha$ then show that

$$\cos^2 \alpha - \sin^2 \alpha = \tan^2 \theta.$$

Solution:

$$\cos^2 \theta - \sin^2 \theta = \tan^2 \alpha$$

$$\Rightarrow \cos 2\theta = \tan^2 \alpha$$

$$\Rightarrow \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \tan^2 \alpha$$

$$\Rightarrow \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1}{\tan^2 \alpha}$$

$$\Rightarrow \frac{1 + \tan^2 \theta + 1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha}$$

$$\Rightarrow \frac{2 \tan^2 \theta}{2} = \cos 2\alpha$$

$$\Rightarrow \cos 2\alpha = \tan^2 \theta$$

$$\Rightarrow \cos^2 \alpha - \sin^2 \alpha = \tan^2 \theta \quad (\text{Proved})$$

46. If $1 + 2 \tan^2 \phi = \tan^2 \theta$, show that
 $\cos 2\phi = 1 + 2 \cos 2\theta$.

Solution:

$$\begin{aligned} 1 + 2 \tan^2 \phi &= \tan^2 \theta \\ \Rightarrow 1 + 2 \frac{1 - \cos 2\phi}{1 + \cos 2\phi} &= \frac{1 - \cos 2\theta}{1 + \cos 2\theta} \\ \Rightarrow \frac{1 + \cos 2\phi + 2 - 2 \cos 2\phi}{1 + \cos 2\phi} &= \frac{1 - \cos 2\theta}{1 + \cos 2\theta} \\ \Rightarrow \frac{3 - \cos 2\phi}{1 + \cos 2\phi} &= \frac{1 - \cos 2\theta}{1 + \cos 2\theta} \\ \Rightarrow \frac{3 - \cos 2\phi + 1 + \cos 2\phi}{3 - \cos 2\phi - 1 - \cos 2\phi} &= \frac{1 - \cos 2\theta + 1 + \cos 2\theta}{1 - \cos 2\theta - 1 - \cos 2\theta} \\ \Rightarrow \frac{4}{2 - 2 \cos 2\phi} &= \frac{2}{-2 \cos 2\theta} \\ \Rightarrow \frac{2}{1 - \cos 2\phi} &= \frac{1}{-\cos 2\theta} \\ \Rightarrow \frac{2}{1 - \cos 2\phi} &= \frac{1}{-\cos 2\theta} \\ \Rightarrow -2 \cos 2\theta &= 1 - \cos 2\phi \\ \Rightarrow \cos 2\phi &= 1 + 2 \cos 2\theta \text{ (Proved)} \end{aligned}$$

47. If $2 \tan \alpha = 3 \tan \beta$, show that

$$\tan(\alpha - \beta) = \frac{\sin 2\beta}{5 - \cos 2\beta}.$$

Solution:

$$\text{Given, } 2 \tan \alpha = 3 \tan \beta \Rightarrow \tan \alpha = \frac{3}{2} \tan \beta$$

LHS

$$\begin{aligned} &\frac{\tan(\alpha - \beta)}{\tan \alpha - \tan \beta} \\ &= \frac{1 + \tan \alpha \tan \beta}{\frac{3}{2} \tan \beta - \tan \beta} \\ &= \frac{1 + \frac{3}{2} \tan \beta \tan \beta}{\tan \beta} \\ &= \frac{2 + 3 \tan^2 \beta}{\frac{\sin \beta}{\cos \beta}} \\ &= \frac{2 + 3 \frac{\sin^2 \beta}{\cos^2 \beta}}{\frac{\sin \beta \cos \beta}{\cos^2 \beta}} \\ &= \frac{2 \cos^2 \beta + 3 \sin^2 \beta}{2 \sin \beta \cos \beta} \\ &= \frac{4 \cos^2 \beta + 6 \sin^2 \beta}{\sin 2\beta} \\ &= \frac{4 \cos^2 \beta + 4 \sin^2 \beta + 2 \sin^2 \beta}{\sin 2\beta} \\ &= \frac{4(\cos^2 \beta + \sin^2 \beta) + 1 - \cos 2\beta}{\sin 2\beta} \\ &= \frac{4 + 1 - \cos 2\beta}{\sin 2\beta} \\ &= \frac{\sin 2\beta}{5 - \cos 2\beta} = \text{RHS (Proved)} \end{aligned}$$

48. If $x \cos \alpha + y \sin \alpha = x \cos \beta + y \sin \beta$ show that

$$\tan\left(\frac{\alpha + \beta}{2}\right) = \frac{y}{x}.$$

Solution:

$$\begin{aligned} \text{Given, } x \cos \alpha + y \sin \alpha &= x \cos \beta + y \sin \beta \\ \Rightarrow x \cos \alpha - x \cos \beta &= y \sin \beta - y \sin \alpha \\ \Rightarrow x(\cos \alpha - \cos \beta) &= y(\sin \beta - \sin \alpha) \\ \Rightarrow x \cdot 2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\beta - \alpha}{2}\right) &= y \cdot 2 \cos\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\beta - \alpha}{2}\right) \\ \Rightarrow \frac{\sin\left(\frac{\alpha + \beta}{2}\right)}{\cos\left(\frac{\alpha + \beta}{2}\right)} &= \frac{y}{x} \\ \Rightarrow \tan\left(\frac{\alpha + \beta}{2}\right) &= \frac{y}{x} \text{ (Proved)} \end{aligned}$$

49. If $\sin \theta + \sin \phi = a$ and $\cos \theta + \cos \phi = b$ then prove that,

$$\tan\left(\frac{\theta - \phi}{2}\right) = \pm \sqrt{\frac{4 - a^2 - b^2}{a^2 + b^2}}$$

Solution:

$$\begin{aligned} \sin \theta + \sin \phi &= a \text{ and } \cos \theta + \cos \phi = b \\ \Rightarrow (\sin \theta + \sin \phi)^2 + (\cos \theta + \cos \phi)^2 &= a^2 + b^2 \\ \Rightarrow \sin^2 \theta + \cos^2 \theta + \sin^2 \phi + \cos^2 \phi + &2(\sin \theta \sin \phi + \cos \theta \cos \phi) = a^2 + b^2 \\ \Rightarrow 1 + 1 + 2 \cos(\theta - \phi) &= a^2 + b^2 \\ \Rightarrow 2 + 2 \cos(\theta - \phi) &= a^2 + b^2 \\ \Rightarrow 1 + \cos(\theta - \phi) &= \frac{a^2 + b^2}{2} \\ \Rightarrow 2 \cos^2\left(\frac{\theta - \phi}{2}\right) &= \frac{a^2 + b^2}{2} \\ \Rightarrow \cos^2\left(\frac{\theta - \phi}{2}\right) &= \frac{a^2 + b^2}{4} \\ \Rightarrow \sec^2\left(\frac{\theta - \phi}{2}\right) &= \frac{4}{a^2 + b^2} \\ \Rightarrow \tan^2\left(\frac{\theta - \phi}{2}\right) + 1 &= \frac{4}{a^2 + b^2} \\ \Rightarrow \tan^2\left(\frac{\theta - \phi}{2}\right) &= \frac{4}{a^2 + b^2} - 1 \\ \Rightarrow \tan^2\left(\frac{\theta - \phi}{2}\right) &= \frac{4 - a^2 - b^2}{a^2 + b^2} \\ \Rightarrow \tan\left(\frac{\theta - \phi}{2}\right) &= \pm \sqrt{\frac{4 - a^2 - b^2}{a^2 + b^2}} \text{ (Proved)} \end{aligned}$$

50. If $\tan \frac{\theta}{2} = \tan^3 \frac{\phi}{2}$ and $\tan \phi = 2 \tan \alpha$, then prove that $\theta + \phi = 2\alpha$.

Solution:

$$\begin{aligned} \tan\left(\frac{\theta}{2} + \frac{\phi}{2}\right) &= \frac{\tan \frac{\theta}{2} + \tan \frac{\phi}{2}}{1 - \tan \frac{\theta}{2} \tan \frac{\phi}{2}} \\ &= \frac{\tan^3 \frac{\phi}{2} + \tan \frac{\phi}{2}}{1 - \tan^3 \frac{\phi}{2} \tan \frac{\phi}{2}} \quad \left[\because \tan \frac{\theta}{2} = \tan^3 \frac{\phi}{2} \right] \\ &= \frac{\tan \frac{\phi}{2} (1 + \tan^2 \frac{\phi}{2})}{1 - \tan^4 \frac{\phi}{2}} \\ &= \frac{\tan \frac{\phi}{2} (1 + \tan^2 \frac{\phi}{2})}{(1 + \tan^2 \frac{\phi}{2})(1 - \tan^2 \frac{\phi}{2})} \\ &= \frac{1}{2} \cdot \frac{2 \tan \frac{\phi}{2}}{(1 - \tan^2 \frac{\phi}{2})} \\ &= \frac{1}{2} \cdot \tan \phi \\ &= \frac{1}{2} \cdot 2 \tan \alpha \quad [\because \tan \phi = 2 \tan \alpha] \\ &= \tan \alpha \end{aligned}$$

$$\therefore \tan\left(\frac{\theta}{2} + \frac{\phi}{2}\right) = \tan \alpha$$

$$\Rightarrow \frac{\theta}{2} + \frac{\phi}{2} = \alpha$$

$$\Rightarrow \theta + \phi = 2\alpha \text{ (Proved)}$$

51. If $\cos \theta = \frac{a \cos \varphi + b}{a + b \cos \varphi}$ then prove that

$$\tan \frac{\theta}{2} = \pm \sqrt{\frac{a-b}{a+b}} \tan \frac{\varphi}{2}$$

Solution:

Now,

$$\begin{aligned} \tan^2 \frac{\theta}{2} &= \frac{1 - \cos \theta}{1 + \cos \theta} \\ &= \frac{1 - \frac{a \cos \varphi + b}{a + b \cos \varphi}}{1 + \frac{a \cos \varphi + b}{a + b \cos \varphi}} \\ &= \frac{a + b \cos \varphi - a \cos \varphi - b}{a + b \cos \varphi + a \cos \varphi + b} \\ &= \frac{(a-b) - \cos \varphi (a-b)}{(a+b) + \cos \varphi (a+b)} \\ &= \frac{(a-b)(1 - \cos \varphi)}{(a+b)(1 + \cos \varphi)} \\ &= \frac{(a-b)}{(a+b)} \cdot \tan^2 \frac{\varphi}{2} \\ \therefore \tan^2 \frac{\theta}{2} &= \frac{(a-b)}{(a+b)} \cdot \tan^2 \frac{\varphi}{2} \\ \Rightarrow \tan \frac{\theta}{2} &= \pm \sqrt{\frac{a-b}{a+b}} \tan \frac{\varphi}{2} \text{ (Proved)} \end{aligned}$$

52. If $\cos \theta + \sin \theta = 2$, then find $\sin 2\theta$.

Solution:

Given, $\cos \theta + \sin \theta = 2$

$$\Rightarrow (\cos \theta + \sin \theta)^2 = 4$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta + 2 \cos \theta \sin \theta = 4$$

$$\Rightarrow 1 + \sin 2\theta = 4$$

$$\Rightarrow \sin 2\theta = 4 - 1 = 3 \text{ (Ans)}$$

53. If $\tan \theta = \sec 2\alpha$ then prove that, $\sin 2\theta = \frac{1 - \tan^4 \alpha}{1 + \tan^4 \alpha}$

Solution:

$$\text{LHS} = \sin 2\theta$$

$$\begin{aligned} &= \frac{2 \tan \theta}{1 + \tan^2 \theta} \\ &= \frac{2 \sec 2\alpha}{1 + \sec^2 2\alpha} \\ &= \frac{2 \cos 2\alpha}{\cos^2 2\alpha + 1} \\ &= \frac{2 \frac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha}}{\frac{(1 - \tan^2 \alpha)^2}{(1 + \tan^2 \alpha)^2} + 1} \\ &= \frac{2(1 - \tan^2 \alpha)(1 + \tan^2 \alpha)}{(1 - \tan^2 \alpha)^2 + (1 + \tan^2 \alpha)^2} \\ &= \frac{2(1 - \tan^4 \alpha)}{2(1 + \tan^4 \alpha)} = \frac{1 - \tan^4 \alpha}{1 + \tan^4 \alpha} = \text{RHS (Proved)} \end{aligned}$$

54. Find the minimum value of $(16 \cos^2 \theta + 9 \sec^2 \theta)$.

Solution:

$$\begin{aligned} &16 \cos^2 \theta + 9 \sec^2 \theta \\ &= (4 \cos \theta)^2 + (3 \sec \theta)^2 \\ &= (4 \cos \theta - 3 \sec \theta)^2 + 24 \cos \theta \cdot 3 \sec \theta \\ &= (4 \cos \theta - 3 \sec \theta)^2 + 24 \\ &\because \text{Minimum value of } (4 \cos \theta - 3 \sec \theta)^2 \text{ is zero.} \\ &\therefore \text{The minimum value of } (16 \cos^2 \theta + 9 \sec^2 \theta) \\ &\text{is 24. (Ans)} \end{aligned}$$

55. Solve: $\tan x + \tan 2x + \tan x \tan 2x = 1$

Solution:

$$\tan x + \tan 2x + \tan x \tan 2x = 1$$

$$\Rightarrow \tan x + \tan 2x = 1 - \tan x \tan 2x$$

$$\Rightarrow \frac{\tan x + \tan 2x}{1 - \tan x \tan 2x} = 1$$

$$\Rightarrow \tan(x + 2x) = \tan \frac{\pi}{4}$$

$$\Rightarrow \tan 3x = \tan \frac{\pi}{4}$$

Using general solution formula we can say

$$3x = n\pi + \frac{\pi}{4} \text{ [When } n = 0, \pm 1, \pm 2 \dots]$$

$$\Rightarrow x = \frac{1}{3} \left(n\pi + \frac{\pi}{4} \right) \text{ (Ans)}$$

56. Solve: $\sin x + \sin 5x = \sin 3x$.

Solution:

$$\sin x + \sin 5x = \sin 3x$$

$$\Rightarrow 2 \sin \left(\frac{x+5x}{2} \right) \cos \left(\frac{x-5x}{2} \right) = \sin 3x$$

$$\Rightarrow 2 \sin 3x \cos(-2x) - \sin 3x = 0$$

$$\Rightarrow \sin 3x (2 \cos 2x - 1) = 0$$

Either,

$$\sin 3x = 0$$

$$\therefore 3x = n\pi \text{ [When } n = 0, \pm 1, \pm 2 \dots]$$

$$\Rightarrow x = \frac{n\pi}{3} \text{ (Ans)}$$

Or,

$$2 \cos 2x - 1 = 0$$

$$\Rightarrow 2 \cos 2x = 1$$

$$\Rightarrow \cos 2x = \frac{1}{2}$$

$$\Rightarrow \cos 2x = \cos \frac{\pi}{3}$$

$$\therefore 2x = 2n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{6} \text{ (Ans)}$$

57. Solve: $\sqrt{3} \cos x + \sin x = 1$ [$0 < x < 2\pi$]

Solution:

$$\sqrt{3} \cos x + \sin x = 1$$

$$\Rightarrow \frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x = \frac{1}{2}$$

$$\Rightarrow \cos \frac{\pi}{6} \cos x + \cos \frac{\pi}{6} \sin x = \cos \frac{\pi}{3}$$

$$\Rightarrow \cos \left(x - \frac{\pi}{6} \right) = \cos \frac{\pi}{3}$$

Using general solution formula we can say

$$x - \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow x - \frac{\pi}{6} = 2n\pi + \frac{\pi}{3} \text{ [Taking (+ve) sign]}$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{3} + \frac{\pi}{6}$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{2}$$

$$\Rightarrow x = \frac{\pi}{2} \text{ [Putting } n = 0]$$

$$\text{Again, } x - \frac{\pi}{6} = 2n\pi - \frac{\pi}{3} \text{ [Taking (-ve) sign]}$$

$$\Rightarrow x = 2n\pi - \frac{\pi}{3} + \frac{\pi}{6}$$

$$\Rightarrow x = 2n\pi - \frac{\pi}{6}$$

$$\Rightarrow x = 2\pi - \frac{\pi}{6} \text{ [Putting } n = 1]$$

$$\Rightarrow x = \frac{11\pi}{6}$$

$$\therefore \text{The general solution is } x = \frac{\pi}{2}, \frac{11\pi}{6} \text{ (Ans)}$$

58. Solve: $3 \sin^2 x - 8 \sin x + 5 = 0$, $0 \leq x \leq 2\pi$

Solution:

$$\begin{aligned} 3 \sin^2 x - 8 \sin x + 5 &= 0 \\ \Rightarrow 3 \sin^2 x - 3 \sin x - 5 \sin x + 5 &= 0 \\ \Rightarrow 3 \sin x (\sin x - 1) - 5(\sin x - 1) &= 0 \\ \Rightarrow (\sin x - 1)(3 \sin x - 5) &= 0 \end{aligned}$$

Either,

$$\begin{aligned} \sin x - 1 &= 0 \\ \Rightarrow \sin x &= 1 \\ \Rightarrow x &= (4n + 1) \frac{\pi}{2} \text{ [Putting } n = 0] \\ \Rightarrow x &= \frac{\pi}{2} \text{ (Ans.)} \end{aligned}$$

Or,

$$\begin{aligned} 3 \sin x - 5 &= 0 \\ \Rightarrow 3 \sin x &= 5 \\ \Rightarrow \sin x &= \frac{5}{3} > 1 \text{ (Not acceptable)} \end{aligned}$$

59. Find the value of $\sin\left(\frac{1}{2} \cos^{-1} \frac{1}{2}\right)$.

Solution:

$$\begin{aligned} \sin\left(\frac{1}{2} \cos^{-1} \frac{1}{2}\right) \\ = \sin\left(\frac{1}{2} \times \frac{\pi}{3}\right) &= \sin \frac{\pi}{6} = \frac{1}{2} \text{ (Ans)} \end{aligned}$$

60. Find the value of $\sin\left[\cos^{-1}\left(-\frac{1}{2}\right)\right]$.

Solution:

$$\begin{aligned} \text{Let, } \theta &= \cos^{-1}\left(-\frac{1}{2}\right) \\ \Rightarrow \cos \theta &= -\frac{1}{2} \\ \Rightarrow \cos \theta &= \cos 120^\circ \\ \Rightarrow \theta &= 120^\circ \\ \therefore \sin\left[\cos^{-1}\left(-\frac{1}{2}\right)\right] \\ &= \sin \theta \\ &= \sin 120^\circ \\ &= \sin(2 \times 90^\circ - 60^\circ) \\ &= \sin 60^\circ = \frac{\sqrt{3}}{2} \text{ (Ans)} \end{aligned}$$

61. Find the value of $\sin \cot^{-1} x$.

Solution:

$$\begin{aligned} \text{Let, } \theta &= \cot^{-1} x \\ \Rightarrow \cot \theta &= x \\ \Rightarrow \cot^2 \theta &= x^2 \\ \Rightarrow \operatorname{cosec}^2 \theta - 1 &= x^2 \\ \Rightarrow \operatorname{cosec}^2 \theta &= 1 + x^2 \\ \Rightarrow \sin^2 \theta &= \frac{1}{1+x^2} \\ \Rightarrow \sin \theta &= \frac{1}{\sqrt{1+x^2}} \\ \Rightarrow \sin \cot^{-1} x &= \frac{1}{\sqrt{1+x^2}} \text{ (Ans.)} \end{aligned}$$

62. Find the value of $\sin\left(\tan^{-1} x + \tan^{-1} \frac{1}{x}\right)$.

Solution:

$$\begin{aligned} \sin\left(\tan^{-1} x + \tan^{-1} \frac{1}{x}\right) \\ = \sin(\tan^{-1} x + \cot^{-1} x) \\ = \sin \frac{\pi}{2} &= 1 \text{ (Ans.)} \end{aligned}$$

63. Find the value of $\left(\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{2}\right)$.

Solution:

$$\begin{aligned} \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{2} \\ = \tan^{-1} \left(\frac{\frac{1}{3} + \frac{1}{2}}{1 - \frac{1}{3} \times \frac{1}{2}} \right) \\ = \tan^{-1} \left(\frac{2+3}{6-1} \right) \\ = \tan^{-1} \left(\frac{5}{5} \right) = \tan^{-1}(1) = \frac{\pi}{4} \text{ (Ans)} \end{aligned}$$

64. Find the value of $\cos\left(\sin^{-1} \frac{1}{2} + \cos^{-1} \frac{1}{2}\right)$

Solution:

$$\cos\left(\sin^{-1} \frac{1}{2} + \cos^{-1} \frac{1}{2}\right) = \cos \frac{\pi}{2} = 0 \text{ [Ans.]}$$

65. If $\cos\left(\sin^{-1} \frac{1}{5} + \cos^{-1} x\right) = 0$, then the value of x is-

(a) 0

(b) 1

(c) $\frac{4}{5}$

(d) $\frac{1}{5}$

Solution:

$$\begin{aligned} \cos\left(\sin^{-1} \frac{1}{5} + \cos^{-1} x\right) &= 0 \\ \Rightarrow \cos\left(\sin^{-1} \frac{1}{5} + \cos^{-1} x\right) &= \cos \frac{\pi}{2} \\ \Rightarrow \sin^{-1} \frac{1}{5} + \cos^{-1} x &= \frac{\pi}{2} \\ \Rightarrow \sin^{-1} \frac{1}{5} &= \frac{\pi}{2} - \cos^{-1} x \\ \Rightarrow \sin^{-1} x &= \sin^{-1} \frac{1}{5} \\ \Rightarrow x &= \frac{1}{5} \end{aligned}$$

Answer: (d)

66. Solve: $\sin^{-1} \cos \sin^{-1} x = \frac{\pi}{3}$

Solution:

$$\begin{aligned} \sin^{-1} \cos \sin^{-1} x &= \frac{\pi}{3} \\ \Rightarrow \cos \sin^{-1} x &= \sin \frac{\pi}{3} \\ \Rightarrow \cos \sin^{-1} x &= \frac{\sqrt{3}}{2} \\ \Rightarrow \sin^{-1} x &= \cos^{-1} \frac{\sqrt{3}}{2} \\ \Rightarrow \cos^{-1} \sqrt{1-x^2} &= \cos^{-1} \frac{\sqrt{3}}{2} \\ \Rightarrow \sqrt{1-x^2} &= \frac{\sqrt{3}}{2} \\ \Rightarrow 1-x^2 &= \frac{3}{4} \\ \Rightarrow x^2 &= 1 - \frac{3}{4} = \frac{1}{4} \\ \Rightarrow x &= \pm \frac{1}{2} \text{ (Ans)} \end{aligned}$$

67. Prove that

$$\sec^2(\tan^{-1} \sqrt{5}) + \operatorname{cosec}^2(\cot^{-1} 5) = 32$$

Solution:

We know,

$$\tan^{-1} x = \sec^{-1} \sqrt{x^2 + 1}$$

$$\therefore \tan^{-1} \sqrt{5} = \sec^{-1} \sqrt{(\sqrt{5})^2 + 1} = \sec^{-1} \sqrt{6}$$

$$\text{And, } \cot^{-1} x = \operatorname{cosec}^{-1} \sqrt{x^2 + 1}$$

$$\therefore \cot^{-1} 5 = \operatorname{cosec}^{-1} \sqrt{5^2 + 1} = \operatorname{cosec}^{-1} \sqrt{26}$$

$$\therefore \sec^2(\tan^{-1} \sqrt{5}) + \operatorname{cosec}^2(\cot^{-1} 5)$$

$$\Rightarrow \{\sec(\sec^{-1} \sqrt{6})\}^2 + \{\operatorname{cosec}(\operatorname{cosec}^{-1} \sqrt{26})\}^2$$

$$\Rightarrow (\sqrt{6})^2 + (\sqrt{26})^2$$

$$\Rightarrow 6 + 26 = 32 \quad (\text{Proved})$$

68. If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}$, then prove

that, $xy + yz + zx = 1$

Solution:

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1} \frac{x+y}{1-xy} = \frac{\pi}{2} - \tan^{-1} z$$

$$\Rightarrow \tan^{-1} \frac{x+y}{1-xy} = \cot^{-1} z$$

$$\Rightarrow \tan^{-1} \frac{x+y}{1-xy} = \tan^{-1} \frac{1}{z}$$

$$\Rightarrow \frac{x+y}{1-xy} = \frac{1}{z}$$

$$\Rightarrow zx + yz = 1 - xy$$

$$\Rightarrow xy + zx + yz = 1 \quad (\text{Proved})$$

69. If $\tan^{-1} \left(\frac{yz}{xr} \right) + \tan^{-1} \left(\frac{zx}{yr} \right) + \tan^{-1} \left(\frac{xy}{zr} \right) = \frac{\pi}{2}$

Then prove that, $x^2 + y^2 + z^2 = r^2$

Solution:

$$\tan^{-1} \left(\frac{yz}{xr} \right) + \tan^{-1} \left(\frac{zx}{yr} \right) + \tan^{-1} \left(\frac{xy}{zr} \right) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1} \frac{\frac{yz}{xr} + \frac{zx}{yr}}{1 - \frac{yz}{xr} \cdot \frac{zx}{yr}} = \frac{\pi}{2} - \tan^{-1} \left(\frac{xy}{zr} \right)$$

$$\Rightarrow \tan^{-1} \frac{y^2 zr + zx^2 r}{xyr^2 - xy z^2} = \cot^{-1} \left(\frac{xy}{zr} \right)$$

$$\Rightarrow \tan^{-1} \frac{y^2 zr + zx^2 r}{xyr^2 - xy z^2} = \tan^{-1} \left(\frac{zr}{xy} \right)$$

$$\Rightarrow \frac{y^2 zr + zx^2 r}{xyr^2 - xy z^2} = \frac{zr}{xy}$$

$$\Rightarrow \frac{zr(x^2 + y^2)}{xy(r^2 - z^2)} = \frac{zr}{xy}$$

$$\Rightarrow \frac{(x^2 + y^2)}{(r^2 - z^2)} = 1$$

$$\Rightarrow x^2 + y^2 = r^2 - z^2$$

$$\Rightarrow x^2 + y^2 + z^2 = r^2 \quad (\text{Proved})$$

{Prepared by [NatiTute](#) YouTube Channel}

1. If $f(x) = e^{2x}$, then $f(a) \times f(b)$:

(a) e^{2a+2b} (b) e^{2ab} (c) e^{a+b} (d) None

Solution:

$$f(x) = e^{2x} \therefore f(a) = e^{2a}, f(b) = e^{2b}$$

$$\therefore f(a) \times f(b) = e^{2a} \cdot e^{2b} = e^{2a+2b}$$

Answer: (a)

2. If $f(x-1) = 7x-5$ then find $f(x)$

Solution:

$$f(x-1) = 7x-5$$

Putting, $x = x+1$

$$f(x+1-1) = 7(x+1)-5$$

$$\Rightarrow f(x) = 7x+7-5 = 7x+2 \text{ (Ans)}$$

3. If $f(x-2) = 2x^2 + 3x - 5$ then find $f(x)$

Solution:

$$f(x-2) = 2x^2 + 3x - 5$$

$$\Rightarrow f(x+2-2) = 2(x+2)^2 + 3(x+2) - 5$$

$$\Rightarrow f(x) = 2(x^2 + 4x + 4) + 3x + 6 - 5$$

$$= 2x^2 + 11x + 9 \text{ (Ans)}$$

4. If $f(x-2) = 2x^2 + 3x - 5$ then $f(-1) = ?$

(a) 0

(b) 1

(c) -1

(d) 2

Solution:

$$f(x-2) = 2x^2 + 3x - 5$$

Putting $x = 1$ we get,

$$f(1-2) = 2 \cdot 1^2 + 3 \cdot 1 - 5$$

$$\Rightarrow f(-1) = 2 + 3 - 5 = 0$$

Answer: (a)

5. If $y = f(x) = \frac{lx+m}{nx-l}$, then find $f(y)$

Solution:

$$y = f(x) = \frac{lx+m}{nx-l}$$

$$\Rightarrow y = \frac{lx+m}{nx-l}$$

$$\Rightarrow f(x) = \frac{lx+m}{nx-l}$$

$$\Rightarrow f(y) = \frac{ly+m}{ny-l} = \frac{l \times \frac{lx+m}{nx-l} + m}{n \times \frac{lx+m}{nx-l} - l} = \frac{\frac{l^2x+ml}{nx-l} + m}{\frac{lnx+mn}{nx-l} - l}$$

$$= \frac{\frac{l^2x+ml+mnx-ml}{nx-l}}{\frac{lnx+mn-l(nx-l)}{nx-l}}$$

$$= \frac{l^2x+ml+mnx-ml}{lnx+mn-lnx+l^2}$$

$$= \frac{l^2x+mnx}{mn+l^2}$$

$$= \frac{x(mn+l^2)}{(mn+l^2)} = x \text{ (Ans)}$$

6. If $f(x) = \frac{ax-b}{bx-a}$ then find $f(x) \times f\left(\frac{1}{x}\right)$.

Solution:

$$f(x) = \frac{ax-b}{bx-a}$$

$$\therefore f\left(\frac{1}{x}\right) = \frac{\frac{a}{x}-b}{\frac{b}{x}-a} = \frac{\frac{a-bx}{x}}{\frac{b-ax}{x}} = \frac{a-bx}{b-ax} = \frac{bx-a}{ax-b}$$

$$\therefore f(x) \times f\left(\frac{1}{x}\right) = \frac{ax-b}{bx-a} \times \frac{bx-a}{ax-b} = 1 \text{ (Ans)}$$

7. If $f(x) = \frac{4^x}{4^{x+2}}$, then find $f(x) + f(1-x)$

Solution:

$$f(x) = \frac{4^x}{4^{x+2}}$$

$$\Rightarrow f(1-x) = \frac{4^{1-x}}{4^{1-x+2}} = \frac{4}{4^{1-x+2}} = \frac{4}{4^{1-x+2}} = \frac{4}{4^{1-x+2}} = \frac{4}{2(2+4^x)}$$

$$\Rightarrow f(1-x) = \frac{2}{4^{x+2}}$$

$$\therefore f(x) + f(1-x) = \frac{4^x}{4^{x+2}} + \frac{2}{4^{x+2}} = \frac{4^x+2}{4^{x+2}} = 1 \text{ (Ans)}$$

8. If $f(x) = \frac{1-x}{1+x}$, then find $f\left(\frac{1}{x}\right) + f(x)$.

Solution:

$$f(x) = \frac{1-x}{1+x}$$

$$\Rightarrow f\left(\frac{1}{x}\right) = \frac{1-\frac{1}{x}}{1+\frac{1}{x}}$$

$$\Rightarrow f\left(\frac{1}{x}\right) = \frac{x-1}{x+1}$$

$$\Rightarrow f\left(\frac{1}{x}\right) = -\frac{1-x}{1+x}$$

$$\Rightarrow f\left(\frac{1}{x}\right) = -f(x)$$

$$\Rightarrow f\left(\frac{1}{x}\right) + f(x) = 0 \text{ (Ans)}$$

9. If $f(x) = 5^x$, then prove that

$$f(x+y) = f(x) \cdot f(y)$$

Solution:

$$f(x) = 5^x \Rightarrow f(y) = 5^y$$

$$\therefore f(x+y) = 5^x \cdot 5^y = f(x) \cdot f(y)$$

$$\Rightarrow f(x+y) = f(x) \cdot f(y) \text{ (Proved)}$$

10. If $\varphi(x) = \log(\sin x)$ and $\psi(x) = \log(\cos x)$ then find $e^{2\varphi(x)} + e^{2\psi(x)}$

Solution:

$$\varphi(x) = \log(\sin x)$$

$$\Rightarrow 2\varphi(x) = 2\log(\sin x)$$

$$\Rightarrow 2\varphi(x) = \log(\sin^2 x)$$

And

$$\psi(x) = \log(\cos x)$$

$$\Rightarrow 2\psi(x) = 2\log(\cos x)$$

$$\Rightarrow 2\psi(x) = \log(\cos^2 x)$$

$$\therefore e^{2\varphi(x)} + e^{2\psi(x)}$$

$$= e^{\log(\sin^2 x)} + e^{\log(\cos^2 x)}$$

$$= \sin^2 x + \cos^2 x = 1 \text{ (Ans)}$$

11. What is the period of $f(x) = \tan x$

Solution: $f(x) = \tan x$
 $\Rightarrow f(\pi + x) = \tan(\pi + x)$
 $\Rightarrow f(\pi + x) = \tan x$
 $\therefore f(x) = \tan x$
 $\therefore$ Period of the function is π (Ans.)

12. If $f(x) = \log\left(\frac{1+x}{1-x}\right)$ then $f\left(\frac{2x}{1+x^2}\right) = ?$

Solution:
 $f(x) = \log\left(\frac{1+x}{1-x}\right)$
 $\Rightarrow f\left(\frac{2x}{1+x^2}\right) = \log\left(\frac{1+\frac{2x}{1+x^2}}{1-\frac{2x}{1+x^2}}\right)$
 $= \log\left(\frac{1+x^2+2x}{1+x^2-2x}\right)$
 $= \log\left(\frac{1+x^2+2x}{1+x^2-2x}\right)^2$
 $= \log\left(\frac{1+x}{1-x}\right)^2$
 $= 2 \log\left(\frac{1+x}{1-x}\right) = 2f(x)$ (Ans.)

13. If $f(x) = \log_3 x$ and $\varphi(x) = x^2$ then find $f\{\varphi(3)\}$

Solution:
 $\varphi(x) = x^2 \Rightarrow \varphi(3) = 3^2$
 And,
 $f(x) = \log_3 x$
 $\Rightarrow f\{\varphi(3)\} = \log_3 \varphi(3) = \log_3 3^2$
 $= 2 \log_3 3 = 2$ (Ans)

14. If $\varphi(x) = a^x$ then find $[\varphi(p)]^3$

Solution:
 $\varphi(x) = a^x$
 $\Rightarrow \varphi(p) = a^p$
 $\Rightarrow [\varphi(p)]^3 = (a^p)^3$
 $\Rightarrow [\varphi(p)]^3 = a^{3p}$
 $\left[\begin{array}{l} \because \varphi(x) = a^x \\ \Rightarrow \varphi(3p) = a^{3p} \end{array} \right]$
 $\Rightarrow [\varphi(p)]^3 = \varphi(3p)$ (Ans)

15. Find the domain of the function $f(x) = \frac{1}{\sqrt{6-x}}$

Solution:
 For real value of $f(x)$
 $6 - x > 0$
 $\Rightarrow 6 > x$
 $\Rightarrow x < 6$
 $\therefore$ The domain is $-\infty < x < 6$ (Ans)

16. Find the domain of function $f(x) = \sqrt{1-x^2}$

Solution:
 For real value of $f(x)$
 $1 - x^2 \geq 0$
 $\Rightarrow (1+x)(1-x) \geq 0$
 $\Rightarrow -(1+x)(x-1) \geq 0$
 $\Rightarrow (1+x)(x-1) \leq 0$
 $\Rightarrow -1 \leq x \leq 1$
 $\therefore$ The domain of function is
 $-1 \leq x \leq 1$ (Ans.)

17. Find the domain of function

$$f(x) = \sqrt{x^2 - 7x + 10}$$

Solution:

For real value of $f(x)$ $x^2 - 7x + 10 \geq 0$
 $\Rightarrow x^2 - 5x - 2x + 10 \geq 0$
 $\Rightarrow x(x-5) - 2(x-5) \geq 0$
 $\Rightarrow (x-5)(x-2) \geq 0$
 $\therefore$ The domain of function is
 $x \leq 2$ and $x \geq 5$ (Ans)

18. Find the domain of function

$$f(x) = \frac{1}{\sqrt{x^2 - 5x + 6}}$$

Solution:

For real value of $f(x)$
 $\Rightarrow x^2 - 5x + 6 > 0$
 $\Rightarrow x^2 - 3x - 2x + 6 > 0$
 $\Rightarrow x(x-3) - 2(x-3) > 0$
 $\Rightarrow (x-3)(x-2) > 0$
 $\therefore$ The domain of function is $2 < x < 3$ (Ans)

19. The domain of the function $\frac{1}{\sqrt{(x-2)(3-x)}}$

- (a) $2 \leq x \leq 3$
 (b) $2 < x \leq 3$
 (c) $2 \leq x < 3$
 (d) $2 < x < 3$

Solution:

For real value of given function $\frac{1}{\sqrt{(x-2)(3-x)}}$
 $(x-2)(3-x) > 0$
 $\Rightarrow -(x-2)(x-3) > 0$
 $\Rightarrow (x-2)(x-3) < 0$
 $\therefore$ Domain of the function $2 < x < 3$
 Answer: (d)

20. Find range of function $y = \frac{1}{3 - \cos 2x}$

Solution:

$y = \frac{1}{3 - \cos 2x}$
 $\Rightarrow 1 = 3y - y \cos 2x$
 $\Rightarrow y \cos 2x = 3y - 1$
 $\Rightarrow \cos 2x = 3 - \frac{1}{y}$
 Now, for real values of x , we must have
 $-1 \leq \cos 2x \leq 1$
 $\Rightarrow -1 \leq 3 - \frac{1}{y} \leq 1$
 $\Rightarrow -4 \leq -\frac{1}{y} \leq -2$
 $\Rightarrow 4 \geq \frac{1}{y} \geq 2$
 $\Rightarrow \frac{1}{4} \leq y \leq \frac{1}{2}$
 $\therefore$ The required range of y is $\frac{1}{4} \leq y \leq \frac{1}{2}$ (Ans.)

21. Find the domain of function

$$f(x) = \sqrt{x-1} + \sqrt{2-x}$$

Solution:

For real value of $f(x)$

$$x \geq 1 \text{ and } x \leq 2$$

∴ The domain of function is

$$1 \leq x \leq 2 \text{ (Ans)}$$

22. Find the range of function $y = \frac{x}{1+x^2}$

Solution:

$$y = \frac{x}{1+x^2}$$

$$\Rightarrow yx^2 + y = x$$

$$\Rightarrow yx^2 - x + y = 0$$

$$\therefore x = \frac{1 \pm \sqrt{1-4y^2}}{2y}$$

∴ x is real and definite

$$\therefore 1 - 4y^2 \geq 0$$

$$\Rightarrow 1 - (2y)^2 \geq 0$$

$$\Rightarrow (1+2y)(1-2y) \geq 0$$

$$\Rightarrow -\frac{1}{2} \leq y \leq \frac{1}{2}$$

∴ rang of $y = \frac{x}{1+x^2}$ is

$$\left[y: y \in \mathbb{R} \text{ and } -\frac{1}{2} \leq y \leq \frac{1}{2} \right] \text{ (Ans)}$$

23. The function $f(x) = \sqrt{1+x+x^2} - \sqrt{1-x+x^2}$ is even function or odd function?

Solution:

$$f(x) = \sqrt{1+x+x^2} - \sqrt{1-x+x^2}$$

$$\Rightarrow f(-x) = \sqrt{1-x+(-x)^2} - \sqrt{1+x+(-x)^2}$$

$$\Rightarrow f(-x) = \sqrt{1-x+x^2} - \sqrt{1+x+x^2}$$

$$\Rightarrow f(-x) = -\{\sqrt{1+x+x^2} - \sqrt{1-x+x^2}\}$$

$$\Rightarrow f(-x) = -f(x)$$

∴ $f(x)$ is an odd function (Ans)

24. $f(x) = \frac{\cos x + \sec x}{x}$ determine the function even or odd.

Solution:

$$f(x) = \frac{\cos x + \sec x}{x}$$

$$\Rightarrow f(-x) = \frac{\cos(-x) + \sec(-x)}{-x}$$

$$\Rightarrow f(-x) = -\frac{\cos x + \sec x}{x}$$

$$\Rightarrow f(-x) = -f(x)$$

∴ $f(x)$ is an odd function (Ans)

25. Find the period of function $f(x) = \cos\left(\frac{x}{3}\right)$.

Solution:

$$f(x) = \cos\left(\frac{x}{3}\right)$$

$$\Rightarrow f(6\pi + x) = \cos\left(\frac{6\pi + x}{3}\right)$$

$$\Rightarrow f(6\pi + x) = \cos\left(2\pi + \frac{x}{3}\right)$$

$$\Rightarrow f(6\pi + x) = \cos\left(\frac{x}{3}\right)$$

∴ Period of $f(x)$ is 6π (Ans)

26. $f(x) = x^3 \cdot \frac{3^x+1}{3^x-1}$ determine the function even or odd.

Solution:

$$f(-x) = (-x)^3 \cdot \frac{3^{-x}+1}{3^{-x}-1}$$

$$\Rightarrow f(-x) = -x^3 \cdot \frac{\frac{1}{3^x}+1}{\frac{1}{3^x}-1}$$

$$\Rightarrow f(-x) = -x^3 \cdot \frac{\frac{1+3^x}{3^x}}{\frac{1-3^x}{3^x}}$$

$$\Rightarrow f(-x) = -x^3 \cdot \frac{1+3^x}{1-3^x}$$

$$\Rightarrow f(-x) = x^3 \cdot \frac{3^x+1}{3^x-1}$$

$$\Rightarrow f(-x) = f(x)$$

∴ $f(x)$ is an even function (Ans)

27. Prove that the function $\log(x + \sqrt{x^2 + 1})$ is an odd function.

Solution:

$$\text{Let, } f(x) = \log(x + \sqrt{x^2 + 1})$$

$$\therefore f(-x) = \log(-x + \sqrt{(-x)^2 + 1})$$

$$\Rightarrow f(-x) = \log(\sqrt{x^2 + 1} - x)$$

$$\therefore f(x) + f(-x)$$

$$= \log(x + \sqrt{x^2 + 1}) + \log(\sqrt{x^2 + 1} - x)$$

$$= \log(x + \sqrt{x^2 + 1})(\sqrt{x^2 + 1} - x)$$

$$= \log(x^2 + 1 - x^2) = \log 1 = 0$$

$$\therefore f(x) + f(-x) = 0$$

$$\Rightarrow f(-x) = -f(x)$$

So, $f(x)$ is an odd function. (Proved)

28. If $f(x) = |x - 5|$, find the value of $f(3)$.

Solution:

$$f(x) = |x - 5|$$

$$\Rightarrow f(3) = |3 - 5|$$

$$= |-2|$$

$$= 2 \text{ (Ans)}$$

29. If $f(x) = x - |x|$ then find $f(-4)$

Solution:

$$f(x) = x - |x|$$

$$\Rightarrow f(-4) = (-4) - 4$$

$$\Rightarrow f(-4) = -4 - 4 \left[\begin{array}{l} \because |x| = -x, \text{ When } x < 0 \\ |x| = x, \text{ When } x \geq 0 \end{array} \right]$$

$$\Rightarrow f(-4) = -8 \text{ (Ans)}$$

30. Determine period of function $f(x) = \sin 3x$

Solution:

$$f(x) = \sin 3x$$

$$\Rightarrow f(2\pi + x) = \sin 3(2\pi + x)$$

$$\Rightarrow f(2\pi + x) = \sin(6\pi + 3x)$$

$$\Rightarrow f(2\pi + x) = \sin 3x$$

∴ Period of $f(x)$ is 2π (Ans)

1. The value of $\lim_{x \rightarrow 0} \frac{\tan x}{x}$:

(a) 3 (b) 2 (c) 1 (d) None

Solution:

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \frac{1}{\cos x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \lim_{x \rightarrow 0} \frac{1}{\cos x} = 1 \times \frac{1}{\cos 0} = 1$$

Answer: (c)

2. Evaluate: $\lim_{x \rightarrow 0} \frac{\log(1+\sin x)}{x}$

Solution:

$$\lim_{x \rightarrow 0} \frac{\log(1+\sin x)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\log(1+\sin x)}{\sin x} \cdot \frac{\sin x}{x} \quad [\because x \rightarrow 0, \Rightarrow \sin x \rightarrow 0]$$

$$= \lim_{\sin x \rightarrow 0} \frac{\log(1+\sin x)}{\sin x} \cdot \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$= 1.1 = 1 \text{ (Ans)}$$

3. Evaluate: $\lim_{x \rightarrow 0} \frac{\log(1+2x)}{\sin 3x}$

Solution:

$$\lim_{x \rightarrow 0} \frac{\log(1+2x)}{\sin 3x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\log(1+2x)}{2x} \cdot 2}{\frac{\sin 3x}{3x} \cdot 3} = \frac{\lim_{x \rightarrow 0} \frac{\log(1+2x)}{2x} \cdot 2}{\lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot 3} = \frac{1.2}{1.3} = \frac{2}{3} \text{ (Ans)}$$

4. Find the value of $\lim_{x \rightarrow 0} \frac{e^{ax}-1}{\log(1+ax)}$

Solution:

$$\lim_{x \rightarrow 0} \frac{e^{ax}-1}{\log(1+ax)} = \lim_{x \rightarrow 0} \frac{\frac{e^{ax}-1}{ax}}{\frac{\log(1+ax)}{ax}}$$

$$= \frac{\lim_{ax \rightarrow 0} \frac{e^{ax}-1}{ax}}{\lim_{ax \rightarrow 0} \frac{\log(1+ax)}{ax}} \quad [\because x \rightarrow 0, \Rightarrow ax \rightarrow 0]$$

$$= \frac{1}{1} = 1 \text{ (Ans)}$$

5. If $\lim_{x \rightarrow 0} \frac{\log(1+mx)}{3x} = 1$ then find the value of m .

Solution:

$$\lim_{x \rightarrow 0} \frac{\log(1+mx)}{3x} = 1$$

$$\Rightarrow \lim_{mx \rightarrow 0} \frac{\log(1+mx)}{mx} \times \frac{m}{3} = 1 \quad [\because x \rightarrow 0, \Rightarrow mx \rightarrow 0]$$

$$\Rightarrow 1 \times \frac{m}{3} = 1, \Rightarrow m = 3 \text{ (Ans)}$$

6. Evaluate: $\lim_{x \rightarrow 1} \frac{\sqrt{x}-1}{x-1}$

Solution:

$$\lim_{x \rightarrow 1} \frac{(\sqrt{x}-1)}{x-1} = \lim_{x \rightarrow 1} \frac{(\sqrt{x}-1)}{(\sqrt{x})^2-1^2}$$

$$= \lim_{x \rightarrow 1} \frac{(\sqrt{x}-1)}{(\sqrt{x}+1)(\sqrt{x}-1)}$$

$$= \lim_{x \rightarrow 1} \frac{1}{(\sqrt{x}+1)} = \frac{1}{1+1} = \frac{1}{2} \text{ (Ans)}$$

7. Evaluate: $\lim_{x \rightarrow 4} \frac{x^2-16}{x-4}$

Solution:

$$\lim_{x \rightarrow 4} \frac{x^2-16}{x-4} = \lim_{x \rightarrow 4} \frac{x^2-4^2}{x-4}$$

$$= \lim_{x \rightarrow 4} \frac{(x+4)(x-4)}{(x-4)}$$

$$= \lim_{x \rightarrow 4} (x+4) = 4+4 = 8 \text{ (Ans)}$$

8. Evaluate: $\lim_{x \rightarrow \pi} \frac{\sin x}{x-\pi}$

Solution:

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x-\pi}$$

$$= -\lim_{x \rightarrow \pi} \frac{-\sin x}{x-\pi} = -\lim_{x \rightarrow \pi} \frac{\sin(x-\pi)}{x-\pi} = -1 \text{ (Ans)}$$

9. Find the value of $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cos x}{\frac{\pi}{2}-x} \right)$

Solution:

$$\text{Let, } z = \frac{\pi}{2} - x$$

$$\Rightarrow x = \frac{\pi}{2} - z \text{ and } x \rightarrow \frac{\pi}{2}, \Rightarrow z \rightarrow 0$$

$$\text{Now, } \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cos x}{\frac{\pi}{2}-x} \right)$$

$$= \lim_{z \rightarrow 0} \left\{ \frac{\cos\left(\frac{\pi}{2}-z\right)}{z} \right\} = \lim_{z \rightarrow 0} \left\{ \frac{\sin z}{z} \right\} = 1 \text{ (Ans)}$$

10. Find the value of $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cot x}{\frac{\pi}{2}-x} \right)$

Solution:

$$\text{Let, } z = \frac{\pi}{2} - x$$

$$\Rightarrow x = \frac{\pi}{2} - z \text{ and } x \rightarrow \frac{\pi}{2}, \Rightarrow z \rightarrow 0$$

$$\text{Now, } \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cot x}{\frac{\pi}{2}-x} \right)$$

$$= \lim_{z \rightarrow 0} \left\{ \frac{\cot\left(\frac{\pi}{2}-z\right)}{z} \right\}$$

$$= \lim_{z \rightarrow 0} \left\{ \frac{\tan z}{z} \right\}$$

$$= \lim_{z \rightarrow 0} \left\{ \frac{\sin z}{z} \right\} \times \lim_{z \rightarrow 0} \frac{1}{\cos z}$$

$$= 1 \times \frac{1}{\cos 0} = 1 \text{ (Ans)}$$

11. Evaluate: $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x}$

Solution:

$$\text{Let, } \sin^{-1} x = z \Rightarrow x = \sin z$$

$$\Rightarrow x \rightarrow 0 \Rightarrow z \rightarrow 0$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x}$$

$$= \lim_{z \rightarrow 0} \frac{z}{\sin z} = \frac{1}{\lim_{z \rightarrow 0} \frac{\sin z}{z}} = \frac{1}{1} = 1 \text{ (Ans)}$$

12. Evaluate: $\lim_{x \rightarrow 0} \frac{e^{\alpha x} - e^{\beta x}}{x}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{\alpha x} - e^{\beta x}}{x} &= \lim_{x \rightarrow 0} \frac{e^{\alpha x} - 1 - e^{\beta x} + 1}{x} \\ &= \lim_{x \rightarrow 0} \frac{(e^{\alpha x} - 1) - (e^{\beta x} - 1)}{x} \\ &= \lim_{x \rightarrow 0} \frac{e^{\alpha x} - 1}{x} - \lim_{x \rightarrow 0} \frac{e^{\beta x} - 1}{x} \\ &= \lim_{\alpha x \rightarrow 0} \frac{e^{\alpha x} - 1}{\alpha x} \cdot \alpha - \lim_{\beta x \rightarrow 0} \frac{e^{\beta x} - 1}{\beta x} \cdot \beta \\ &= 1 \cdot \alpha - 1 \cdot \beta = \alpha - \beta \text{ (Ans)} \end{aligned}$$

13. Evaluate: $\lim_{x \rightarrow 0} \frac{e^{ax} - 1}{\sin bx}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{ax} - 1}{\sin bx} \quad [\because x \rightarrow 0, \Rightarrow ax \rightarrow 0, \Rightarrow bx \rightarrow 0] \\ = \frac{\lim_{ax \rightarrow 0} \frac{e^{ax} - 1}{ax} \cdot ax}{\lim_{bx \rightarrow 0} \frac{\sin bx}{bx} \cdot bx} = \frac{1 \cdot a}{1 \cdot b} = \frac{a}{b} \text{ (Ans)} \end{aligned}$$

14. Evaluate: $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\sin 3x}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\sin 3x} \quad [\because x \rightarrow 0, \Rightarrow 2x \rightarrow 0, \Rightarrow 3x \rightarrow 0] \\ = \frac{\lim_{2x \rightarrow 0} \frac{e^{2x} - 1}{2x} \cdot 2x}{\lim_{3x \rightarrow 0} \frac{\sin 3x}{3x} \cdot 3x} = \frac{1 \cdot 2}{1 \cdot 3} = \frac{2}{3} \text{ (Ans)} \end{aligned}$$

15. Evaluate: $\lim_{x \rightarrow 0} \frac{\sin x(1 - \cos x)}{x^3}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x(1 - \cos x)}{x^3} &= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \cdot \frac{1 - \cos x}{x^2} \right) \\ &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x^2} \quad [\because x \rightarrow 0 \Rightarrow \frac{x}{2} \rightarrow 0] \\ &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \lim_{\frac{x}{2} \rightarrow 0} \frac{\sin^2 \frac{x}{2}}{\frac{x^2}{4} \cdot 2} \\ &= 1 \times \left(\lim_{\frac{x}{2} \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \right)^2 \cdot \frac{1}{2} = 1 \times 1^2 \times \frac{1}{2} = \frac{1}{2} \text{ (Ans)} \end{aligned}$$

16. Evaluate: $\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\log(1 + 3x)}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\log(1 + 3x)} &= \lim_{x \rightarrow 0} \frac{\frac{e^{\sin x} - 1}{\sin x} \times \sin x}{\frac{\log(1 + 3x)}{3x} \times 3x} \\ &= \frac{\lim_{\sin x \rightarrow 0} \frac{e^{\sin x} - 1}{\sin x} \times \lim_{x \rightarrow 0} \sin x}{\lim_{3x \rightarrow 0} \frac{\log(1 + 3x)}{3x} \times 3} \quad \left[\begin{array}{l} \because x \rightarrow 0, \\ \Rightarrow \sin x \rightarrow 0 \\ \text{and, } x \rightarrow 0 \\ \Rightarrow 3x \rightarrow 0 \end{array} \right] \\ &= \frac{1 \times 1}{1 \times 3} = \frac{1}{3} \text{ (Ans)} \end{aligned}$$

17. Evaluate: $\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos 2x}{x^2}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{x^2} - \cos 2x}{x^2} &= \lim_{x \rightarrow 0} \frac{e^{x^2} - 1 + 1 - \cos 2x}{x^2} \\ &= \lim_{x \rightarrow 0} \left(\frac{e^{x^2} - 1}{x^2} + \frac{1 - \cos 2x}{x^2} \right) \\ &= \lim_{x^2 \rightarrow 0} \frac{e^{x^2} - 1}{x^2} + \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} \\ &= 1 + 2 \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right)^2 \\ &= 1 + 2 \cdot 1^2 = 3 \text{ (Ans)} \end{aligned}$$

18. Evaluate: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} &= \lim_{x \rightarrow 0} \frac{2 \sin^2 \left(\frac{x}{2} \right)}{x^2} \\ &= 2 \lim_{\frac{x}{2} \rightarrow 0} \left\{ \frac{\sin \left(\frac{x}{2} \right)}{\frac{x}{2}} \right\}^2 \cdot \frac{1}{4} = 2 \cdot 1^2 \cdot \frac{1}{4} = \frac{1}{2} \text{ (Ans)} \end{aligned}$$

19. Evaluate: $\lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1} \\ &= \lim_{x \rightarrow 0} \frac{(2^x - 1)(\sqrt{1+x} + 1)}{(\sqrt{1+x} - 1)(\sqrt{1+x} + 1)} \\ &= \lim_{x \rightarrow 0} \frac{(2^x - 1)(\sqrt{1+x} + 1)}{(\sqrt{1+x})^2 - 1^2} \\ &= \lim_{x \rightarrow 0} \frac{(2^x - 1)(\sqrt{1+x} + 1)}{1 + x - 1} \\ &= \lim_{x \rightarrow 0} \frac{(2^x - 1)}{x} \cdot \lim_{x \rightarrow 0} (\sqrt{1+x} + 1) \\ &= \log_e 2 \cdot (\sqrt{1+0} + 1) = 2 \log_e 2 \text{ (Ans)} \end{aligned}$$

20. Evaluate: $\lim_{x \rightarrow 0} \frac{3^x - 1}{\sqrt{9+x} - 3}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{3^x - 1}{\sqrt{9+x} - 3} &= \lim_{x \rightarrow 0} \frac{(3^x - 1)(\sqrt{9+x} + 3)}{(\sqrt{9+x} - 3)(\sqrt{9+x} + 3)} \\ &= \lim_{x \rightarrow 0} \frac{(3^x - 1)(\sqrt{9+x} + 3)}{(\sqrt{9+x})^2 - 3^2} \\ &= \lim_{x \rightarrow 0} \frac{(3^x - 1)(\sqrt{9+x} + 3)}{9 + x - 9} \\ &= \lim_{x \rightarrow 0} \frac{(3^x - 1)}{x} \cdot \lim_{x \rightarrow 0} (\sqrt{9+x} + 3) \\ &= \log_e 3 \cdot (\sqrt{9+0} + 3) = 6 \log_e 3 \text{ (Ans)} \end{aligned}$$

21. Evaluate: $\lim_{x \rightarrow 0} \frac{\tan px}{\tan qx}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan px}{\tan qx} &= \lim_{x \rightarrow 0} \frac{\frac{\tan px}{px} \cdot px}{\frac{\tan qx}{qx} \cdot qx} \\ &= \frac{\lim_{px \rightarrow 0} \frac{\tan px}{px} \cdot p}{\lim_{qx \rightarrow 0} \frac{\tan qx}{qx} \cdot q} = \frac{1 \cdot p}{1 \cdot q} = \frac{p}{q} \text{ (Ans)} \end{aligned}$$

22. Evaluate: $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 2x}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 2x} &= \lim_{x \rightarrow 0} \frac{\frac{\sin 3x}{3x} \cdot 3x}{\frac{\sin 2x}{2x} \cdot 2x} \\ &= \frac{\lim_{x \rightarrow 0} \frac{\sin 3x}{3x}}{\lim_{x \rightarrow 0} \frac{\sin 2x}{2x}} \times \frac{3}{2} \left[\begin{array}{l} \because x \rightarrow 0, \Rightarrow 3x \rightarrow 0 \\ \Rightarrow 2x \rightarrow 0 \end{array} \right] \\ &= \frac{1}{1} \times \frac{3}{2} = \frac{3}{2} \text{ (Ans)} \end{aligned}$$

23. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right)$

Solution:

$$\begin{aligned} &\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{\cos x}{\sin x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{\sin x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{2 \sin^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} \right) = \lim_{x \rightarrow 0} \tan \frac{x}{2} = 0 \text{ (Ans)} \end{aligned}$$

24. Evaluate : $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1 - \sin^3 x}{\cos^2 x} \right)$.

Solution:

$$\begin{aligned} &\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1 - \sin^3 x}{\cos^2 x} \right) \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \left\{ \frac{(1 - \sin x)(1 + \sin x + \sin^2 x)}{1 - \sin^2 x} \right\} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \left\{ \frac{(1 - \sin x)(1 + \sin x + \sin^2 x)}{(1 + \sin x)(1 - \sin x)} \right\} \\ &= \frac{1 + \sin \frac{\pi}{2} + \sin^2 \frac{\pi}{2}}{1 + \sin \frac{\pi}{2}} = \frac{1 + 1 + 1^2}{1 + 1} = \frac{3}{2} \text{ (Ans)} \end{aligned}$$

25. Evaluate: $\lim_{x \rightarrow a} \frac{x \sin a - a \sin x}{x - a}$.

Solution:

$$\begin{aligned} &\lim_{x \rightarrow a} \frac{x \sin a - a \sin x}{x - a} \quad \left[\frac{0}{0} \text{ form} \right] \\ &= \lim_{x \rightarrow a} \frac{\sin a - a \cos x}{1} \quad [\text{by L' Hospital rule}] \\ &= \sin a - a \cos a \text{ (Ans)} \end{aligned}$$

26. Evaluate: $\lim_{x \rightarrow 0} \frac{2^{2x} - 3^{3x}}{x}$

Solution:

$$\begin{aligned} &\lim_{x \rightarrow 0} \frac{2^{2x} - 3^{3x}}{x} \\ &= \lim_{x \rightarrow 0} \frac{(2^2)^x - (3^3)^x}{x} \\ &= \lim_{x \rightarrow 0} \frac{4^x - 27^x}{x} \\ &= \lim_{x \rightarrow 0} \frac{(4^x - 1) - (27^x - 1)}{x} \\ &= \lim_{x \rightarrow 0} \left(\frac{4^x - 1}{x} - \frac{27^x - 1}{x} \right) \\ &= \lim_{x \rightarrow 0} \frac{4^x - 1}{x} - \lim_{x \rightarrow 0} \frac{27^x - 1}{x} \\ &= \log_e 4 - \log_e 27 = \log_e \left(\frac{4}{27} \right) \text{ (Ans)} \end{aligned}$$

27. Determine continuity of $f(x)$ at $x = 1$

$$\begin{aligned} \text{Where, } f(x) &= -x \quad [\text{for } x \leq 0] \\ &= x \quad [\text{for } 0 < x < 1] \\ &= 2 - x \quad [\text{for } x \geq 1] \end{aligned}$$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 1+0} f(x) &= \lim_{x \rightarrow 1+0} (2 - x) = 2 - 1 = 1 \\ \lim_{x \rightarrow 1-0} f(x) &= \lim_{x \rightarrow 1-0} (x) = 1 \\ f(x) &= 2 - x, \\ \Rightarrow f(1) &= 2 - 1 = 1 \\ \therefore \lim_{x \rightarrow 1+0} f(x) &= \lim_{x \rightarrow 1-0} f(x) = f(1) \\ \therefore \text{at } x = 1 \text{ } f(x) &\text{ is continuous. (Ans)} \end{aligned}$$

28. Discuss the continuity of the following function at $x = 2$.

$$\begin{aligned} f(x) &= x^2 + 2 \quad [0 \leq x \leq 2] \\ &= x + 4 \quad [2 \leq x \leq 3] \\ &= 6 \quad [x = 2] \end{aligned}$$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 2+} f(x) &= \lim_{x \rightarrow 2+} (x + 4) = 2 + 4 = 6 \\ \lim_{x \rightarrow 2-} f(x) &= \lim_{x \rightarrow 2-} (x^2 + 2) = 2^2 + 2 = 6 \\ f(x) &= 6 \quad [\text{When } x = 2] \Rightarrow f(2) = 6 \\ \therefore \lim_{x \rightarrow 2+} f(x) &= \lim_{x \rightarrow 2-} f(x) = f(2) \\ \therefore f(x) &\text{ is continuous at } x = 2. \text{ (Ans)} \end{aligned}$$

29. Determine continuity of $f(x) = |x|$ at $x = 0$

Solution:

$$\begin{aligned} &\text{We know,} \\ |x| &= x, \text{ for } x \geq 0 \\ |x| &= -x, \text{ for } x < 0 \\ \therefore \lim_{x \rightarrow 0+} f(x) &= \lim_{x \rightarrow 0+} |x| = \lim_{x \rightarrow 0+} x = 0 \\ \lim_{x \rightarrow 0-} f(x) &= \lim_{x \rightarrow 0-} |x| = \lim_{x \rightarrow 0-} x = 0 \\ \text{and } f(0) &= 0 \\ \therefore \lim_{x \rightarrow 0+} f(x) &= \lim_{x \rightarrow 0-} f(x) = f(0) \\ \therefore f(x) &\text{ is continuous at } x = 0 \text{ (Ans)} \end{aligned}$$

30. Find whether the function $f(x) = \frac{|x|}{x}$ is continuous at $x = 0$.

Solution:

$$\begin{aligned} f(x) &= \frac{|x|}{x} = \frac{x}{x} = 1 \quad [\text{When } x \geq 0] \\ f(x) &= \frac{|x|}{x} = \frac{-x}{x} = -1 \quad [\text{When } x < 0] \\ \text{Now,} \\ \lim_{x \rightarrow 0+} f(x) &= 1 \text{ and } \lim_{x \rightarrow 0-} f(x) = -1 \\ \therefore \lim_{x \rightarrow 0+} f(x) &\neq \lim_{x \rightarrow 0-} f(x) \\ \therefore f(x) &\text{ is not continuous at } x = 0. \text{ (Ans)} \end{aligned}$$

$$\begin{aligned} 31. f(x) &= 2x + 1 \quad [\text{for } x < 2] \\ &= K \quad [\text{for } x = 2] \\ &= 3x - 1 \quad [\text{for } x > 2] \end{aligned}$$

Find the value of k for which $f(x)$ is continuous at $x = 2$

Solution:

According to question

$$\begin{aligned} \lim_{x \rightarrow 2+} f(x) &= f(x) \\ \Rightarrow \lim_{x \rightarrow 2+} (3x - 1) &= f(2) \\ \Rightarrow (3 \cdot 2 - 1) &= k \\ \Rightarrow k &= 5 \\ \lim_{x \rightarrow 2-} f(x) &= f(x) \\ \Rightarrow \lim_{x \rightarrow 2-} (2x + 1) &= f(2) \\ \Rightarrow (2 \cdot 2 + 1) &= k \\ \Rightarrow k &= 5 \quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned} 32. f(x) &= 1 + 2x \quad \left[\text{When, } -\frac{1}{2} \leq x < 0 \right] \\ &= 1 - 2x \quad \left[\text{When, } 0 \leq x < \frac{1}{2} \right] \\ &= 4 \quad \left[\text{When, } x \geq \frac{1}{2} \right] \end{aligned}$$

show that $f(x)$ is continuous at $x = 0$ and discontinuous at $x = \frac{1}{2}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0+} f(x) &= \lim_{x \rightarrow 0+} (1 - 2x) = 1 - 2 \cdot 0 = 1 \\ \lim_{x \rightarrow 0-} f(x) &= \lim_{x \rightarrow 0-} (1 + 2x) = 1 + 2 \cdot 0 = 1 \\ f(0) &= 1 - 2 \cdot 0 = 1 \\ \therefore \lim_{x \rightarrow 0+} f(x) &= \lim_{x \rightarrow 0-} f(x) = f(0) \\ \therefore f(x) &\text{ is continuous at } x = 0 \quad (\text{Proved}) \\ \text{Again,} \\ \lim_{x \rightarrow \frac{1}{2}+} f(x) &= \lim_{x \rightarrow \frac{1}{2}+} 4 = 4 \\ \lim_{x \rightarrow \frac{1}{2}-} f(x) &= \lim_{x \rightarrow \frac{1}{2}-} (1 - 2x) = 1 - 2 \times \frac{1}{2} = 0 \\ f\left(\frac{1}{2}\right) &= 4 \\ \therefore \lim_{x \rightarrow \frac{1}{2}+} f(x) &\neq \lim_{x \rightarrow \frac{1}{2}-} f(x) \neq f\left(\frac{1}{2}\right) \\ \therefore f(x) &\text{ is discontinuous at } x = \frac{1}{2} \quad (\text{Proved}) \end{aligned}$$

{Prepared by NatiTute YouTube Channel}

33. Find the value of k for which

$$\begin{aligned} f(x) &= \frac{\sin 5x}{3x} \quad [\text{When } x \neq 0] \\ &= k \quad [\text{When } x = 0] \end{aligned}$$

is continuous at $x = 0$.

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{\sin 5x}{3x} \\ &= \lim_{5x \rightarrow 0} \frac{\sin 5x}{5x} \times \frac{5}{3} = 1 \times \frac{5}{3} = \frac{5}{3} \\ \text{Given, } f(x) &= k \quad [\text{When } x = 0] \\ \Rightarrow f(0) &= k \\ \therefore f(x) &\text{ is continuous at } x = 0 \\ \therefore \lim_{x \rightarrow 0} f(x) &= f(0) \\ \Rightarrow \frac{5}{3} &= k, \Rightarrow k = \frac{5}{3} \quad (\text{Ans}) \end{aligned}$$

34. Let $f(x) = \frac{x^2 - 4}{x - 2}$ [When $x \neq 0$]. What should be the value of $f(x)$ at $x = 2$ such that $f(x)$ be continuous at $x = 2$.

Solution:

$$\begin{aligned} \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{x^2 - 2^2}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{(x + 2)(x - 2)}{x - 2} = 2 + 2 = 4 \\ f(x) &\text{ is said to be continuous at } x = 2 \text{ when} \\ \lim_{x \rightarrow 2} f(x) &= f(2) \\ \Rightarrow f(2) &= 4 \quad [\because \lim_{x \rightarrow 2} f(x) = 4] \\ \Rightarrow f(x) &= 4 \quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned} 35. f(x) &= x^2 \quad [\text{for } -1 \leq x \leq 2] \\ &= \frac{3(x^4 - 16)}{2(x^3 - 8)} \quad [\text{for } 2 < x \leq 5] \end{aligned}$$

$\lim_{x \rightarrow 2} f(x)$ exist or not?

Solution:

$$\begin{aligned} \lim_{x \rightarrow 2+} f(x) &= \lim_{x \rightarrow 2+} \frac{3(x^4 - 16)}{2(x^3 - 8)} \\ &= \lim_{x \rightarrow 2+} \frac{3(x^2 + 4)(x + 2)(x - 2)}{2(x - 2)(x^2 + 2x + 4)} \\ &= \frac{3(4 + 4)(2 + 2)}{2(4 + 4 + 4)} = \frac{3 \times 8 \times 4}{2 \times 12} = 4 \\ \lim_{x \rightarrow 2-} f(x) &= \lim_{x \rightarrow 2-} x^2 = 2^2 = 4 \\ f(2) &= 4 \\ \therefore \lim_{x \rightarrow 2+} f(x) &= \lim_{x \rightarrow 2-} f(x) = f(2) \\ \therefore f(x) &\text{ is exist at } x = 2 \quad (\text{Ans}) \end{aligned}$$

1. If $y = x^3 e^{2x}$, then $\frac{dy}{dx}$:

(a) $3x^2 e^{2x} + 2x^3 e^{2x}$ (b) $3x^3 e^{2x} + 2x^2 e^{2x}$

(c) $3x e^{2x} + 2x e^{2x}$ (d) None

Solution:

$$y = x^3 e^{2x}$$

$$\therefore \frac{dy}{dx} = e^{2x} \frac{d}{dx}(x^3) + x^3 \frac{d}{dx}(e^{2x})$$

$$= 3x^2 e^{2x} + 2x^3 e^{2x}$$

Answer: (a)

2. If $y = a \cos x + b \sin x$, then $\frac{d^2y}{dx^2}$:

(a) $\cos x$ (b) $a \cos x + b \sin x$ (c) $-y$ (d) None

Solution:

$$y = a \cos x + b \sin x$$

$$\therefore \frac{dy}{dx} = -a \sin x + b \cos x$$

$$\therefore \frac{d^2y}{dx^2} = -a \cos x - b \sin x$$

$$= -(a \cos x + b \sin x) = -y$$

Answer: (c)

3. If $y = \log_a x$ then find the value of $\frac{dy}{dx}$

Solution:

$$y = \log_a x$$

$$\Rightarrow y = \log_a e \times \log_e x$$

$$\therefore \frac{dy}{dx} = \log_a e \times \frac{d}{dx}(\log_e x)$$

$$= \log_a e \times \frac{1}{x} = \frac{1}{x \log_e a} \quad (\text{Ans})$$

4. If $y = \sin x^\circ$, then find $\frac{dy}{dx}$.

Solution:

$$y = \sin x^\circ$$

$$\Rightarrow y = \sin\left(\frac{\pi}{180} x\right)$$

$$\therefore \frac{dy}{dx} = \frac{\pi}{180} \cos\left(\frac{\pi}{180} x\right)$$

$$= \frac{\pi}{180} \cos x^\circ \quad (\text{Ans})$$

5. If $y = \log x^2$, then find the value of $\frac{d^2y}{dx^2}$.

Solution:

$$y = \log x^2$$

$$\Rightarrow y = 2 \log x$$

Differentiating both side with respect to x

$$\frac{dy}{dx} = 2 \frac{d}{dx}(\log x) = 2 \times \frac{1}{x} = \frac{2}{x}$$

Again differentiating both side with respect to x

$$\frac{d^2y}{dx^2} = 2 \frac{d}{dx}\left(\frac{1}{x}\right) = 2 \times \left(-\frac{1}{x^2}\right) = -\frac{2}{x^2} \quad (\text{Ans})$$

6. If $y = \sin 2x$ then, find $\frac{d^2y}{dx^2}$ when $x = 0$.

Solution:

$$y = \sin 2x$$

Differentiating both side with respect to x

$$\frac{dy}{dx} = 2 \cos 2x$$

Again differentiating both side with respect to x

$$\frac{d^2y}{dx^2} = -4 \sin 2x$$

$$\Rightarrow \left[\frac{d^2y}{dx^2}\right]_{x=0} = -4 \sin 0 = 0 \quad (\text{Ans})$$

7. If $y = \log(\tan x)$ then find the value of $\frac{dy}{dx}$

Solution:

$$y = \log(\tan x)$$

$$\therefore \frac{dy}{dx} = \frac{1}{\tan x} \cdot \frac{d}{dx}(\tan x)$$

$$= \frac{\cos x}{\sin x} \cdot \sec^2 x$$

$$= \frac{\cos x}{\sin x} \cdot \frac{1}{\cos^2 x}$$

$$= \frac{2}{2 \sin x \cos x}$$

$$= \frac{2}{\sin 2x} = 2 \operatorname{cosec} 2x \quad (\text{Ans})$$

8. If $y = \log\left(\tan \frac{x}{2}\right)$ then find the value of $\frac{d^2y}{dx^2}$

Solution:

$$y = \log\left(\tan \frac{x}{2}\right)$$

$$\therefore \frac{dy}{dx} = \frac{1}{\tan \frac{x}{2}} \cdot \frac{d}{dx}\left(\tan \frac{x}{2}\right)$$

$$= \frac{1}{2} \cdot \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} \cdot \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} \cdot \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} \cdot \frac{1}{\cos^2 \frac{x}{2}}$$

$$= \frac{1}{2 \sin \frac{x}{2} \cos \frac{x}{2}}$$

$$= \frac{1}{\sin x} = \operatorname{cosec} x$$

$$\therefore \frac{d^2y}{dx^2} = -\operatorname{cosec} x \cot x \quad (\text{Ans})$$

9. If $y = \log(\sec x + \tan x)$ then find the value of $\frac{dy}{dx}$

Solution:

$$y = \log(\sec x + \tan x)$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sec x + \tan x} \cdot \frac{d}{dx}(\sec x + \tan x)$$

$$= \frac{1}{\sec x + \tan x} \cdot (\sec x \tan x + \sec^2 x)$$

$$= \frac{1}{(\sec x + \tan x)} \cdot \sec x (\sec x + \tan x)$$

$$= \sec x \quad (\text{Ans})$$

10. A ball travels s cm. in t sec. If $s = 8t + 15t^2$. Find the velocity and acceleration of the ball when $t = 2$.

Solution:

Given, $s = 8t + 15t^2$

$\therefore$ velocity $v = \frac{ds}{dt} = 8 + 15 \times 2t = 8 + 30t$

So, Velocity at $t = 2$

$= \left[\frac{ds}{dt} \right]_{t=2} = 8 + 30 \times 2 = 68 \text{ cm/s}$

And, Acceleration $= \frac{dv}{dt} = 0 + 30$
 $= 30 \text{ cm/s}^2$ (Ans)

11. Differentiate x^4 with respect to x^2 .

Solution:

Let,

$u = x^2$ $\therefore \frac{du}{dx} = 2x$ $\therefore$ Derivative of x^4 with respect to x^2 $= \frac{dv}{du} = \frac{\frac{dv}{dx}}{\frac{du}{dx}} = \frac{4x^3}{2x} = 2x^2$ (Ans)	$v = x^4$ $\therefore \frac{dv}{dx} = 4x^3$
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12. Differentiate x^2 with respect to $\log x$.

Solution:

Let,

$u = x^2$ $\therefore \frac{du}{dx} = 2x$ $\therefore$ Derivative of x^2 with respect to $\log x$ is $= \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{2x}{\frac{1}{x}} = 2x^2$ (Ans)	$v = \log x$ $\therefore \frac{dv}{dx} = \frac{1}{x}$
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13. If $y = \sin^{-1} x + \cos^{-1} x$ then find the value of $\frac{dy}{dx}$

Solution:

$y = \sin^{-1} x + \cos^{-1} x$
 $\Rightarrow y = \frac{\pi}{2}$
 $\therefore \frac{dy}{dx} = \frac{d}{dx} \left(\frac{\pi}{2} \right) = 0$ (Ans)

14. If $y = \sin^{-1} x + \sin^{-1} \sqrt{1-x^2}$ then find the value of $\frac{dy}{dx}$

Solution:

$y = \sin^{-1} x + \sin^{-1} \sqrt{1-x^2}$
 $\Rightarrow y = \sin^{-1} x + \cos^{-1} x$
 $\Rightarrow y = \frac{\pi}{2}$
 $\therefore \frac{dy}{dx} = \frac{d}{dx} \left(\frac{\pi}{2} \right) = 0$ (Ans)

15. If $y = \tan^{-1} x + \cot^{-1} x$ then find the value of $\frac{dy}{dx}$.

Solution:

$y = \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
 $\therefore \frac{dy}{dx} = \frac{d}{dx} \left(\frac{\pi}{2} \right) = 0$ (Ans)

16. If $y = \tan^{-1} x + \tan^{-1} \frac{1}{x}$ then find the value of $\frac{dy}{dx}$

Solution:

$y = \tan^{-1} x + \tan^{-1} \frac{1}{x}$
 $y = \tan^{-1} x + \cot^{-1} x$
 $y = \frac{\pi}{2}$
 $\therefore \frac{dy}{dx} = \frac{d}{dx} \left(\frac{\pi}{2} \right) = 0$ (Ans)

17. If $f(x) = x|x|$ then find the value of $f'(x)$

Solution:

$f(x) = x|x|$
 $f'(x) = 2|x|$ (Ans)

18. If $f(x) = e^{\sin^{-1} x}$ then find the value of $f'(x)$

Solution:

$f(x) = e^{\sin^{-1} x}$
 $\therefore f'(x) = e^{\sin^{-1} x} \cdot \frac{1}{\sqrt{1-x^2}} = \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$ (Ans)

19. If $y = e^{\sin^{-1} t}$ and $x = e^{-\cos^{-1} t}$, prove that $\frac{dy}{dx}$ is constant.

Solution:

Given, $y = e^{\sin^{-1} t}$
 $\therefore \frac{dy}{dt} = e^{\sin^{-1} t} \cdot \frac{d}{dt} (\sin^{-1} t) = e^{\sin^{-1} t} \times \frac{1}{\sqrt{1-t^2}}$
 And, $x = e^{-\cos^{-1} t}$
 $\therefore \frac{dx}{dt} = e^{-\cos^{-1} t} \cdot \frac{d}{dt} (-\cos^{-1} t) = e^{-\cos^{-1} t} \times \frac{1}{\sqrt{1-t^2}}$
 $\therefore \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{e^{\sin^{-1} t} \times \frac{1}{\sqrt{1-t^2}}}{e^{-\cos^{-1} t} \times \frac{1}{\sqrt{1-t^2}}} = e^{(\sin^{-1} t + \cos^{-1} t)}$
 $\Rightarrow \frac{dy}{dx} = e^{\left(\frac{\pi}{2} \right)} = \text{Constant}$ (Proved)

20. If $f(x) = e^{\tan^{-1} x}$ then find the value of $f'(x)$

Solution:

$f(x) = e^{\tan^{-1} x}$
 $\therefore f'(x) = e^{\tan^{-1} x} \cdot \frac{1}{1+x^2} = \frac{e^{\tan^{-1} x}}{1+x^2}$ (Ans.)

21. If $f(x) = \log e^x + e^{\log x}$ then find the value of $f'(x)$

Solution:

$$\begin{aligned} f(x) &= \log e^x + e^{\log x} \\ \Rightarrow f(x) &= x + x \\ \Rightarrow f(x) &= 2x \\ \Rightarrow f'(x) &= 2 \quad (\text{Ans}) \end{aligned}$$

22. If $x = at^2$ and $y = 2at$, then find $\frac{d^2y}{dx^2}$.

Solution:

$$x = at^2 \therefore \frac{dx}{dt} = 2at$$

$$y = 2at \therefore \frac{dy}{dt} = 2a$$

$$\text{So, } \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2a}{2at} = \frac{1}{t}$$

$$\begin{aligned} \therefore \frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx} \\ &= \frac{d}{dt} \left(\frac{1}{t} \right) \cdot \frac{1}{2at} \\ &= -\frac{1}{t^2} \cdot \frac{1}{2at} \\ &= -\frac{1}{2at^3} \quad (\text{Ans}) \end{aligned}$$

23. If $y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$ then find the value of $\frac{dy}{dx}$

Solution:

$$\text{Let, } x = \tan \theta$$

$$\Rightarrow \theta = \tan^{-1} x$$

$$\therefore y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$\Rightarrow y = \sin^{-1} \left(\frac{2 \tan \theta}{1+\tan^2 \theta} \right)$$

$$\Rightarrow y = \sin^{-1}(\sin 2\theta)$$

$$\Rightarrow y = 2\theta$$

$$\Rightarrow y = 2 \tan^{-1} x \therefore \frac{dy}{dx} = \frac{2}{1+x^2} \quad (\text{Ans})$$

24. If $y = \tan^{-1} \sqrt{\frac{1+x}{1-x}}$ then find the value of $\frac{dy}{dx}$

Solution:

$$\text{Let, } x = \cos 2\theta$$

$$\Rightarrow 2\theta = \cos^{-1} x, \Rightarrow \theta = \frac{1}{2} \cos^{-1} x$$

$$\therefore y = \tan^{-1} \sqrt{\frac{1+x}{1-x}}$$

$$\Rightarrow y = \tan^{-1} \sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}}$$

$$\Rightarrow y = \tan^{-1} \sqrt{\cot^2 \theta}$$

$$\Rightarrow y = \tan^{-1} \cot \theta$$

$$\Rightarrow y = \tan^{-1} \tan \left(\frac{\pi}{2} - \theta \right)$$

$$\Rightarrow y = \frac{\pi}{2} - \theta$$

$$\Rightarrow y = \frac{\pi}{2} - \frac{1}{2} \cos^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}} \quad (\text{Ans})$$

25. If $y = \tan^{-1} \left(\frac{a \cos x - b \sin x}{b \cos x + a \sin x} \right)$, then find $\frac{dy}{dx}$.

Solution:

$$y = \tan^{-1} \left(\frac{a \cos x - b \sin x}{b \cos x + a \sin x} \right)$$

$$= \tan^{-1} \left(\frac{\frac{a \cos x - b \sin x}{b \cos x}}{\frac{b \cos x + a \sin x}{b \cos x}} \right)$$

$$= \tan^{-1} \left(\frac{\frac{a}{b} - \tan x}{1 + \frac{a}{b} \tan x} \right)$$

$$= \tan^{-1} \left(\frac{a}{b} \right) - \tan^{-1}(\tan x)$$

$$= \tan^{-1} \left(\frac{a}{b} \right) - x \therefore \frac{dy}{dx} = 0 - 1 = -1 \quad (\text{Ans})$$

26. Differentiate $\tan^{-1} \left(\frac{2x}{1-x^2} \right)$ with respect to $\sin^{-1} \left(\frac{2x}{1+x^2} \right)$

Solution: Let, $x = \tan \theta, \Rightarrow \theta = \tan^{-1} x$

$$\text{And, } u = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$\Rightarrow u = \sin^{-1} \left(\frac{2 \tan \theta}{1+\tan^2 \theta} \right)$$

$$\Rightarrow u = \sin^{-1}(\sin 2\theta), \Rightarrow u = 2\theta$$

$$\Rightarrow u = 2 \tan^{-1} x \therefore \frac{du}{dx} = \frac{2}{1+x^2}$$

Again,

$$v = \sin^{-1} \left(\frac{2x}{1-x^2} \right)$$

$$\Rightarrow v = \sin^{-1} \left(\frac{2 \tan \theta}{1-\tan^2 \theta} \right)$$

$$\Rightarrow v = \sin^{-1}(\sin 2\theta)$$

$$\Rightarrow v = 2\theta$$

$$\Rightarrow v = 2 \tan^{-1} x \therefore \frac{dv}{dx} = \frac{2}{1+x^2}$$

$\therefore$ The required derivative is

$$= \frac{dv}{du} = \frac{\frac{dv}{dx}}{\frac{du}{dx}} = \frac{\frac{2}{1+x^2}}{\frac{2}{1+x^2}} = 1 \quad (\text{Ans})$$

27. Differentiate $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ with respect to $\tan^{-1} x$.

Solution:

$$\text{Let, } z = \tan^{-1} x, \Rightarrow x = \tan z$$

$$\text{and } y = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$$

$$= \tan^{-1} \left(\frac{\sqrt{1+\tan^2 z}-1}{\tan z} \right)$$

$$= \tan^{-1} \left(\frac{\sqrt{\sec^2 z}-1}{\tan z} \right)$$

$$= \tan^{-1} \left\{ (\sec z - 1) \times \frac{\cos z}{\sin z} \right\}$$

$$= \tan^{-1} \left\{ \left(\frac{1}{\cos z} - 1 \right) \times \frac{\cos z}{\sin z} \right\}$$

$$= \tan^{-1} \left\{ \left(\frac{1-\cos z}{\cos z} \right) \times \frac{\cos z}{\sin z} \right\}$$

$$= \tan^{-1} \left(\frac{2 \sin^2 \frac{z}{2}}{2 \sin \frac{z}{2} \cos \frac{z}{2}} \right) = \tan^{-1} \left(\tan \frac{z}{2} \right) = \frac{z}{2}$$

$$\Rightarrow y = \frac{z}{2} \therefore \frac{dy}{dz} = \frac{1}{2} \quad (\text{Ans})$$

28. Differentiate $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$ with respect to $\sqrt{1-x^2}$

Solution:

$$\text{Let, } x = \cos \theta, \Rightarrow \cos^{-1} x = \theta$$

$$\text{Now, } \sec^{-1}\left(\frac{1}{2x^2-1}\right)$$

$$= \sec^{-1}\left(\frac{1}{2\cos^2\theta-1}\right)$$

$$= \sec^{-1}\left(\frac{1}{\cos 2\theta}\right)$$

$$= \sec^{-1} \sec 2\theta = 2\theta$$

$$\text{and, } \sqrt{1-x^2} = \sqrt{1-\cos^2\theta} = \sin \theta$$

$\therefore$ Differentiation of 2θ with respect to $\sin \theta$.

$$= \frac{\frac{d}{d\theta}(2\theta)}{\frac{d}{d\theta}(\sin \theta)} = \frac{2}{\cos \theta} = \frac{2}{x} \quad (\text{Ans})$$

29. Find $\frac{dy}{dx}$. If $y = \cot^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$.

Solution:

$$y = \cot^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$$

$$= \cot^{-1}\left(\frac{1 - \tan x}{1 + \tan x}\right)$$

$$= \cot^{-1}\left(\frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x}\right)$$

$$= \cot^{-1} \tan\left(\frac{\pi}{4} - x\right)$$

$$= \cot^{-1} \cot\left(\frac{\pi}{2} - \frac{\pi}{4} + x\right) = \frac{\pi}{4} + x$$

$$\Rightarrow y = \frac{\pi}{4} + x \quad \therefore \frac{dy}{dx} = 0 + 1 = 1 \quad (\text{Ans})$$

30. If $y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, then find $\frac{dy}{dx}$.

Solution:

$$\text{Let, } x = \tan \theta, \Rightarrow \theta = \tan^{-1} x$$

$$\text{Now, } y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

$$= \sin^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)$$

$$= \sin^{-1}(\cos 2\theta)$$

$$= \sin^{-1} \sin\left(\frac{\pi}{2} - 2\theta\right) = \frac{\pi}{2} - 2\theta$$

$$\Rightarrow y = \frac{\pi}{2} - 2\theta$$

$$\Rightarrow y = \frac{\pi}{2} - 2 \tan^{-1} x$$

Differentiating both side with respect to x

$$\frac{dy}{dx} = 0 - 2 \cdot \frac{1}{1+x^2} = -\frac{2}{1+x^2} \quad (\text{Ans})$$

31. If $y = x^5$ and $x^2 \frac{d^2y}{dx^2} = ay$ find the value of a .

Solution:

$$\text{Given, } y = x^5$$

Differentiating both side with respect to x we get,

$$\frac{dy}{dx} = 5x^4$$

Differentiating both side with respect to x we get,

$$\frac{d^2y}{dx^2} = 20x^3$$

$$\text{Given, } x^2 \frac{d^2y}{dx^2} = ay$$

$$\Rightarrow x^2 \cdot 20x^3 = ax^5 \quad [\because y = x^5]$$

$$\Rightarrow a = 20 \quad (\text{Ans})$$

32. If $x = y \log(xy)$ then find the value of $\frac{dy}{dx}$

Solution:

$$x = y \log(xy), \Rightarrow \frac{x}{y} = \log(xy)$$

$$\Rightarrow y = \frac{x}{\log(xy)}$$

Differentiating both side with respect to x

$$\frac{dy}{dx} = \frac{\log(xy) \cdot 1 - x \cdot \frac{1}{xy} \left(y \cdot 1 + x \cdot \frac{dy}{dx}\right)}{\{\log(xy)\}^2}$$

$$\Rightarrow \{\log(xy)\}^2 \cdot \frac{dy}{dx} = \log(xy) - \frac{1}{y} \left(y + x \cdot \frac{dy}{dx}\right)$$

$$\Rightarrow \left(\frac{x}{y}\right)^2 \cdot \frac{dy}{dx} = \frac{x}{y} - 1 - \frac{x}{y} \cdot \frac{dy}{dx} \quad \left[\because \frac{x}{y} = \log(xy)\right]$$

$$\Rightarrow \frac{x}{y} \cdot \frac{dy}{dx} = 1 - \frac{y}{x} - \frac{dy}{dx}$$

$$\Rightarrow \frac{x}{y} \cdot \frac{dy}{dx} + \frac{dy}{dx} = 1 - \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{x}{y} + 1\right) = \frac{x-y}{x}$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{x+y}{y}\right) = \frac{x-y}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(x-y)}{x(x+y)} \quad (\text{Ans})$$

33. If $x^y = e^{x-y}$ then find the value of $\frac{dy}{dx}$

Solution:

$$x^y = e^{x-y}$$

$$\Rightarrow \log x^y = \log e^{x-y}$$

$$\Rightarrow y \log x = (x-y) \log e$$

$$\Rightarrow y \log x = (x-y)$$

$$\Rightarrow y \log x + y = x$$

$$\Rightarrow y(\log x + 1) = x, \Rightarrow y = \frac{x}{1+\log x}$$

Differentiating both side with respect to x

$$\frac{dy}{dx} = \frac{(1+\log x) \cdot 1 - x \cdot \frac{1}{x}}{(1+\log x)^2}$$

$$= \frac{1+\log x - 1}{(\log e + \log x)^2} = \frac{\log x}{(\log ex)^2} \quad (\text{Ans})$$

34. If $y = x^x$ then find $\frac{dy}{dx}$.

Solution:

$$y = x^x$$

Taking logarithm both side we get,

$$\log y = \log x^x$$

$$\Rightarrow \log y = x \log x$$

Differentiating both side with respect to x .

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = x \cdot \frac{1}{x} + \log x$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = 1 + \log x$$

$$\Rightarrow \frac{dy}{dx} = y(1 + \log x)$$

$$\Rightarrow \frac{dy}{dx} = x^x(1 + \log x) \quad (\text{Ans.})$$

35. If $v = \frac{A}{r} + B$ then show that $\frac{d^2v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = 0$

Solution:

$$v = \frac{A}{r} + B$$

Differentiating v with respect to r we get,

$$\frac{dv}{dr} = -\frac{A}{r^2} + 0$$

$$\Rightarrow r \frac{dv}{dr} = -\frac{A}{r}$$

Again, Differentiating with respect to r we get,

$$\Rightarrow r \frac{d^2v}{dr^2} + \frac{dv}{dr} = \frac{A}{r^2}$$

$$\Rightarrow r \frac{d^2v}{dr^2} + \frac{dv}{dr} = -\left(-\frac{A}{r^2}\right)$$

$$\Rightarrow r \frac{d^2v}{dr^2} + \frac{dv}{dr} = -\frac{dv}{dr}$$

$$\Rightarrow r \frac{d^2v}{dr^2} + 2 \frac{dv}{dr} = 0$$

$$\Rightarrow \frac{d^2v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = 0 \quad (\text{Proved})$$

36. If $x^y \cdot y^x = 1$ then find the value of $\frac{dy}{dx}$

Solution:

$$x^y \cdot y^x = 1$$

$$\Rightarrow \log(x^y \cdot y^x) = \log 1$$

$$\Rightarrow \log x^y + \log y^x = 0$$

$$\Rightarrow y \log x + x \log y = 0$$

Differentiating both side with respect to x

$$y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx} + x \cdot \frac{1}{y} \cdot \frac{dy}{dx} + \log y = 0$$

$$\Rightarrow \log x \cdot \frac{dy}{dx} + \frac{x}{y} \cdot \frac{dy}{dx} = -\left(\log y + \frac{y}{x}\right)$$

$$\Rightarrow \frac{dy}{dx} \left(\log x + \frac{x}{y}\right) = -\left(\log y + \frac{y}{x}\right)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{\left(\log y + \frac{y}{x}\right)}{\left(\log x + \frac{x}{y}\right)} \quad (\text{Ans})$$

37. If $x^y \cdot y^x = x + y$ then find the value of $\frac{dy}{dx}$

Solution:

$$x^y \cdot y^x = x + y$$

$$\Rightarrow \log(x^y \cdot y^x) = \log(x + y)$$

$$\Rightarrow \log x^y + \log y^x = \log(x + y)$$

$$\Rightarrow y \log x + x \log y = \log(x + y)$$

Differentiating both side with respect to x

$$y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx} + x \cdot \frac{1}{y} \cdot \frac{dy}{dx} + \log y = \frac{1}{x+y} \left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow \frac{y}{x} + \log x \cdot \frac{dy}{dx} + \frac{x}{y} \cdot \frac{dy}{dx} + \log y = \frac{1}{x+y} + \frac{1}{x+y} \cdot \frac{dy}{dx}$$

$$\Rightarrow \log x \cdot \frac{dy}{dx} + \frac{x}{y} \cdot \frac{dy}{dx} - \frac{1}{x+y} \cdot \frac{dy}{dx} = \frac{1}{x+y} - \log y - \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} \left(\log x + \frac{x}{y} - \frac{1}{x+y}\right) = \left(\frac{1}{x+y} - \log y - \frac{y}{x}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\log x + \frac{x}{y} - \frac{1}{x+y}}{\frac{1}{x+y} - \log y - \frac{y}{x}} \quad (\text{Ans})$$

38. Find $\frac{dy}{dx}$. If $y = (\tan x)^{\sin x}$ at $x = \frac{\pi}{4}$.

Solution:

$$y = (\tan x)^{\sin x}$$

$$\Rightarrow \log y = \log(\tan x)^{\sin x}$$

$$\Rightarrow \log y = \sin x \log(\tan x)$$

Differentiating both side with respect to x

$$\frac{1}{y} \cdot \frac{dy}{dx} = \cos x \log(\tan x) + \sin x \times \frac{1}{\tan x} \times \sec^2 x$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \cos x \log(\tan x) + \sin x \times \frac{\cos x}{\sin x} \times \frac{1}{\cos^2 x}$$

$$\Rightarrow \frac{dy}{dx} = y[\cos x \log(\tan x) + \sec x]$$

$$\Rightarrow \frac{dy}{dx} = (\tan x)^{\sin x} [\cos x \log(\tan x) + \sec x]$$

$$\Rightarrow \left[\frac{dy}{dx}\right]_{x=\frac{\pi}{4}} = \left(\tan \frac{\pi}{4}\right)^{\sin \frac{\pi}{4}} \left[\cos \frac{\pi}{4} \log\left(\tan \frac{\pi}{4}\right) + \sec \frac{\pi}{4}\right]$$

$$= 1[0 + \sqrt{2}] = \sqrt{2} \quad (\text{Ans.})$$

39. If $y = \log_{\cot x} \tan x$, prove that $\frac{dy}{dx} = 0$.

Solution:

$$\text{Given, } y = \log_{\cot x} \tan x$$

$$\Rightarrow y = \log_{\cot x} \left(\frac{1}{\cot x}\right)$$

$$\Rightarrow y = \log_{\cot x} (\cot x)^{-1}$$

$$\Rightarrow y = -\log_{\cot x} (\cot x)$$

$$\Rightarrow y = -1$$

Differentiating both side with respect to x we get,

$$\frac{dy}{dx} = \frac{d}{dx} (-1) = 0 \quad (\text{Proved})$$

40. If $y = (\cos x)^{\sin x}$ then find the value of $\frac{dy}{dx}$

Solution:

$$y = (\cos x)^{\sin x}$$

$$\Rightarrow \log y = \log(\cos x)^{\sin x}$$

$$\Rightarrow \log y = \sin x \log(\cos x)$$

Differentiating both side with respect to x

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \log(\cos x) \cdot \cos x - \sin x \cdot \frac{1}{\cos x} \sin x$$

$$\Rightarrow \frac{dy}{dx} = y \left(\cos x \log(\cos x) - \frac{\sin^2 x}{\cos x}\right)$$

$$\Rightarrow \frac{dy}{dx} = (\cos x)^{\sin x} \left[\log(\cos x) \cos x - \frac{\sin^2 x}{\cos x}\right] \quad (\text{Ans})$$

41. If $x = a \sin^2 \theta$ and $y = b \cos^2 \theta$ then find the value of $\frac{dy}{dx}$

Solution:

$$x = a \sin^2 \theta$$

$$\therefore \frac{dx}{d\theta} = a 2 \sin \theta \cos \theta$$

$$y = b \cos^2 \theta$$

$$\therefore \frac{dy}{d\theta} = -b 2 \cos \theta \sin \theta$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{-b 2 \cos \theta \sin \theta}{a 2 \sin \theta \cos \theta} = -\frac{b}{a} \quad (\text{Ans})$$

42. If $x^m \cdot y^n = (x + y)^{m+n}$ then prove that $\frac{dy}{dx} = \frac{y}{x}$

Solution:

$$x^m \cdot y^n = (x + y)^{m+n}$$

$$\Rightarrow \log(x^m \cdot y^n) = \log(x + y)^{m+n}$$

$$\Rightarrow \log x^m + \log y^n = (m + n) \log(x + y)$$

$$\Rightarrow m \log x + n \log y = (m + n) \log(x + y)$$

Differentiating both side with respect to x

$$\frac{m}{x} + \frac{n}{y} \cdot \frac{dy}{dx} = \frac{(m+n)}{(x+y)} \left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{m+n}{x+y} - \frac{n}{y}\right) = \frac{m}{x} - \frac{m+n}{x+y}$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{mx+my-nx-mx}{y(x+y)}\right) = \left(\frac{my+ny-nx-ny}{x(x+y)}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{(my-nx)}{y} = \frac{(my-nx)}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} \text{ (Proved)}$$

43. Find $\frac{dy}{dx}$. If $e^{xy} - 4xy = 4$

Solution: $e^{xy} - 4xy = 4$

Differentiating both side with respect to x

$$\frac{d}{dx}(e^{xy}) - \frac{d}{dx}(4xy) = \frac{d}{dx}(4)$$

$$\Rightarrow e^{xy} \frac{d}{dx}(xy) - 4 \frac{d}{dx}(xy) = 0$$

$$\Rightarrow \frac{d}{dx}(xy)(e^{xy} - 4) = 0$$

$$\Rightarrow \frac{d}{dx}(xy) = 0 \text{ [Assuming } e^{xy} \neq 4]$$

$$\Rightarrow x \frac{dy}{dx} + y = 0$$

$$\Rightarrow x \frac{dy}{dx} = -y, \Rightarrow \frac{dy}{dx} = -\frac{y}{x} \text{ (Ans)}$$

44. If $x = a(\theta - \sin \theta)$ and $y = a(1 + \cos \theta)$ then find the value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}$

Solution:

$$x = a(\theta - \sin \theta) \quad \left| \quad y = a(1 + \cos \theta)\right.$$

$$\therefore \frac{dx}{d\theta} = a(1 - \cos \theta) \quad \left| \quad \therefore \frac{dy}{d\theta} = -a \sin \theta\right.$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{-a \sin \theta}{a(1 - \cos \theta)} = -\frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}} = -\cot \frac{\theta}{2}$$

$$\therefore \left[\frac{dy}{dx}\right]_{\theta=\frac{\pi}{2}} = -\cot \frac{\pi}{4} = -1 \text{ (Ans)}$$

45. If $\sin y = x \sin(a + y)$ then prove that

$$\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$$

Solution:

$$\sin y = x \sin(a + y)$$

$$\Rightarrow x = \frac{\sin y}{\sin(a+y)}$$

$$\therefore \frac{dx}{dy} = \frac{\sin(a+y) \cdot \cos y - \sin y \cos(a+y)}{\sin^2(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a+y-y)}{\sin^2(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin a}{\sin^2(a+y)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a} \text{ (Proved)}$$

46. $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$, find $\frac{d^2y}{dx^2}$.

Solution:

$$x = a(\theta + \sin \theta), \Rightarrow \frac{dx}{d\theta} = a(1 + \cos \theta)$$

$$y = a(1 - \cos \theta), \Rightarrow \frac{dy}{d\theta} = a \sin \theta$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a \sin \theta}{a(1 + \cos \theta)} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = \tan \frac{\theta}{2}$$

$$\therefore \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx}\right) = \frac{d}{d\theta} \left(\frac{dy}{dx}\right) \cdot \frac{d\theta}{dx}$$

$$= \frac{d}{d\theta} \left(\tan \frac{\theta}{2}\right) \cdot \frac{1}{a(1 + \cos \theta)}$$

$$= \frac{1}{2} \sec^2 \frac{\theta}{2} \times \frac{1}{a \cdot 2 \cos^2 \frac{\theta}{2}}$$

$$= \frac{1}{4a} \sec^2 \frac{\theta}{2} \times \sec^2 \frac{\theta}{2}$$

$$= \frac{1}{4a} \sec^4 \frac{\theta}{2} \text{ (Ans)}$$

47. If $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ then find the value of

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Solution:

$$x = a \cos^3 \theta$$

$$\therefore \frac{dx}{d\theta} = a \cdot 3 \cos^2 \theta (-\sin \theta) = -3a \sin \theta \cos^2 \theta$$

$$y = a \sin^3 \theta$$

$$\therefore \frac{dy}{d\theta} = a \cdot 3 \sin^2 \theta \cos \theta = 3a \sin^2 \theta \cos \theta$$

$$\text{So, } \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3a \sin^2 \theta \cos \theta}{-3a \sin \theta \cos^2 \theta} = -\tan \theta$$

$$\therefore \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + (-\tan \theta)^2}$$

$$= \sqrt{1 + \tan^2 \theta}$$

$$= \sec \theta$$

$$\text{Given, } x = a \cos^3 \theta$$

$$\Rightarrow \cos^3 \theta = \frac{x}{a}, \Rightarrow \cos \theta = \left(\frac{x}{a}\right)^{\frac{1}{3}}$$

$$\Rightarrow \sec \theta = \left(\frac{a}{x}\right)^{\frac{1}{3}}$$

$$\text{So, } \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \left(\frac{a}{x}\right)^{\frac{1}{3}} \text{ (Ans)}$$

48. If $\cos y = x \cos(a + y)$ then prove that

$$\frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}$$

Solution:

$$\sin y = x \sin(a + y)$$

$$\Rightarrow x = \frac{\sin y}{\sin(a+y)}$$

$$\therefore \frac{dx}{dy} = \frac{-\cos(a+y) \cdot \sin y + \cos y \sin(a+y)}{\sin^2(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a+y-y)}{\sin^2(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin a}{\sin^2(a+y)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a} \text{ (Proved)}$$

49. If $y = (\sin^{-1} x)^2$ then show that

$$(1 - x^2)y_2 - xy_1 = 2$$

Solution:

$$y = (\sin^{-1} x)^2$$

Differentiating both side with respect to x

$$\frac{dy}{dx} = 2 \sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2} \cdot y_1 = 2 \sin^{-1} x \left[\because \frac{dy}{dx} = y_1 \right]$$

$$\Rightarrow (1-x^2)y_1^2 = (2 \sin^{-1} x)^2$$

$$\Rightarrow (1-x^2)y_1^2 = 4(\sin^{-1} x)^2$$

$$\Rightarrow (1-x^2)y_1^2 = 4y \left[\because y = (\sin^{-1} x)^2 \right]$$

Differentiating both side with respect to x

$$(1-x^2) \cdot 2y_1y_2 + y_1^2(-2x) = 4y_1$$

$$\Rightarrow (1-x^2) \cdot y_2 - xy_1 = 2 \quad (\text{Proved})$$

50. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$ then find the value of $\frac{dy}{dx} = \sqrt{\frac{1-x^2}{1-y^2}}$

Solution:

$$\text{Let, } x = \cos \alpha, y = \cos \beta$$

$$\Rightarrow \alpha = \cos^{-1} x, \beta = \cos^{-1} y$$

$$\because \sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$$

$$\therefore \sqrt{1-\cos^2 \alpha} + \sqrt{1-\cos^2 \beta} = a(\cos \alpha - \cos \beta)$$

$$\Rightarrow \sin \alpha + \sin \beta = a(\cos \alpha - \cos \beta)$$

$$\Rightarrow 2 \sin \left(\frac{\alpha+\beta}{2} \right) \cos \left(\frac{\alpha-\beta}{2} \right) = a \cdot 2 \sin \left(\frac{\alpha+\beta}{2} \right) \cdot \sin \left(\frac{\beta-\alpha}{2} \right)$$

$$\Rightarrow \cos \left(\frac{\alpha-\beta}{2} \right) = -a \sin \left(\frac{\alpha-\beta}{2} \right)$$

$$\Rightarrow \cot \left(\frac{\alpha-\beta}{2} \right) = -a$$

$$\Rightarrow \frac{\alpha-\beta}{2} = \cot^{-1}(-a)$$

$$\Rightarrow \alpha - \beta = 2 \cot^{-1}(-a)$$

$$\Rightarrow \cos^{-1} x - \cos^{-1} y = 2 \cot^{-1}(-a)$$

Differentiating both side with respect to x

$$-\frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-y^2}} \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{1}{\sqrt{1-y^2}} \cdot \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}} \quad (\text{Proved})$$

51. If $y = e^{m \sin^{-1} x}$ then show that

$$(1-x^2)y_2 - xy_1 - m^2y = 0$$

Solution:

$$y = e^{m \sin^{-1} x}$$

Differentiating both side with respect to x

$$y_1 = me^{m \sin^{-1} x} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1(\sqrt{1-x^2}) = me^{m \sin^{-1} x}$$

$$\Rightarrow y_1(\sqrt{1-x^2}) = my$$

$$\Rightarrow y_1^2(1-x^2) = m^2y^2$$

Differentiating both side with respect to x

$$(1-x^2) \cdot 2y_1y_2 + y_1^2(-2x) = m^2 \cdot 2yy_1$$

$$\Rightarrow (1-x^2)y_2 - xy_1 - m^2y = 0 \quad (\text{Proved})$$

52. If $\sin^{-1} x = \log_e y$ then show that

$$(1-x^2)y_2 - xy_1 - y = 0$$

Solution:

$$\sin^{-1} x = \log_e y, \Rightarrow y = e^{\sin^{-1} x}$$

Differentiating both side with respect to x

$$y_1 = e^{\sin^{-1} x} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1(\sqrt{1-x^2}) = e^{\sin^{-1} x}$$

$$\Rightarrow y_1(\sqrt{1-x^2}) = y$$

$$\Rightarrow y_1^2(1-x^2) = y^2$$

Differentiating both side with respect to x

$$(1-x^2) \cdot 2y_1y_2 + y_1^2(-2x) = 2yy_1$$

$$\Rightarrow (1-x^2)y_2 - xy_1 - y = 0 \quad (\text{Proved})$$

53. If $x^y = e^{x-y}$, then prove that: $\frac{dy}{dx} = \frac{\log x}{(\log ex)^2}$

Solution:

$$\text{Given, } x^y = e^{x-y}$$

$$\Rightarrow \log_e x^y = \log_e e^{x-y}$$

$$\Rightarrow y \log_e x = (x-y) \log_e e$$

$$\Rightarrow y \log_e x = x - y$$

$$\Rightarrow y + y \log_e x = x$$

$$\Rightarrow y(1 + \log_e x) = x$$

$$\Rightarrow y(\log_e e + \log_e x) = x$$

$$\Rightarrow y \log_e ex = x$$

$$\Rightarrow y = \frac{x}{\log_e ex}$$

Differentiating both side with respect to x

$$\frac{dy}{dx} = \frac{\log_e ex \cdot 1 - x \cdot \frac{1}{x}}{(\log_e ex)^2}$$

$$= \frac{\log_e ex + \log_e x - 1}{(\log_e ex)^2}$$

$$= \frac{1 + \log_e x - 1}{(\log_e ex)^2}$$

$$= \frac{\log_e x}{(\log_e ex)^2} = \frac{\log x}{(\log ex)^2} \quad (\text{Proved})$$

54. If $y = (x + \sqrt{1+x^2})^m$, then prove that

$$(1+x^2)y_2 + xy_1 - m^2y = 0$$

Solution:

$$y = (x + \sqrt{1+x^2})^m$$

$$\Rightarrow \log y = \log(x + \sqrt{1+x^2})^m$$

$$\Rightarrow \log y = m \log(x + \sqrt{1+x^2})$$

Differentiating both side with respect to x

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{m}{x + \sqrt{1+x^2}} \left(1 + \frac{1}{2\sqrt{1+x^2}} \times 2x \right)$$

$$= \frac{m}{x + \sqrt{1+x^2}} \left(1 + \frac{x}{\sqrt{1+x^2}} \right)$$

$$= \frac{m}{x + \sqrt{1+x^2}} \left(\frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2}} \right) = \frac{m}{\sqrt{1+x^2}}$$

$$\Rightarrow \frac{1}{y} \times y_1 = \frac{m}{\sqrt{1+x^2}}$$

$$\Rightarrow y_1 \sqrt{1+x^2} = my$$

$$\Rightarrow (y_1 \sqrt{1+x^2})^2 = m^2y^2$$

$$\Rightarrow y_1^2(1+x^2) = m^2y^2$$

Differentiating both side with respect to x

$$(1+x^2) \cdot 2y_1y_2 + y_1^2 \cdot 2x = m^2 \cdot 2yy_1$$

$$\Rightarrow (1+x^2)y_2 + xy_1 = m^2y$$

$$\Rightarrow (1+x^2)y_2 + xy_1 - m^2y = 0 \quad (\text{Proved})$$

55. If $y = \sin(m \sin^{-1} x)$ then show that
 $(1 - x^2)y_2 - xy_1 + m^2y = 0$

Solution:

$$y = \sin(m \sin^{-1} x)$$

$$\therefore y_1 = m \cos(m \sin^{-1} x) \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2} \cdot y_1 = m \cos(m \sin^{-1} x)$$

$$\Rightarrow (1-x^2)y_1^2 = m^2 \cos^2(m \sin^{-1} x)$$

$$\Rightarrow (1-x^2)y_1^2 = m^2(1 - \sin^2(m \sin^{-1} x))$$

$$\Rightarrow (1-x^2)y_1^2 = m^2(1 - y^2)$$

Differentiating both side with respect to x

$$(1-x^2) \cdot 2y_1y_2 + y_1^2(-2x) = -m^2 2yy_1$$

$$\Rightarrow (1-x^2) \cdot y_2 - xy_1 + m^2y = 0 \text{ (Proved)}$$

56. If $y = \sin(2 \sin^{-1} x)$ then show that
 $(1 - x^2)y_2 - xy_1 + 4y = 0$

Solution:

$$y = \sin(2 \sin^{-1} x)$$

$$\therefore y_1 = 2 \cos(2 \sin^{-1} x) \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2} \cdot y_1 = 2 \cos(2 \sin^{-1} x)$$

$$\Rightarrow (1-x^2)y_1^2 = 4 \cos^2(2 \sin^{-1} x)$$

$$\Rightarrow (1-x^2)y_1^2 = 4(1 - \sin^2(2 \sin^{-1} x))$$

$$\Rightarrow (1-x^2)y_1^2 = 4(1 - y^2)$$

Differentiating both side with respect to x

$$(1-x^2) \cdot 2y_1y_2 + y_1^2(-2x) = -42yy_1$$

$$\Rightarrow (1-x^2) \cdot y_2 - xy_1 + 4y = 0 \text{ (Proved)}$$

57. If $y = ae^{mx} + be^{-mx}$ then show that

$$\frac{d^2y}{dx^2} - m^2y = 0$$

Solution:

$$y = ae^{mx} + be^{-mx}$$

Differentiating both side with respect to x

$$\frac{dy}{dx} = ame^{mx} - bme^{-mx}$$

Differentiating both side with respect to x

$$\frac{d^2y}{dx^2} = am^2e^{mx} + bm^2e^{-mx}$$

$$\Rightarrow \frac{d^2y}{dx^2} = m^2(ae^{mx} + be^{-mx})$$

$$\Rightarrow \frac{d^2y}{dx^2} = m^2y$$

$$\Rightarrow \frac{d^2y}{dx^2} - m^2y = 0 \text{ (Proved)}$$

58. The maximum value of the function,

$$f(x) = 3x - x^2:$$

(a) -2 (b) 9/4 (c) 2 (d) None

Solution:

$$f(x) = 3x - x^2, \Rightarrow f'(x) = 3 - 2x$$

$$\Rightarrow f''(x) = -2 < 0$$

For max. value of x , $f'(x) = 0$

$$\Rightarrow 3 - 2x = 0, 2x = 3, \Rightarrow x = \frac{3}{2}$$

$\therefore$ Max. value of $f(x)$ is

$$= 3 \times \frac{3}{2} - \left(\frac{3}{2}\right)^2 = \frac{9}{2} - \frac{9}{4} = \frac{9}{4}$$

Answer: (b)

59. The function $(3 - x)(x - 1)$ is maximum for $x = ?$

(a) 1

(b) 2

(c) 3

(d) 4

Solution:

$$\text{Let, } f(x) = (3 - x)(x - 1)$$

$$\Rightarrow f(x) = 3x - 3 - x^2 + x = -x^2 + 4x - 3$$

$$\therefore f'(x) = -2x + 4$$

$$f(x) \text{ is maximum for } f'(x) = 0$$

$$\therefore -2x + 4 = 0$$

$$\Rightarrow 2x = 4, \Rightarrow x = 2$$

Answer: (b)

60. For real value of x , find the maximum value of

$$f(x) = 5 + 4x - 4x^2.$$

Solution:

$$f(x) = 5 + 4x - 4x^2$$

$$\Rightarrow f'(x) = 0 + 4 - 8x$$

For maximum value of x , $f'(x) = 0$

$$\Rightarrow 4 - 8x = 0$$

$$\Rightarrow 8x = 4$$

$$\Rightarrow x = \frac{4}{8} = \frac{1}{2}$$

$\therefore$ The maximum value of $f(x)$

$$= f\left(\frac{1}{2}\right) = 5 + 4 \times \frac{1}{2} - 4\left(\frac{1}{2}\right)^2$$

$$= 5 + 2 - 1$$

$$= 6 \text{ (Ans)}$$

61. Show that $(\sin x + \cos x)$ will maximum at $x = \frac{\pi}{4}$.

Solution:

$$\text{Let, } f(x) = \sin x + \cos x$$

$$\therefore f'(x) = \cos x - \sin x$$

$$\therefore f''(x) = -\sin x - \cos x = -(\sin x + \cos x)$$

For maximum or minimum value of $f(x)$

$$f'(x) = 0$$

$$\Rightarrow \cos x - \sin x = 0$$

$$\Rightarrow \cos x = \sin x$$

$$\Rightarrow \tan x = 1 = \tan \frac{\pi}{4}$$

$$\Rightarrow x = n\pi + \frac{\pi}{4} \quad [n = 0, \pm 1, \pm 2, \pm 3, \dots]$$

$$\Rightarrow x = \frac{\pi}{4} \quad [\text{When } n = 0]$$

$$\text{Now, } f''\left(\frac{\pi}{4}\right) = -\left(\sin \frac{\pi}{4} + \cos \frac{\pi}{4}\right)$$

$$= -\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)$$

$$= -\frac{2}{\sqrt{2}} = -\sqrt{2} < 0$$

So, $f(x)$ has maximum value at $x = \frac{\pi}{4}$. (Proved)

62. Find the minimum value of $(x - 1)(x - 2)$.

Solution:

$$\text{Let, } f(x) = (x - 1)(x - 2)$$

$$\Rightarrow f(x) = x^2 - 2x - x + 2$$

$$\Rightarrow f(x) = x^2 - 3x + 2$$

$$\therefore f'(x) = 2x - 3$$

$$\text{and } f''(x) = 2 > 0$$

For maximum or minimum value of $f(x)$

$$f'(x) = 0$$

$$\Rightarrow 2x - 3 = 0$$

$$\Rightarrow 2x = 3, \Rightarrow x = \frac{3}{2}$$

$\therefore$ The minimum value of $f(x)$

$$= f\left(\frac{3}{2}\right) = \left(\frac{3}{2} - 1\right)\left(\frac{3}{2} - 2\right)$$

$$= \frac{3-2}{2} \times \frac{3-4}{2}$$

$$= \frac{1}{2} \times \frac{-1}{2} = -\frac{1}{4} \text{ (Ans)}$$

63. Find the maximum and minimum value of

$$f(x) = 2x^3 - 21x^2 + 36x - 20$$

Solution:

$$f(x) = 2x^3 - 21x^2 + 36x - 20$$

$$\therefore f'(x) = 6x^2 - 42x + 36$$

$$\text{and, } f''(x) = 12x - 42$$

For maximum and minimum value $f'(x) = 0$

$$\Rightarrow 6x^2 - 42x + 36 = 0$$

$$\Rightarrow x^2 - 7x + 6 = 0$$

$$\Rightarrow x^2 - 6x - x + 6 = 0$$

$$\Rightarrow x(x - 6) - 1(x - 6) = 0$$

$$\Rightarrow (x - 6)(x - 1) = 0$$

$$\Rightarrow x = 6 \text{ or } x = 1$$

$$\text{Now, } f''(6) = 12 \times 6 - 42 = 30 > 0$$

$$\text{and } f''(1) = 12 \times 1 - 42 = -30 < 0$$

So, $f(x)$ has minimum value at $x = 6$

and maximum value at $x = 1$

$$\therefore f_{\max} = f(1) = 2 - 21 + 36 - 20 = -3 \text{ (Ans.)}$$

$$\text{and } f_{\min} = f(6) = 2 \cdot 6^3 - 21 \cdot 6^2 + 36 \cdot 6 - 20$$

$$= 432 - 756 + 216 - 20$$

$$= -128 \text{ (Ans.)}$$

64. Divide 20 into two parts such that their product to be maximum.

Solution:

Let, one part is x , so other part be $(20 - x)$.

$$\text{and } f(x) = x(20 - x)$$

$$\Rightarrow f(x) = 20x - x^2$$

$$\Rightarrow f'(x) = 20 - 2x$$

$$\Rightarrow f''(x) = -2 < 0$$

For maximum value of $f(x)$

$$\Rightarrow f'(x) = 20 - 2x = 0$$

$$\Rightarrow 20 = 2x, \Rightarrow x = 10$$

$\therefore$ For maximum value of product the required two parts will be 10 and 10. (Ans)

65. Find the maximum and minimum value of

$$\left(x^3 + \frac{1}{x^3}\right).$$

Solution:

$$\text{Let, } f(x) = x^3 + \frac{1}{x^3} = x^3 + x^{-3}$$

$$\Rightarrow f'(x) = 3x^2 - 3x^{-4} = 3\left(x^2 - \frac{1}{x^4}\right)$$

$$\therefore f''(x) = 6x + 12x^{-5} = 6\left(x + \frac{2}{x^5}\right)$$

For maximum and minimum value $f'(x) = 0$

$$\Rightarrow 3\left(x^2 - \frac{1}{x^4}\right) = 0$$

$$\Rightarrow x^2 - \frac{1}{x^4} = 0$$

$$\Rightarrow x^2 = \frac{1}{x^4}$$

$$\Rightarrow x^6 = 1, \Rightarrow x = \pm 1$$

$$\text{Now, } f''(1) = 6\left(1 + \frac{2}{1^5}\right) = 18 > 0$$

$$\text{and } f''(-1) = 6\left(-1 + \frac{2}{(-1)^5}\right) = -18 < 0$$

So, $f(x)$ has minimum value at $x = 1$ and maximum value at $x = -1$

$$\therefore f_{\max} = f(-1) = (-1)^3 + \frac{1}{(-1)^3} = -2 \text{ (Ans)}$$

$$\text{and } f_{\min} = f(1) = (1)^3 + \frac{1}{(1)^3} = 2 \text{ (Ans)}$$

66. A parachutist falls through a distance $x = \log_e(6 - 5e^{-t})$ in t^{th} second of its motion. Find $\frac{dx}{dt}$ at $t = 0$.

Solution:

$$\text{Given, } x = \log_e(6 - 5e^{-t})$$

Differentiating both side with respect to t we get.

$$\frac{dx}{dt} = \frac{1}{6-5e^{-t}} \times \frac{d}{dt}(6 - 5e^{-t})$$

$$= \frac{1}{6-5e^{-t}} \times 5e^{-t}$$

$$\therefore \left[\frac{dx}{dt}\right]_{t=0} = \frac{1}{6-5e^0} \times 5e^0 = \frac{5}{6-5} = 5 \text{ unit. (Ans)}$$

67. If the area of a circle changes uniformly with respect to time, show that the rate of change of its circumference varies inversely as its radius.

Solution:

Let, r be the radius of the circle.

Then by the problem, $\frac{d}{dt}(\pi r^2) = k$ (Constant)

$$\Rightarrow 2\pi r \cdot \frac{dr}{dt} = k$$

$$\Rightarrow \frac{dr}{dt} = \frac{k}{2\pi r}$$

Now, circumference of the circle, $c = 2\pi r$

$$\therefore \frac{dc}{dt} = 2\pi \frac{dr}{dt} = 2\pi \times \frac{k}{2\pi r} = \frac{k}{r}$$

$$\Rightarrow \frac{dc}{dt} \propto \frac{1}{r}$$

$\therefore$ The rate of change of its circumference varies inversely as its radius. (Proved)

68. A particle is moving in a straight line such that its distance "x" at any time "t" is given by

$$x = \frac{1}{4}t^4 - 2t^3 + 4t^2 - 7$$

Find the value of "t" when its velocity is maximum and acceleration is minimum.

Solution:

$$\text{Given, } x = \frac{1}{4}t^4 - 2t^3 + 4t^2 - 7$$

$$\therefore \text{Velocity } v = \frac{dx}{dt} = t^3 - 6t^2 + 8t$$

$$\text{Acceleration } a = \frac{dv}{dt} = 3t^2 - 12t + 8$$

$$\text{For max. velocity } 3t^2 - 12t + 8 = 0$$

$$\Rightarrow t = \frac{12 \pm \sqrt{(-12)^2 - 4 \cdot 3 \cdot 8}}{2 \cdot 3}$$

$$\therefore t = \frac{12 + \sqrt{48}}{6} = 3.155$$

$$\text{Or, } t = \frac{12 - \sqrt{48}}{6} = 0.845$$

$$\text{Now, } \frac{da}{dt} = 6t - 12$$

$$\Rightarrow \left[\frac{da}{dt} \right]_{t=3.155} = 6 \times 3.155 - 12 = 6.93 > 0$$

$$\Rightarrow \left[\frac{da}{dt} \right]_{t=0.845} = 6 \times 0.845 - 12 = -6.93 < 0$$

$\therefore$ Velocity is max. at $t = 0.845 \text{ sec.}$ (Ans)

$$\text{For max. or min. acceleration } \frac{da}{dt} = 6t - 12 = 0$$

$$\Rightarrow 6t = 12, \Rightarrow t = 2 > 0$$

$\therefore$ Acceleration is min. at $t = 2 \text{ sec.}$ (Ans)

69. A particle is moving along a straight line, after time t the distance of the particle s is given by an equation $s = a \cos nt + b \sin nt$. Prove that the acceleration of the particle is proportional to its distance. (a, b, n is constant)

Solution:

$$\text{Given, } s = a \cos nt + b \sin nt$$

Differentiating both side with respect to t we get,

$$\frac{ds}{dt} = -an \sin nt + bn \cos nt$$

Again, differentiating side with respect to t ,

$$\frac{d^2s}{dt^2} = -an^2 \cos nt - bn^2 \sin nt$$

$$\Rightarrow \frac{d^2s}{dt^2} = -n^2(a \cos nt + b \sin nt)$$

$$\Rightarrow \frac{d^2s}{dt^2} = -n^2s$$

$$\Rightarrow \frac{d^2s}{dt^2} \propto s \quad [\because n \text{ constant}]$$

i.e acceleration is proportional to distance.

(Proved)

70. An open tank with square base is to be made by metallic sheet of 48 sq.mt. What will be the dimensions of the tank so that it may contain maximum quantity of water?

Solution:

$$\text{Let, Side of square base} = a \text{ m}$$

$$\text{Height of tank} = h \text{ m}$$

$$\text{Volume of the tank} = v \text{ m}^3$$

Since, the upper side of the tank is open.

$$\therefore \text{The metallic surface of the tank} = a^2 + 4ah$$

$$\text{By the problem, } a^2 + 4ah = 48$$

$$\Rightarrow 4ah = 48 - a^2$$

$$\Rightarrow h = \frac{48 - a^2}{4a}$$

$$\text{Now, } v = a^2 h = a^2 \left(\frac{48 - a^2}{4a} \right) = 12a - \frac{a^3}{4}$$

$$\therefore \frac{dv}{da} = 12 - \frac{3}{4}a^2 \quad \text{and} \quad \frac{d^2v}{da^2} = -\frac{3}{2}a$$

$$\text{For maximum value of } v, \frac{dv}{da} = 0$$

$$\Rightarrow 12 - \frac{3}{4}a^2 = 0$$

$$\Rightarrow \frac{3}{4}a^2 = 12$$

$$\Rightarrow a^2 = 12 \times \frac{4}{3} = 4^2$$

$$\Rightarrow a = \pm 4$$

$$\text{Now, } \left[\frac{d^2v}{da^2} \right]_{a=4} = -\frac{3}{2} \times 4 \quad [\text{Taking +ve value of } a]$$

$$= -6 < 0$$

$$\therefore v \text{ is maximum when } a = 4 \text{ m}$$

$$\text{and } h = \frac{48 - a^2}{4a} = \frac{48 - 4^2}{4 \times 4} = \frac{32}{16} = 2 \text{ m}$$

$$\text{So, each side of the square tank is } = 4 \text{ m. (Ans)}$$

$$\text{and height of the tank is } = 2 \text{ m (Ans)}$$

{Prepared by NatiTute YouTube Channel}